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Learning-Based Security Analysis: Contextual Representations and Adversarial Evaluation

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Technische Universität München

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Abstract

This dissertation examines two questions in learning-based security analysis. The first concerns representation: which observations of a program or a network a detector should receive, and how they should be structured. The second concerns evaluation, namely how a defence can be measured when an adversary may choose the test data.

In a collaborative study of IoT microservices, each node stores the service relations observed at start-up as a whitelist that the site owner can revise; detection added about 6 ms to a round trip of 24 ms. In industrial control systems, the modular framework NADICS places the learning stage behind fixed interfaces. A random forest trained on an S7-300 corpus from the IUNO project reached an attack F_1 of 94.50 per cent at a false-positive rate of 0.484 per cent under a session-disjoint split. A GAN-based detector at the same interface flagged 92.00 per cent of anomalous SWaT timestamps at a false-positive rate of 5.12 per cent on later test data.

Two Android systems form the main contribution. Hybroid combines an embedding of the function-call graph with 13 features of the network flows that an application produces. On CICAndMal2017, fusion added one or two points of detection F_1 over the better single modality, and between minus two and plus three points in malware-category classification. HGANN-Mal lifts the function-call graph to a hypergraph whose hyperedges include the backward slices of security-sensitive call sites, and it learns hyperedge membership weights by attention. Across Drebin and CICMalDroid, every hypergraph operator exceeded both pairwise graph operators in all four task and corpus combinations. HGANN-Mal led the non-attentive hypergraph operators by 1.2 to 4.1 points of accuracy in three combinations and was level in the fourth. Since construction and attention changed together, an ablation that isolates attention remains open.

Within the SPARTA project, an adversary model was translated into a contest on a hidden test set and then into a modular benchmark that combines attacks and defences by configuration. The accuracy-loss score used for ranking assigns zero loss to any constant classifier under label-preserving attacks; under PGD, the defence with zero loss had an attacked accuracy 12.9 points below that of adversarial training. The four contest results do not establish a general relation between clean performance and robustness, so the two should be reported together.

Kurzfassung

Diese Dissertation untersucht zwei Fragen der lernbasierten Sicherheitsanalyse. Die erste betrifft die Repräsentation: Welche Beobachtungen eines Programms oder Netzwerks soll ein Detektor erhalten, und wie sollen sie strukturiert sein? Die zweite betrifft die Evaluation, also die Messung eines Abwehrverfahrens, wenn ein Angreifer die Testdaten wählen kann.

In einer kollaborativen Studie zu IoT-Microservices speichert jeder Knoten die beim Start beobachteten Dienstbeziehungen als vom Betreiber anpassbare Positivliste; die Erkennung kostete etwa 6 ms bei einer Umlaufzeit von 24 ms. In industriellen Steuerungssystemen kapselt das modulare Framework NADICS die Lernstufe hinter festen Schnittstellen. Ein Random Forest erreichte auf einem S7-300-Korpus aus dem Projekt IUNO unter sitzungsdisjunkter Aufteilung einen F_1 -Wert der Angriffsklasse von 94,50 % bei einer Falsch-Positiv-Rate von 0,484 %. Ein GAN-basierter Detektor an derselben Schnittstelle erkannte auf späteren Testdaten 92,00 % der anomalen SWaT-Zeitstempel bei einer Falsch-Positiv-Rate von 5,12 %.

Den Hauptbeitrag bilden zwei Android-Systeme. Hybroid kombiniert eine Einbettung des Funktionsaufrufgraphen mit 13 Merkmalen der Netzwerkflüsse einer Anwendung. Auf CI-CAndMal2017 erhöhte die Fusion den F_1 -Wert der Erkennung um ein bis zwei Prozentpunkte gegenüber der besseren Einzelmodalität; bei der Klassifikation nach Schadsoftwarekategorien lag der Effekt zwischen minus zwei und plus drei Punkten. HGANN-Mal überführt den Funktionsaufrufgraphen in einen Hypergraphen, dessen Hyperkanten auch Rückwärts-Slices sicherheitsrelevanter Aufrufstellen umfassen, und lernt Zugehörigkeitsgewichte mit Attention. Auf Drebin und CICMalDroid übertraf jeder Hypergraphoperator in allen vier Kombinationen aus Aufgabe und Korpus beide paarweisen Operatoren. HGANN-Mal lag in drei Kombinationen um 1,2 bis 4,1 Prozentpunkte Genauigkeit vor den Hypergraphoperatoren ohne Attention und in der vierten gleichauf. Da Konstruktion und Attention gemeinsam verändert wurden, fehlt noch eine Ablation, welche die Attention isoliert.

Im Projekt SPARTA wurde ein Angreifermodell in einen Wettbewerb mit verdecktem Testdatensatz und dann in einen modularen Benchmark überführt, der Angriffe und Abwehrverfahren per Konfiguration kombiniert. Der zur Rangbildung genutzte Genauigkeitsverlust weist jedem konstanten Klassifikator unter labelerhaltenden Angriffen den Wert null zu; unter PGD lag das verlustfreie Abwehrverfahren in der Genauigkeit unter Angriff 12,9 Prozentpunkte hinter adversarialem Training. Aus den vier Wettbewerbsergebnissen lässt sich kein allgemeiner Zusammenhang zwischen Genauigkeit und Robustheit ableiten; beide sollten gemeinsam berichtet werden.

Acknowledgements

Replace this text with your acknowledgements.

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Introduction

1.1 Research Context

Detection systems built on machine learning are ordinary practice in security engineering. Drebin classified Android applications from statically extracted feature sets in 2014 [1], MaMaDroid modelled sequences of abstracted API calls as Markov chains soon afterwards [2], and a systematic review published in 2023 identified 132 studies applying deep learning to Android malware defence between 2014 and 2021 [3]. The same shift took place in network defence, where anomaly detection moved from statistical baselining [4] towards deep architectures trained directly on traffic [5], and in program analysis, where graph neural networks are now applied to code structure [6, 7, 8]. Android alone reports more than three billion active devices [9], and the number of applications submitted for analysis each year is far larger than any team of analysts can inspect by hand. That imbalance is what makes automated classification attractive.

Industrial and Internet of Things (IoT) networks reached the same point by a different route. Their traffic is comparatively regular, since the devices are few and their interactions are prescribed by the process they control, which is what makes learned models of normal behaviour plausible there at all. Testbeds built for the purpose supply the labelled attack data that operational plants cannot [10]. The difficulty is that a model of normal behaviour has to be kept current as the deployment changes, and in a service-oriented architecture it changes often.

The methods were developed for settings that differ from security in two ways that matter. A classifier trained on one sample of data and deployed on another assumes that both come from the same distribution. Malware violates the assumption by construction: families rise and fall, developers repackage applications specifically to defeat the detectors of the previous year, and the label an antivirus engine assigns to a file may change as engines are updated. Jordaney et al. measured this drift directly and showed that a classifier's confidence gives little warning of it [11]. The second difference is that the data are chosen by someone who benefits when classification fails.

How much reported performance survives a stricter protocol has been measured. Published F1 scores for Android malware classification reach 0.99, which leaves little room for improvement and invites the question of what the number means. Pendlebury et al. traced the figure to two forms of experimental bias: training and test data whose composition does not match a real deployment, and time splits that let a classifier learn from samples that appeared after the ones it is tested on [12]. Arp et al. then examined thirty papers from the four leading security conferences against a list of ten methodological pitfalls. Every paper exhibited at least three of

them [13]. Artefacts in the datasets add to the effect. Drebin, the most widely used Android corpus, carries duplicate samples produced by repackaging [14], although the systematic study of what those duplicates cost found the effect limited for supervised classification and considerably larger for unsupervised family clustering [15]. None of these findings concerns a single weak paper. They concern the standard protocol.

A second body of evidence concerns what happens once the adversary is treated as active. Small, deliberately chosen perturbations flip the output of trained models [16], and the observation predates deep learning: Biggio et al. demonstrated gradient-based evasion against detectors at test time in 2013 [17], with a line of work on adversarial classification running back further still [18]. The defensive response has not held up well. Athalye et al. re-examined the nine non-certified defences accepted at ICLR 2018 and found that seven owed their apparent robustness to obfuscated gradients, of which six were then circumvented completely [19]. Tramèr et al. broke a further thirteen defences published at leading machine-learning venues, all of which had already been evaluated against adaptive attacks by their own authors [20]. Carlini et al. responded by writing down what an adequate robustness evaluation requires [21]. In the malware setting the constraint is tighter than in image classification, because an adversarial example must be a file that still installs and still runs [22].

Two questions follow from this, and they organise the present work. The first is a question about representation: what should be observed about a program or a network, and how should those observations be structured, so that a model is given the evidence that separates malicious from benign behaviour? The second is a question about evaluation: how should a defence be measured when the party generating the test data is trying to defeat it? The two are usually pursued by different communities. This dissertation treats them together, because a representation that is never tested against an adaptive opponent and a benchmark that never varies the representation are weak in complementary ways.

1.2 Problem Statement and Scope

The representation problem

An analysis system sees only what its observation model admits. Static analysis reads the application package and reasons about code that may never execute, which gives broad coverage at the cost of accuracy against obfuscation and dynamic loading; the limits are well understood and were formalised early [23]. Dynamic analysis observes what a sample actually does, using instrumentation such as taint tracking [24], which resists obfuscation but reaches only the paths that the harness triggers [25]. Neither observation model dominates the other, so systems that combine them should be able to describe behaviour that neither describes alone. What such a combination buys, measured on the same samples under one protocol, is the first open question this dissertation addresses.

Structure raises a second problem. Program behaviour is commonly encoded as a graph, with functions or basic blocks as nodes and calls or transfers as edges, and graph neural networks then learn over that structure. An edge relates exactly two elements. Behaviour that

is security-relevant frequently involves more: a permission request, a sequence of API calls, and a network write may be innocuous separately and malicious in combination. Encoding such a group as a set of pairwise edges loses the fact that the members belong to one group. Hypergraphs remove that restriction by allowing an edge to join any number of nodes [26], and hypergraph neural networks provide the learning machinery [27, 28]. Whether the higher-order encoding pays for itself on Android program structure, and whether weighting hyperedges by attention adds anything beyond the encoding itself, is the second open question. The question is worth asking sceptically, since attention mechanisms in graph learning have been shown to be less expressive than their formulation suggests [29].

The evaluation problem

Robustness claims are only as good as the attacks used to test them, and the record shows that attacks chosen by the defender's own authors are too weak [19, 21]. An evaluation that is credible has to begin from an explicit statement of what the adversary knows and what the adversary may change, and it has to be reproducible by a third party. Building such an evaluation as a piece of infrastructure, rather than as a section of a paper, is the third open question treated here.

Scope

The empirical work concerns Android. Applications are analysed from the package as distributed, using Dalvik bytecode and, where the design calls for it, traffic recorded while the application runs. Native code, kernel-level instrumentation, and iOS lie outside the scope. The industrial and IoT material concerns communication behaviour observed at the network level and, in the SWaT study, sensor and actuator values of the physical process. It was recorded in testbeds and project deployments, not at operator scale.

Three limits on the claims should be stated here rather than discovered later. First, the work on communication graphs for IoT services was disseminated as a two-page contribution to PAM 2018 on which the candidate is the fourth of four authors. The full paper that reports anomaly detection on that system was written by two of his co-authors without him [30], as was the companion short paper on graph-based access control [31]. Chapter 3 identifies both and scopes the claim accordingly. Second, the graph work on IoT services did not lead to the Android work. The two are compared as representations in Chapter 7, and no developmental sequence between them is claimed. Third, the adversarial material is a contribution to evaluation methodology and to benchmark construction. No new attack algorithm and no certified defence is proposed.

1.3 Research Questions

Four questions organise the dissertation. The first concerns the representation of behaviour in networked and industrial systems and is divided into two subquestions. The second and third concern the representation of Android program behaviour. The fourth concerns evaluation.

Each question takes up one of the four gaps stated in Section 2.7, and Chapter 8 answers each of them in the same words in which they are asked here.

RQ1: Behaviour in networked and industrial systems

How can behaviour in networked and industrial systems be represented to support anomaly detection?

RQ1 takes up gap G1. Its two subquestions organise three empirical strands over service relations, network records, and industrial process data. The strands are complementary rather than cumulative, and no experiment compares all of them.

RQ1.1: Communication graphs

How can service relationships be represented as communication graphs to identify unexpected interactions?

Chapter 3 answers RQ1.1 through a relation model for service-to-service communication, a procedure for keeping the graph current as services change, and measurements of the overhead this imposes. Detection accuracy against a labelled attack set is not reported, and the chapter states why.

RQ1.2: Modular industrial anomaly detection

How can a modular learning framework analyse network and process features for anomaly detection in industrial control systems?

Chapter 3 answers RQ1.2 through NADICS, a modular framework for network and process features with interchangeable preprocessing and learning components. The chapter reports its IUNO integration and a multivariate reconstruction detector on SWaT. Here, reconstruction means that the model reproduces its input readings and uses the discrepancies to help detect anomalies.

RQ2: Multimodal Android malware analysis

To what extent does fusing static code-graph representations with dynamic network-flow features improve Android malware detection and category classification compared with either modality alone?

RQ2 takes up gap G2. Chapter 4 answers it through Hybroid, which extracts features from program structure and from recorded traffic and classifies their combination. The chapter reports binary detection and malware-category classification, and it compares each modality against the fusion under one protocol.

RQ3: Higher-order Android malware analysis

To what extent do attention-weighted hypergraph representations of static Android program structure improve binary malware detection and malware classification over pairwise graph models and non-attentive hypergraph models?

RQ3 takes up gap G3. Chapter 5 answers it through HGANN-Mal, which represents an application as a hypergraph over its call structure and applies attention across hyperedges. Two classification tasks are distinguished throughout and are not interchangeable: malware-family classification on Drebin, and malware-category classification on CICMalDroid, in which benign applications form one of the categories. Binary detection is a third, separate task.

RQ4: Adversarial robustness evaluation

How can an explicit adversary model be translated into a reproducible benchmark for adversarial attacks and defences, and what does such a benchmark reveal about the relation between clean predictive performance and adversarial robustness?

RQ4 takes up gap G4. Chapter 6 answers it through a threat analysis of machine-learning systems, a public contest run on a hidden test set, and a modular benchmark in which attacks and defences are combined by configuration. It examines how task definitions, scoring and attack validity affect the interpretation of the reported results, and identifies the experimental records needed for reproduction.

1.4 Contributions

A communication-graph model for IoT service interaction. Service-to-service relations are represented as a graph that is built from observed traffic and updated as the deployment changes, together with a measurement of the cost of maintaining it. The contribution answers RQ1.1 and is reported in Chapter 3. Its scope is limited by the evidence available: the disseminated artefact runs to two pages, and the chapter states what the candidate contributed to it.

A modular framework for anomaly detection in industrial networks. NADICS decomposes packet parsing, feature construction, and classification into interchangeable components, so that learning methods can be exchanged without rebuilding the pipeline, and it fed the visual security control room built by the IUNO project. The later work adds a GAN-based detector for multivariate SWaT process data. The contribution answers RQ1.2 and is reported in Chapter 3.

Multimodal Android malware analysis. Hybroid combines a static representation of program structure with features derived from observed network flows, and evaluates the combination against each modality alone on the same samples. The contribution answers RQ2 and is reported in Chapter 4.

Higher-order Android malware analysis. HGANN-Mal encodes groups of related program elements as hyperedges rather than as sets of pairwise edges, and weights those hyperedges by attention. The contribution answers RQ3 and is reported in Chapter 5, which compares the result against pairwise graph models and against non-attentive hypergraph models. Because the encoding and the weighting were changed together, the chapter does not attribute the gain to attention alone.

Threat-guided evaluation of adversarial defences. An adversary model for machine-learning systems is translated into an evaluation protocol, realised first as a contest with a hidden test set and then as a modular benchmark in which attacks and defences are composed by configuration. The chapter derives the behaviour of accuracy-loss scoring under constant predictions and distinguishes gradient masking from prediction collapse. It analyses the response policy of binary adversarial detectors and compares the published clean and attacked accuracies, with unresolved experimental mechanisms identified as such. The contribution answers RQ4 and is reported in Chapter 6. Where the underlying project deliverables were written collectively, the chapter states what the candidate designed and implemented and what he synthesised from the work of partners.

A comparison across the contributions. Chapter 7 relates the representations to one another along the axes of observability, data-acquisition cost, and validity of evaluation. The comparison is conceptual. Numerical results from the individual studies are not pooled, because their datasets and validation protocols differ.

1.5 Publications, Projects, and Individual Contributions

Parts of this dissertation have been disseminated previously, in accordance with the doctoral regulations of the Technical University of Munich. Table 1.1 lists every such item, with the candidate’s position in the author list and the chapter that draws on it. Where a chapter reuses previously published material, it rewrites it rather than reprinting it, and the corresponding results are restated from the original experimental records.

Table 1.1. Previously disseminated parts of this dissertation.

Item	Venue or type	Year	Position	Chapter
Graph-based Anomaly Detection for IoT Microservices [32]	PAM, two pages	2018	4 of 4	3
IUNO work package 4, visual security control room [33]	Project deliverable	2018	contributor	3
IUNO work package 7, security tools [34]	Project deliverable	2018	contributor	3
Hybroid [35]	ISC, Springer LNCS	2021	1 of 4	4
HGANN-Mal [36]	ISC, Springer LNCS	2025	1 of 2	5
SPARTA D7.1, AI systems threat analysis [37]	Public deliverable	2021	contributor	6
SPARTA D7.5, AI systems security mechanisms [38]	Public deliverable	2021	contributor	6
SPARTA D7.6, Validation and evaluation report [39]	Public deliverable	2022	editor	6
SAFAIR AI Contest development toolkit [40]	Software repository	2021	sole author	6

Individual contribution

The two-page PAM 2018 contribution on graph-based anomaly detection for IoT microservices [32] arose from a collaboration in which the candidate is the last of four authors. The peer-reviewed full paper on anomaly detection in that system was published by two of those

co-authors without him [30], and so was the short paper on graph-based access control in the same architecture [31]. Chapter 3 therefore presents the communication-graph material with its authorship stated explicitly, and it limits the originality claim to the parts the candidate designed and implemented.

Two final deliverables of the IUNO project [33, 34] report work the candidate carried out at the chair. Both are consortium documents whose author lists sit at the level of the document, so neither assigns any individual passage to one contributor. Within the project the candidate designed and implemented NADICS, together with the extractor that derives connection features from S7 and PROFINET traffic. He also produced the labelled attack corpus the deliverables describe, recorded between Siemens S7-300 controllers and their control server and labelled for binary detection as well as for five classes of attack. An earlier internal deliverable of June 2017 [41] documents the first version of that tool and names him among its authors. That document was not published, which is why it is absent from Table 1.1. Chapter 3 states which components are his, and it records the passages in which the deliverable text credits the implementation with functionality it did not yet have.

The candidate also designed, implemented, evaluated, and documented an unpublished extension reported in Chapter 3. It combines a multivariate GAN architecture with reconstruction and critic scoring for SWaT process data. He carried out all of its experiments and wrote the internal research report, which was not disseminated and is therefore absent from Table 1.1.

For Hybrid [35] the candidate is the first of four authors, and the paper records that the first two share first authorship. He designed the pipeline as a whole and the network-traffic analysis within it, together with the framework through which the dataset was analysed, from preprocessing through learning to the reported results. He carried out the experiments and wrote the paper, and he contributed in part to the treatment of the graph data. HGANN-Mal [36] is the work of the candidate with his supervisor. The candidate designed the method, implemented it, carried out the experiments and wrote the paper.

Within the SPARTA project, the candidate contributed to the threat analysis of machine-learning systems reported in [37] and to the contest concept in [38], and he is the editor of [39], in which the benchmark and the contest results are reported. The first two deliverables were written by a consortium and attribute contributors at the level of the document rather than the chapter. Chapter 6 is therefore written afresh, cites each taxonomy it draws on in place, and confines its contribution claim to the evaluation design and to the benchmark implementation, which are the candidate's own.

1.6 Dissertation Structure

Chapter 2 sets out the foundations shared by the later chapters: Android applications and their analysis, graph and hypergraph representations of programs, learning over graphs, detection in networked and cyber-physical systems, adversarial machine learning, and evaluation methodology. It closes with a statement of the gaps that the four research questions address. The work against which each contribution is positioned is treated in the chapter that reports it. Chapter 3 reports communication-graph, network-flow, and industrial process-anomaly

studies and answers RQ1. Chapter 4 presents Hybroid and answers RQ2. Chapter 5 presents HGANN-Mal and answers RQ3. These two chapters carry the main Android empirical weight of the dissertation. Chapter 6 presents the threat-guided evaluation work and answers RQ4. Chapter 7 sets the contributions against one another, and Chapter 8 answers each research question in turn and states what remains open.

Every contribution chapter carries its own account of the work specific to its method, its own evaluation, and its own statement of limitations. There is no shared methodology chapter, because the studies have no common experimental method to share.

Table 1.2 maps each research question to the chapter that answers it, to the material that supports the answer, and to the candidate's role in producing that material. The reader who wants to know what any particular claim rests on should start there.

Table 1.2. Research questions, chapters, and the material each answer rests on.

Question	Chapter	Supporting material	Role
RQ1.1	3	PAM 2018, 2 pp. [32], project artefacts	contributor
RQ1.2	3	NADICS, internal experimental records, IUNO deliverables [33, 34]	design, implementation, evaluation
RQ2	4	ISC 2021 [35]	first author
RQ3	5	ISC 2025 [36]	first author
RQ4	6	SPARTA D7.1, D7.5, D7.6 [37, 38, 39]	contributor, editor

Background

This chapter sets out the material the later chapters share: the artefacts an Android application exposes to analysis, the graph and hypergraph representations used to describe programs, the learning operators defined over those representations, detection in networked and cyber-physical systems, adversarial machine learning, and the methodology by which any of it is assessed. Positioning against competing systems is not attempted here. Every contribution chapter carries its own account of the work specific to its method, and that is where published systems are compared.

Learning-based security analysis has a common shape. Samples are collected and labelled, a representation is derived from each sample, a model is fitted, its output is thresholded, and the result is measured on data held back from training. A methodological defect can enter at any of these stages, and the defects that recur have been catalogued stage by stage [13]. Four assumptions carried over from other application domains do not hold in this setting. Training and test data are taken to come from one stationary distribution, which the evolution of malware denies. The available sample is taken to represent the deployment, which its composition and its timeline usually deny. Errors are treated as symmetric and the classes as roughly balanced, whereas operational traffic is neither [42, 43]. The data are assumed not to be chosen by an opponent. Section 2.6 takes up the first three and Section 2.5 the fourth.

The Android sections open the chapter, because Android malware analysis is the centre of the dissertation and because a hyperedge is easier to read once function-call graphs have been defined. One consequence of that ordering should be signalled here. Section 2.4 supports Chapter 3 although it appears after the Android material, so a reader following chapter order meets the foundations later than the chapter that draws on them. Section 3.1 repeats the pointer.

2.1 Android Applications and Their Analysis

Chapters 4 and 5 both take an Android application package as input and derive features from it. This section fixes what that package contains, what each mode of analysis is able to observe in it, and the two mechanisms that most often prevent observation altogether. Network-flow representation is deferred to Section 2.4, and the graph objects the two chapters build are defined in Section 2.2.

2.1.1 Dalvik Bytecode, Manifest, and Permissions

An Android application is distributed as an application package, a ZIP archive whose security-relevant members are a manifest and one or more files of compiled bytecode [3]. Source written in Java or Kotlin is compiled to bytecode for the Dalvik virtual machine [1] and stored in `classes.dex`, with further numbered files such as `classes2.dex` for larger applications. The container is specified by the platform [44], as is the instruction set [45]. The Android Runtime (ART) replaced Dalvik as the default runtime from Android 5 and executes DEX bytecode [46]. Its instructions encode both opcodes and operands. Operand fields can contain register indices or literal values and can reference other DEX items, such as methods and strings [45]. Section 4.5 describes an embedding that retains the opcode while omitting this operand information.

The manifest declares the components an application provides, the permissions it requests, the intents its components filter, and the hardware features it expects. It is recoverable without any reasoning about code, which is what makes it the cheapest security-relevant metadata available: four of Drebin's eight feature sets are taken from the manifest and the remaining four from the disassembled bytecode [1].

Access to sensitive functionality is mediated by permissions, each carrying a protection level, of which only the level marked *dangerous* guards capability able to cause real harm [47, 48]. Two properties of this system bear on the later chapters. The mapping from a permission to the API calls it guards is documented incompletely and has to be recovered by analysing the platform: PScout extracted it from roughly 3.4 million lines of Android source across four releases and found more than 75 permissions whose guarded API the documentation did not describe [49]. The specification is also unstable, since fine-grained least privilege trades against stability across releases [49], and deprecation on the platform side has been measured as one source of the drift reported for permission-based detection [50]. Section 5.5 relies on both properties, because it selects permissions by protection level and maps flagged API calls back to their call sites.

2.1.2 Static, Dynamic, and Hybrid Analysis

An application may be analysed without running it, by running it, or by combining the two. Figure 2.1 sets the three modes against the artefacts of the previous subsection. The static mode dominates practice: across the studies of deep learning for Android malware defence reviewed by Liu et al., 73 per cent worked statically, 17 per cent dynamically, and 10 per cent combined both [3].

Static analysis reads the package and reasons about code that may never execute, which yields coverage of everything shipped at modest computational cost. The engineering is mature. Disassembly and call-graph construction are supplied by standard toolkits [51, 52], and taint analysis that models Android's component lifecycle is available [53]. What no amount of engineering removes is a ceiling on what static reasoning can decide. An adversary is able to construct obfuscation whose analysis reduces to an NP-hard problem, which defeats semantics-aware static detection [23]. That construction was demonstrated on x86 binaries,

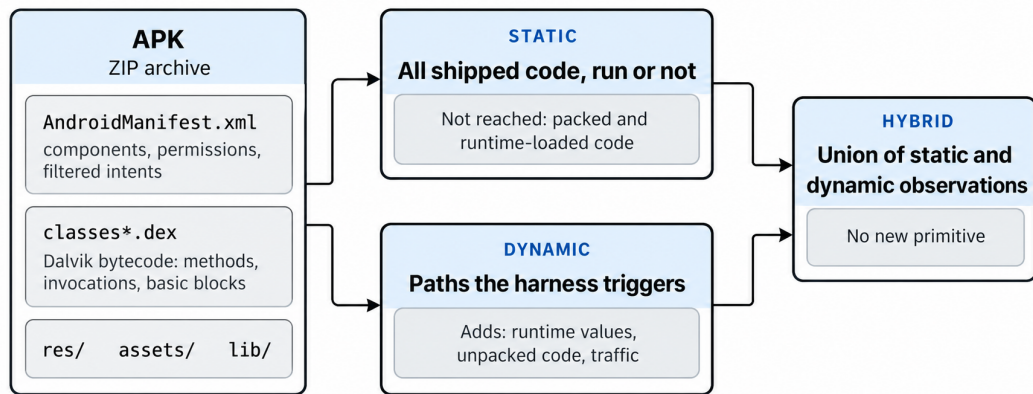


Figure 2.1. What an application package holds and what each mode of analysis reaches. Static observation covers all shipped code and not what the code does; dynamic observation covers what the code did, but only along the paths the harness triggered, and it alone reaches network traffic, which is the channel Chapter 4 uses. The graph objects built from the bytecode are named in Section 2.2.

and the argument carries to Dalvik by the same reasoning rather than having been shown there directly.

Dynamic analysis observes the application while it runs. Instrumentation ranges from taint tracking on the device [24] to whole-system emulation that reconstructs the operating-system and Dalvik views together [54]. Traffic recorded during execution is a further channel, and it is the one Chapter 4 uses. Two limits apply throughout. Only the paths the harness triggers are observed, and an application may detect the analysis environment and withhold the behaviour under investigation [55, 56].

Neither mode dominates the other. Measured under one protocol, static analysis proved at least as effective as dynamic analysis, depending on how the applications were stimulated during execution, while the combination of the two measured best [25]. Hybrid pipelines are consequently an established design [57], and the question Chapter 4 takes up is what such a combination buys on one set of samples under a single protocol.

2.1.3 Obfuscation, Packing, and Repackaging

Obfuscation is ordinary developer practice as much as attacker practice. Of 1.7 million free applications on Google Play, 24.92 per cent had been obfuscated by their developers [58], and a measurement across 114,560 applications drawn from Google Play, third-party markets and malware corpora found string encryption more frequent in malware and packing more frequent outside Google Play [59]. The effect on detection is measurable. Three thousand benign and three thousand malicious applications, expanded into 73,362 obfuscated variants by 29 strategies, left most commercial anti-malware products severely affected, and trivial transformations sufficed [60]. A later study of twelve detectors found that none held its ideal-setting performance once obfuscation, concept drift and adversarial examples were introduced [61].

Packing is a distinct mechanism and is kept separate here. A packed application ships an encrypted original together with a wrapper that decrypts and loads it at runtime, so static extraction fails outright instead of degrading. Commercial packers are used by malware authors, and custom packers are written specifically to defeat the available unpackers [62].

The packaging model also allows an attacker to graft a malicious rider onto a benign carrier [63]. Applications produced this way are near-identical outside the injected payload, and corpora assembled from them therefore contain duplicates. What those duplicates cost depends on the task: the measured effect is limited for supervised classification and substantially larger for unsupervised family clustering [15]. Section 2.6 returns to this as a problem of evaluation rather than of extraction.

2.2 Graph Representations of Programs

Representations are defined here, before the operators that learn over them. Two families are needed. Chapters 4 and 5 both derive a graph from program structure, and Chapter 5 then lifts that graph to a hypergraph. The notation introduced in this section is the notation those chapters use, and neither redefines it.

2.2.1 Control-Flow and Function-Call Graphs

A *control-flow graph* is a directed graph over the basic blocks of a single procedure. Its vertices are maximal instruction sequences that contain no branch except possibly at the end, and a directed edge records that control may pass from one block to the next. The object is intra-procedural by definition [8]. A *call graph* takes the procedures of a program as its vertices, with an edge from f to g recording that a call site in f may invoke g . It is inter-procedural, and a statically recovered call graph over-approximates, since the analysis admits edges that no execution takes. For Android the call graph is taken at method granularity over the recovered bytecode, and this dissertation calls that object a *function-call graph*: vertices are methods, and directed edges are method invocations [64, 36]. The two objects are kept distinct throughout, because their vertex sets and their edge semantics differ.

The two compose. Embeddings computed on per-method control-flow graphs may be attached to the vertices of a function-call graph, which makes intra-procedural and inter-procedural structure available together [8]. Dropping the call edges from that composition leaves a third object: a set of independently embedded control-flow graphs, pooled to application level, with no invocation structure at all. The three are named here so that each contribution chapter can state which of them it builds without introducing new machinery, and so that a claim made about one is not read as a claim about another.

Whichever object is chosen, it is an artefact of the analyser rather than a property of the program. Recovery depends on the toolkit and on how the component lifecycle is modelled [52, 53, 51], and abstracting call targets to package or family level shrinks the vertex set at the cost of the callee's identity [2]. Vertex features may equally be learned rather than hand-built, from the block structure of a procedure [65, 66] or from the body of a method [67].



Figure 2.2. The same six methods represented as (a) a hypergraph, (b) its incidence matrix H , and (c) its unweighted clique expansion. A matrix entry is one precisely when the method belongs to the corresponding hyperedge. Hyperedge e_3 joins two vertices and is therefore an ordinary graph edge. The lower example shows that the triangle over m_1, m_2, m_3 arises both from the single hyperedge e_1 and from three separate pairwise relations. The projection preserves pairwise adjacency but does not uniquely determine the original hyperedge memberships [26].

2.2.2 Hypergraphs and Higher-Order Relations

A hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E}, w)$ has a finite vertex set $\mathcal{V}$ and a family $\mathcal{E}$ of subsets of $\mathcal{V}$, each subset a hyperedge e carrying a weight $w(e) > 0$. Membership is recorded by the incidence matrix $H \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}|}$, where $h(v, e) = 1$ exactly when $v \in e$. The degree of a vertex is $d(v) = \sum_{e \in \mathcal{E}} w(e) h(v, e)$ and the degree of a hyperedge is $\delta(e) = \sum_{v \in \mathcal{V}} h(v, e) = |e|$. Writing D_v and D_e for the diagonal matrices of vertex and hyperedge degrees and W for the diagonal matrix of hyperedge weights completes the apparatus [26]. A graph is the special case in which $\delta(e) = 2$ for every hyperedge, so every graph is a hypergraph and the converse fails.

An edge relates exactly two vertices, and encoding a group of several as a set of pairwise edges loses the fact that its members belonged to one group. The projected graph cannot distinguish a single hyperedge on three vertices from three separate pairwise relations among the same vertices [26]. Two projections are standard. *Clique expansion* replaces each hyperedge by the complete graph on its members, and *star expansion* introduces a new vertex per hyperedge joined to each member [68]. Figure 2.2 shows the hyperedge memberships and the information lost in their pairwise projection.

The sceptical reading of these definitions deserves stating, because it is the stronger position to argue from. Agarwal et al. show that the hypergraph learning formulations of their period reduce to the Laplacian of one expansion or the other, that the two coincide for uniform hypergraphs, and that for non-uniform ones they differ only in how hyperedges of unequal size are weighted; their conclusion is that graphs, not hypergraphs, lie at the heart of the problem [68]. Chitra and Raphael sharpen the boundary. A random walk on a hypergraph whose vertex weights do not vary between the hyperedges a vertex belongs to, the *edge-independent* case, is equivalent to a walk on a weighted clique expansion, so an operator built on such

a walk carries no higher-order information. Only *edge-dependent* vertex weights escape the equivalence [69].

What the projection costs has since been measured rather than asserted. An evaluation framework that expands, learns and then reconstructs reports depressed performance at vertex, hyperedge and hypergraph level across eight datasets [70]; two recurring hyperedge patterns have been identified as the source of the loss in every projection, together with a combinatorial impossibility result on recovering the discarded structure [71]; and an expressivity hierarchy names the invariants a clique expansion cannot represent [72]. Section 2.3 describes the operators built on this apparatus, and Chapter 5 develops the one it uses.

2.3 Learning over Graphs

The operators described here learn representations over a structure whose construction was fixed in Section 2.2. Three families are relevant: neighbourhood aggregation on pairwise graphs, attention-weighted aggregation, and convolution over an incidence structure. This section states what each does and what is known about its limits. The notation stays as defined above, and the operator Chapter 5 builds on is developed there rather than here.

2.3.1 Message Passing

Most graph neural networks share one scheme. Each vertex holds a state, that state is updated from the states of its neighbours by a permutation-invariant aggregation, and the update is applied for a fixed number of rounds [73]. The convolutional instance fixes the aggregation coefficients in advance from the degrees of the two endpoints, so a vertex combines its neighbours in proportions that the graph determines and training does not alter [6].

Two limits on this family are established. The first is a bound on what it can tell apart: no network in the family distinguishes two graphs that one-dimensional Weisfeiler-Leman colour refinement assigns the same colours, and an architecture reaching that bound is as discriminative as the scheme permits [74]. The second concerns depth. Repeated propagation drives the representations within a connected component towards one another, so adding rounds to widen a vertex's receptive field costs discriminability between vertices [75]. Both bear on program graphs, where the structures compared are often large and locally similar.

2.3.2 Attention

Attention replaces fixed coefficients with learned ones. A scoring function is applied to each vertex and its neighbour, the scores across a neighbourhood are normalised, and the resulting weights determine the combination [7]; the mechanism was introduced for sequences before it was carried to graphs [76]. The motivation is that some neighbours matter more than others, and that which ones matter should be learned rather than read off the degree.

Three results qualify that motivation, and they should be kept apart. Brody et al. show that the ranking the standard formulation induces over a vertex's neighbours is the same for every query vertex, so the mechanism selects globally important neighbours rather than neighbours important to a particular vertex; a revised scoring function removes the restriction [29]. Fang et al. bound both scoring functions, the linear form used by the original mechanism and the multi-layer form introduced by its revision, showing that each realises only finitely many neighbour rankings at bounded width and depth [77]. The same work reports that on node-property benchmarks over pairwise graphs, evaluated with GCN, GraphSAGE and GIN as non-attentive baselines, no unmodified attentive model outperformed all of those baselines on any dataset or task [77]. The defensible reading of that last result is negative about the evidence rather than about the mechanism: an advantage for attention over fixed aggregation has not been demonstrated on those benchmarks, and the expressive power of the standard scoring functions is bounded.

Whether either finding transfers to operators defined over hyperedges is not established in the literature, since both analyses concern pairwise operators. Section 5.8 shows that the static-ranking argument carries over to the additive hyperedge score used there. For other scoring functions over hyperedges, and for the finite-ranking bound, the question is open, and Section 2.3.3 returns to it.

A methodological point follows, and it concerns how such architectures are designed rather than how they perform. Where a model changes its relational encoding and its aggregation weighting at the same time, the contribution of each change cannot be recovered from the reported totals. The two have to be varied separately for either to be attributed.

2.3.3 Hypergraph Convolution

Operators over a hypergraph propagate in two stages. Signal moves from the vertices to the hyperedges that contain them, is combined at each hyperedge, and returns to the member vertices, with the incidence structure and the degree matrices of Section 2.2.2 supplying the normalisation [27, 28]. Two design routes exist. One reduces the hypergraph to a graph and applies a pairwise operator to the result [78], which inherits the losses recorded in Section 2.2.2; the other treats a hyperedge as a multiset and learns the functions that aggregate it [79]. Attention has been carried across to this setting by making the incidence structure itself learnable, so that membership is weighted by a real value instead of a binary indicator [80].

Three observations bound what is known. Standard hypergraph convolution averages over the members of a hyperedge and therefore attenuates variation between them, a low-pass bias whose discarded high-frequency component has been characterised [81]. The operator most widely used has since been superseded by a stability-oriented successor from its own authors [82], which is the reference point for any claim about its limitations. No theoretical analysis of attention scoring in hypergraph operators has appeared since 2024; the nearest treatments predate that, in work on equivariant hypergraph networks and on generalising set transformers to hypergraphs [83, 84]. The transfer question raised in Section 2.3.2 therefore remains open on the theoretical side as well as the empirical one.

2.4 Detection in Networked and Cyber-Physical Systems

This section covers the unit of observation used in network detection, the families of method defined over it, the particular character of industrial and IoT deployments, and the corpora on which results are reported. Chapter 3 draws on it, although the Android material precedes it for the reason given at the head of this chapter.

2.4.1 Network Flows as a Unit of Observation

Detection at the flow level summarises the packets sharing a five-tuple into counts, durations and byte totals. Working at that level lowers processing cost and survives payload encryption, at the price of discarding content [85]. Network intrusion detection is conventionally divided into signature matching against descriptions of known attacks and anomaly detection against a learned model of normal behaviour [86], and the second is what the rest of this section concerns. Section 4.7 states the particular flow feature set Chapter 4 uses, rather than drawing it from here.

2.4.2 Anomaly-Detection Method Families

Anomalies are conventionally classed as point, contextual or collective, and detectors differ in whether labels are available for both classes, for one, or for neither [4]. Classical methods score a point by the local density around it [87] or by how few partitions are needed to isolate it [88]. Deep detectors replace a hand-built feature space with a learned one, and the shallow and deep families admit a common formulation [89, 5]. Where the data are relational, the anomaly is a property of structure rather than of any single record, and the object flagged may be a vertex, an edge or a subgraph [90]. That observation is what connects this section to Section 2.2.

The security setting resists the standard formulation. The cost of a false alarm is high, network traffic is more variable than a model of normal behaviour admits, and data adequate for evaluation are scarce [42].

2.4.3 Communication Structure in Industrial and IoT Networks

Industrial and IoT traffic is comparatively regular, because the devices are few and their interactions are prescribed by the process under control. A whitelist of permitted communication relations is therefore a workable model of normal behaviour. Barbosa et al. built such whitelists in SCADA networks from four packet properties and reported both that the whitelists stabilise and that rare legitimate flows continue to raise false positives [91]; Yoon and Ciocarlie modelled communication patterns per host pair for the same purpose [92].

Neither of the two ideas that follow from this originates with service-oriented deployments. Scoring behaviour against a graph of which host talks to which was established for carrier traffic [93, 94], and the dependency graph between services was already being recovered automatically from passively observed traffic in enterprise networks [95, 96, 97].

Where the system is cyber-physical, the observation may be the process rather than the traffic, and detectors are then fitted to sensor and actuator series recorded from testbeds [10, 98]. In IoT deployments the unit modelled is often the individual device. Models of that kind hold under the conditions they were fitted to and lose a substantial part of their measured performance when the deployment changes: reproduced figures fall by roughly a quarter to a half beyond three months, and by a comparable margin between test sites [99].

2.4.4 Datasets and Measurement Limits

The corpora this field reports on are known to be defective. KDD 99 was shown to contain redundant records that bias measured accuracy [100], and a later audit of CICIDS2017, UNSW-NB15 and comparable sets catalogues construction faults that allow trivial features to separate attack traffic from benign [101]. Two further limits bear on any figure reported here. Detectors reporting detection rates above 99 per cent have been shown to be trained and tested on the same attack type, with performance not surviving attack types absent from training [102]. And when 70 published approaches were re-run on one evaluation platform, four deliberately simple tests on process values, using minima, maxima and rate of change, performed on par with them [103]. Section 2.6 treats the protocol-level causes.

2.5 Adversarial Machine Learning

Chapter 1 has already set out what the record shows about published defences. This section supplies the vocabulary that account assumed, and it is the vocabulary Chapters 4, 5 and 6 use to state what they claim and what they do not.

Threat models. A threat model states four things: the attacker’s goal, the attacker’s knowledge of the system, the attacker’s capability to act on it, and the constraints on what may be modified [104, 105]. A claim about an attack or about robustness cannot be interpreted without all four [21]. Knowledge is conventionally divided into three levels. Under *white box* the attacker knows the training data, the feature representation, the learning algorithm, the trained parameters and any deployed defence [18]. Under *black box* the attacker knows neither parameters nor architecture and interacts through the prediction interface, either by transferring from a substitute model trained on its outputs or by searching directly with queries [106]. *Grey box* lies between the two: some components of the white-box list are known and training cannot be influenced. Because the term is used inconsistently in the literature, every use in this dissertation names which components the attacker is assumed to know. One limit on the taxonomy should be recorded with it. A survey of 271 practitioners found that the knowledge assumptions of academic threat models are more generous to the attacker than deployment conditions usually allow [107].

Classes of attack. *Evasion* modifies an input at test time so that a trained model misclassifies it, leaving the model unchanged [17]. *Poisoning* manipulates the training data or the training process so that the resulting model is degraded or carries behaviour the attacker chose [108, 109], and the *backdoor* is its sub-case, in which the implanted behaviour is conditioned on a trigger [110]. Confidentiality attacks form a third class: a functional copy of a deployed model can be recovered through its prediction interface [111]. The word adversarial covers all three and is not a synonym for evasion.

Feature space and problem space. A *feature-space* attack perturbs the feature vector directly, with no requirement that any real input maps to the result. A *problem-space* attack modifies the artefact itself and must satisfy four constraints: the transformation has to be available, semantics have to be preserved, the result has to be plausible, and it has to survive preprocessing. Moving between the two is the inverse feature-mapping problem, and in the malware setting the modified sample must still install and still run [22]. The distinction is load-bearing for Chapters 4 and 5, because attacks on static Android detectors operate at code level, upstream of feature extraction, so a representation derived from that extraction inherits the attacker's influence whatever the learning architecture [112]. Which operators reach which representation depends on what is extracted, and each chapter settles that where it states its threat model. Early work on Android established the feature-space case [113].

Adaptive attacks and robustness. An *adaptive* attack is constructed in knowledge of the defence and tuned against that specific defence, in contrast to a fixed attack applied unchanged [20]. The recurring reason a defence appears to hold and does not is gradient masking, where the defence obstructs the optimisation an attack relies on without reducing the attacker's reach [19]. *Robustness* is therefore defined relative to a stated threat model and a stated attack budget. It is a property of the pair, never of a model alone, and a robustness figure reported without those parameters cannot be compared with another [21]. Gradient-based attacks at fixed perturbation budgets [114, 115] are the reference instruments against which the defences of Chapter 6 are measured.

2.6 Evaluation Methodology and Experimental Bias

Chapter 1 reported what happens to published performance under a stricter protocol. This section states the mechanisms behind those results and fixes the standard this dissertation holds itself to.

Sampling bias in space and time. *Temporal bias* is a split that allows a model to learn from samples postdating the ones it is tested on, which reports a figure no deployment reproduces. *Spatial bias* is a test set whose composition differs from the deployment, which leaves precision uninterpretable. A sound evaluation constrains both: training data must precede test data, and the ratio between classes in the test set must reflect the setting the system is claimed for [12]. Drift is the mechanism that makes a randomly ordered split optimistic [11]. Curation of the corpus decides as much as the split does, and guidance published since those constraints were formalised revises parts of them [116, 117].

Leakage. Any transformation fitted before the split moves information from the test set into training. Resampling, feature selection, normalisation and model selection all have this property when applied to the whole corpus, and the resulting figure measures the procedure rather than the model [13]. Duplicate and near-duplicate samples are a second channel, since repackaging produces applications that differ only in the injected payload and copies of one application may fall on both sides of the split. The measured cost is uneven: limited for supervised classification, and substantially larger for unsupervised family clustering [15, 14].

What a score means. Axelsson derived the constraint that binds every detector deployed against a rare event: because the base rate of intrusions is low, the probability that an alarm signals a genuine intrusion is dominated by the false-alarm rate, and a detector must therefore reach a false-alarm rate that is difficult to attain before its alarms carry information [43]. Accuracy and precision are consequently uninterpretable without the class ratio that produced them [13], and a reported accuracy is read against the floor set by a majority-class or constant-output predictor. In time-series detection a further convention distorts comparison: *point adjustment* marks an entire anomalous window as detected once any single point inside it is flagged, which inflates F1 to the point that randomly generated scores outperform published detectors [118, 119]. Some standard benchmarks also contain mislabelled or trivially separable anomalies, so a high score on them is weak evidence [120].

Baselines and adaptive evaluation. A reported gain is not established until a simple baseline has been run under the same protocol. Where that has been done, the gap has often disappeared: re-implementing eight provenance-based intrusion detectors in one framework left a simple neural network matching the state of the art on most of the datasets tested [121]. For a defence the corresponding rule concerns the attack. A robustness claim is bounded by the strongest attack applied to it, so the defence has to be evaluated against an attack adapted to it [21], and defences evaluated only by their own authors have repeatedly failed under adapted attacks [19, 20].

The standard adopted here. Every empirical chapter of this dissertation states its split policy and uses time-aware splits where the timestamps permit, reports the class ratio alongside any aggregate score, reports no point-adjusted figure, places a trivial baseline beside each headline result, and states preprocessing and random seeds [122]. Results carried over from the source publications cannot be retrofitted to all of this, and each chapter says in its Limitations which parts were not met.

2.7 Summary and Research Gaps

Three threads run through the preceding sections. The first concerns what an analysis system is able to observe. An Android package exposes a manifest and bytecode to static reading, exposes executed behaviour and recorded traffic to dynamic instrumentation, and neither mode dominates the other. Network deployments expose flows, and industrial and IoT systems additionally expose process values and a comparatively regular pattern of who communicates with whom. The second thread concerns how those observations are structured. Program behaviour is encoded as control-flow and function-call graphs, communication as graphs over

hosts or services, and both may be lifted to hypergraphs when a relation holds among more than two elements. The operators that learn over these structures are bounded in what they can distinguish, and the projection from a hypergraph back to a graph discards information that has now been measured. The third thread concerns measurement. Reported performance depends on the split, on the corpus, on the baseline it is compared against, and, where a defence is claimed, on the strength of the attack applied to it.

Four questions are left open by this material, and each is taken up by one of the research questions stated in Section 1.3.

G1. How behaviour in networked and industrial systems should be represented as deployments change. Communication whitelists, service-dependency graphs, flow records, and process time series have each supported anomaly detection under the conditions in which they were fitted [91, 92, 123]. The evidence is split across different observation units and system assumptions, and traffic-based models lose a substantial part of their measured performance once a deployment moves in time or place [99]. The literature lacks a common account of how these representation choices support detection as the observed system changes. RQ1 takes up the question through communication graphs and modular analysis of network and process features.

G2. The value of combining program structure with observed behaviour. MARVIN and the study by Onwuzurike et al. report benefits from combining static and dynamic features in within-system comparisons [57, 25]. Those results concern particular observation types and prediction tasks. Whether the benefit persists when either changes requires comparison of the fused representation with each modality alone on the same samples. RQ2 investigates this question for program-graph and network-flow features in binary detection and malware-category classification.

G3. Whether higher-order representation of program structure pays for itself. Program structure is represented pairwise, and a relation holding among several functions at once is flattened into a set of edges that cannot record it as one group [26]. Hypergraphs remove the restriction and the cost of the projection has been quantified [70, 71], yet the sceptical result also stands: an operator whose vertex weights do not vary between the hyperedges a vertex belongs to is equivalent to one acting on a weighted clique expansion [68, 69]. Whether higher-order encoding, and the attention weighting often applied with it, improves detection and classification of Android malware is open, and no theoretical analysis of attention scoring in hypergraph operators has appeared since 2024 [83, 84]. RQ3 takes up this question.

G4. How an adversary model becomes a reproducible benchmark. Defences are commonly evaluated against attacks chosen by the party proposing them, and a robustness claim is bounded by the strongest attack applied [21]. Threat models are stated inconsistently, and the knowledge assumptions used in academic evaluations are more generous than deployment usually allows [107]. How an explicit adversary model is translated into a benchmark that a third party can reproduce, and what such a benchmark then shows about the relation between clean predictive performance and robustness, has no accepted answer. RQ4 takes up this question.

Anomaly Detection in IoT and Industrial Control Systems

3.1 Scope, Research Question, and Contributions

This chapter answers RQ1.1 and RQ1.2, the two subquestions of RQ1 stated in Section 1.3, and takes up gap G1. Three pieces of work support the answer. The first represents service-to-service communication in an IoT middleware as a graph and flags relations absent from that graph. The second is NADICS, a learning framework for network and process data, together with its S7 and PROFINET extractor and its integration in IUNO. The third studies a multivariate reconstruction detector over SWaT process data by combining elements of MAD-GAN and TadGAN.

Each contribution observes its own setting. The service-graph study represents inter-service messages as graph relations. In the industrial work, the NADICS extractor reduces packet captures to connection records, while the SWaT study represents process measurements as multivariate time windows. No common protocol evaluates all three settings, so the chapter treats them as related but separate experimental strands.

The evidence also differs in kind. The service-graph contribution was disseminated at the 2018 Passive and Active Measurement Conference (PAM) [32]. The original industrial framework is supported by released code and IUNO deliverables [33, 34]. The SWaT study is recorded in an unpublished internal report written by the candidate. Because that report is a research record, it is not a bibliography item. The candidate designed the method, implemented the experimental pipeline, ran and analysed the experiments, and wrote the report. Section 3.5 reports the empirical evaluation of the SWaT detector.

3.2 Observation Models and Datasets

The contribution strands observe different events. In the IoT middleware, the monitor sits at the Knowledge Agent, the entry point of the Virtual State Layer through which services exchange data [124]. It observes an inter-service message rather than a packet. Vertices are service addresses, and an edge records an observed communication relation. The message also exposes metadata such as the operation performed [31].

NADICS receives tabular records. Its network path takes connection features derived from S7, PROFINET, or conventional network traffic. The SWaT path does not model communication.

Table 3.1. Observation units used in Chapter 3. The rows are not numerically comparable because their labels, sampling processes, and validation protocols differ.

Study	Unit observed	Representation	Decision
IoT service graph	inter-service message	vertices and whitelisted relations	known or unknown relation
NADICS in IUNO	packet-derived connection record	protocol and flow features	binary or attack-class label
SWaT GAN study	sensor and actuator sample	multivariate time window	anomalous or normal timestamp

It consumes the sensor and actuator values of a six-stage water-treatment process, sampled once per second [123, 10].

The studies use different decision rules. The service-graph method stores the inter-service relations observed during its learning phase and flags a later relation that is absent from the graph. The GAN study uses mixed normal and attack windows for fitting, combines reconstruction and critic scores, and applies an adaptive threshold that assigns a binary prediction to each timestamp. Both methods depend on representative fitting data: an attack present during learning may be incorporated into the learned representation. The NADICS evaluation on the S7-300 corpus uses labelled classification instead, so it addresses a different decision problem. It holds out whole capture sessions. Its attack share of 4.54 per cent was set by the executed attacks and does not reproduce the class ratio of routine operation.

3.3 Communication Graphs for IoT Services

The lineage set out in Section 2.4.3 places communication whitelisting before service-oriented IoT deployments. Flow whitelists were built for control networks [91], and communication patterns were modelled per host pair soon afterwards [92]. The service-graph contribution acknowledges that lineage and identifies its proposed difference: an IoT deployment changes as services are added or removed, so its communication model must support interactive revision rather than a fixed learning phase [32].

The model is compact. Services are vertices, observed relations are edges, and each node maintains its own graph. A learning phase runs when a service starts and labels every relation observed during that period as normal. A relation absent from the model later raises a notification. The site owner may legitimate it, after which the system relabels the edge as normal. Detection runs on the node hosting the service, so the design does not require central collection [32].

The intended distinguishing property was a portable model attached to service metadata. A service moved to another node would then carry its learned relations with it. The study also states that the implementation identifies services by concrete local addresses, which prevents that portability. Semantic service discovery would have been required for site-independent identities.



Figure 3.1. The two measurements reported for the service-graph design. Panel (a) gives the inter-service round-trip time with and without detection for ten to forty services. Panel (b) gives the vertex and edge counts of one node’s graph over a single run, with the learning phases and the interval of anomalous behaviour shaded. Reproduced from [32], to whose measurement design the candidate contributed.

Two costs were reported. Memory is modelled per vertex as 100 bytes plus 88 bytes for every incident edge, and a factor of 1.33 accounts for the stacked hash tables that hold the graph. That factor enters twice, once inside the per-vertex expression and once over the vertex set, so twenty services with ten relations each occupy 33.7 KB. The second cost is latency: about 6 ms added to an inter-service round trip of 24 ms, measured over configurations of ten to forty services [32].

Figure 3.1 reproduces both measurements. In the latency panel the two distributions overlap heavily. Medians without detection rise from roughly 14 ms at ten services to roughly 27 ms at forty, and every configuration carries a long upper tail past the 160 ms axis limit. Neither the repetition count nor the statistic behind the quoted 6 ms and 24 ms is stated, so both values are best read as an aggregate over the four configurations. The added cost does not grow between ten and forty services.

Panel (b) covers one run of about fourteen minutes. Learning phases cluster in the first three minutes as services start, and another follows at minute seven when a service is added. The graph then settles at fourteen vertices and ten edges. One anomalous interval is recorded shortly after minute ten, and the edge count returns to its earlier value once it ends, so the trace records the detection but no legitimation by the site owner.

No labelled attack set was used, and that interval holds a previously unseen connection rather than an executed attack, so the study reports no detection rate and no false-positive rate. A whitelist emits a verdict, so even a labelled trace yields no threshold curve without a change of method. RQ1.1 is answered here at the level of representation, update procedure, and measured overhead.

3.4 The NADICS Framework and IUNO Integration

NADICS, Network Anomaly Detection for Industrial Control Systems, is a Python framework built at the Chair of IT Security of the Technical University of Munich to detect attacks against PLC and SCADA installations from network traffic. By design, it was limited to detection. The framework is organised in three modules. Preprocessing turns recorded traffic into a labelled feature table, encodes the string columns, scales the numeric ones, and resamples a corpus whose classes are imbalanced. The learning module holds the estimators behind one interface, so that classical models and neural models can be exchanged by name without the pipeline being rewritten: the released repository contains the scikit-learn estimators, the IUNO deliverable additionally reports a deep neural network and a convolutional network [33], and the generative adversarial detector of Section 3.5 was added later at the same interface for multivariate process data. Postprocessing carries the verdict to whoever acts on it. The released code leaves this stage open, because alarm handling depends on the security policy and the network layout of the individual site. The S7 and PROFINET extractor existed as a separate project artefact, so the released repository takes a feature table as its input.

The design follows from the data situation in industrial control. No public corpus of labelled traffic from a real industrial network was available [33], so the estimator that performs well depends on the plant and on how its labels were produced. Fixed interfaces around the learning stage let the models be compared under a single protocol on whatever corpus a site can supply. They also let feature importances be inspected, so that dimensionality and the false-positive rate can be reduced.

Figure 3.2 shows the pipeline as it was deployed in IUNO. Two data artefacts separate its three stages. The first stage produced the corpus. Attacks of five classes (authentication bypass, CPU stop and start, reconnaissance, denial of service and brute force) were executed in a real industrial environment against Siemens S7-300 controllers that exchanged S7comm traffic with a control server, and the traces were labelled both in binary form and by attack class [33]. In the second stage, the public analysers TShark, Bro-IDS and Argus were combined with dissectors written for S7 and PROFINET, and the TUM feature extractor reduced each network connection to one labelled feature vector, written as CSV or JSON [41, 33]. The third stage is the framework proper. It reads that dataset, fits one of the registered estimators, applies the resamplers and the outlier tests, and reports which features carried the decision. Detections reached the visual security control room as JSON event messages over the authenticated message bus of the IUNO aggregation system [33], and acting on them belonged to the control room [34].

The modules map onto these stages at the dataset. Preprocessing spans the second stage and the input side of the third, where encoding, scaling and resampling are applied to the table, while learning and postprocessing both sit in the third stage. Corpus generation lies outside the framework and appears in the figure because the labels had to be produced somewhere. Inside the engine, a layout file declares the full columns, the feature subset, the label column, and the string features that require encoding. Learning methods sit in registries that map a name to an estimator. The original repository holds thirteen classifiers, four novelty or outlier detectors, fifteen resamplers, and two boosting meta-estimators. The framework calls only



Figure 3.2. Architecture of the NADICS framework. Malicious traffic is generated in a real industrial environment, dissected into protocol-characteristic features, and classified by the anomaly detection engine. The stage structure follows the one reported in the IUNO deliverable [33].

fit and predict, following the scikit-learn estimator protocol [125]. It fits min-max scaling on the training partition and applies the fitted transformation to the test partition.

Figure 3.3 shows where the framework ran. The environment was built at the Volkswagen plant in Wolfsburg from hardware of the kind used in regular production [41]. A Hirschmann industrial switch carries three segments, the intranet, the demilitarised zone and the plant network, and manually configured Eagle routers separate them. The first of four servers runs the WinCC and OPC UA clients that originate test traffic, together with a Nagios instance and the database of the aggregation system. The second is the virtualisation host for the detection tools contributed by AISEC, DFKI, SIT and TUM, and the third serves the visual control room to the intranet. The fourth combines the WinCC server with the collector, which gathers traffic from the plant network and passes it, bundled and pre-filtered, to the detection tools, so NADICS was one of four detectors reading the same collector output. Siemens controllers occupy the plant network behind a Scalance switch. The collector observed that network passively at the plant switch, never in line with the traffic, which matches the detection-only design: an alert could reach the control room, but nothing in the path could stop a packet.

The IUNO deliverables describe the tool and its environment but print no measured result for it [33]. The values below come from the experimental records of the project and cover the corpus and its feature schema, the detection results on the held-out session, the alarm burden over a benign monitoring period, and the throughput of the extractor on the prototype collector.

3.4.1 The S7-300 Corpus

The corpus was recorded in 2018 in the environment of Figure 3.3, where six Siemens S7-300 controllers exchanged routine polling and control traffic with WinCC-based services. Complete selected channels were kept, together with ARP and DCP traffic and the connection attempts belonging to those channels. No random packet sampling was applied.

Recording opened with a two-hour clean baseline, which was excluded from the corpus totals. That interval fixed the communication relationships, the polling behaviour and the protocol activity expected under the operating modes present at the time. Six capture sessions followed, with 24 hours of labelled traffic in total. The collector recorded 12,616,875 Ethernet frames in that period, an average of 146.03 frames per second. The captured frames hold 2,317,074,962 bytes, and the six capture files occupy about 2.519 GB in the classic PCAP format.

Twenty-five controlled attack episodes were executed during the sessions, five repetitions of each of the five classes the deliverable names [33]. Background communication continued while they ran. Table 3.3 lists the duration of every episode. The shortest lasted 3 min 07 s and the longest 9 min 55 s, and together they cover 2 h 51 min 05 s. Episodes also differed in the channels they touched and in how often a connection was re-established. Labels were assigned from the execution logs and from inspection of the packet traces, and only the communications an episode actually affected received an attack label, so background traffic concurrent with an attack keeps the normal label. Two attack types were never run against one channel at the same time.



Figure 3.3. The IUNO test environment at the Volkswagen plant in Wolfsburg. Four servers, three network segments and a Siemens plant network reproduce the production setting in which NADICS ran as the TUM detection tool on the virtualisation host. The network TAPs fitted at the relevant interfaces are omitted from the drawing [41].

Table 3.2. Composition of the S7-300 corpus and its partitions. Whole capture sessions were assigned to the partitions, four to training and one to each of the other two, and every attack episode lies entirely within one partition. The episode column counts executed attacks, not records.

Class	Corpus	Train	Validation	Test	Episodes
Normal	113,751	76,556	18,587	18,608	
Authentication bypass	1,058	441	269	348	5
CPU stop and start	593	359	90	144	5
Reconnaissance	1,803	1,070	366	367	5
Denial of service	1,230	765	278	187	5
Brute force	725	445	93	187	5
Total	119,160	79,636	19,683	19,841	25
Capture time (h:min:s)	24:00:00	16:01:57	3:56:27	4:01:36	

Table 3.3. Duration of the 25 attack episodes in minutes and seconds. Each attack type has three training episodes, one validation episode and one test episode.

Class	Training episodes	Validation	Test	All five
Authentication bypass	5:33, 4:01, 5:50	7:16	9:34	32:14
CPU stop and start	7:03, 8:23, 3:07	4:04	7:39	30:16
Reconnaissance	6:42, 9:40, 7:28	9:55	8:27	42:12
Denial of service	6:41, 9:39, 7:07	7:46	5:38	36:51
Brute force	8:05, 3:16, 4:06	4:15	9:50	29:32
Total				171:05

Extraction produced 119,160 feature records. Of these, 113,751 are normal and 5,409 carry one of the five attack labels, an attack share of 4.54 per cent. Table 3.2 gives the composition. The training sessions hold 79,636 records, the validation session 19,683 and the test session 19,841. Each attack type contributes three training episodes and one episode to each of the remaining partitions, and because episodes differ in duration and in the channels they touch, the attack records of a class are not divided between the partitions in a fixed proportion. An episode, its recovery interval and the windows that depend on it stay inside a single partition, and no rolling window crosses a partition boundary. The attack counts must not be read as a count of attacks. A single episode of a few minutes generates many records, and the corpus averages about 216 affected records per episode.

The extraction schema has 63 fields in five groups: 18 flow fields, 13 timing fields, 18 protocol fields, nine relational fields and five process fields. Forty-six are numerical, eleven categorical and six binary, and encoding of the categorical fields expands the input dimensionality of the model. The flow group follows the output of Argus and Bro-IDS, which the extractor wrapped. The UNSW-NB15 intrusion-detection corpus derives most of its features from the same two tools [126], so this group coincides with the flow columns of that corpus. Protocol fields cover S7 message and function codes, addressed memory areas, address entropy and error codes, together with COTP and OPC UA fields. Relational fields describe the communication partners, the age of an edge, and the number of distinct destinations and ports observed in the preceding minute. The process group reads one preconfigured anchor tag per channel, whose identity and units are held as metadata, so a value delta is never computed across unrelated tags. Unobservable protocol fields carry an explicit missing state, and an absent OPC UA security policy is not encoded as the policy *None*.

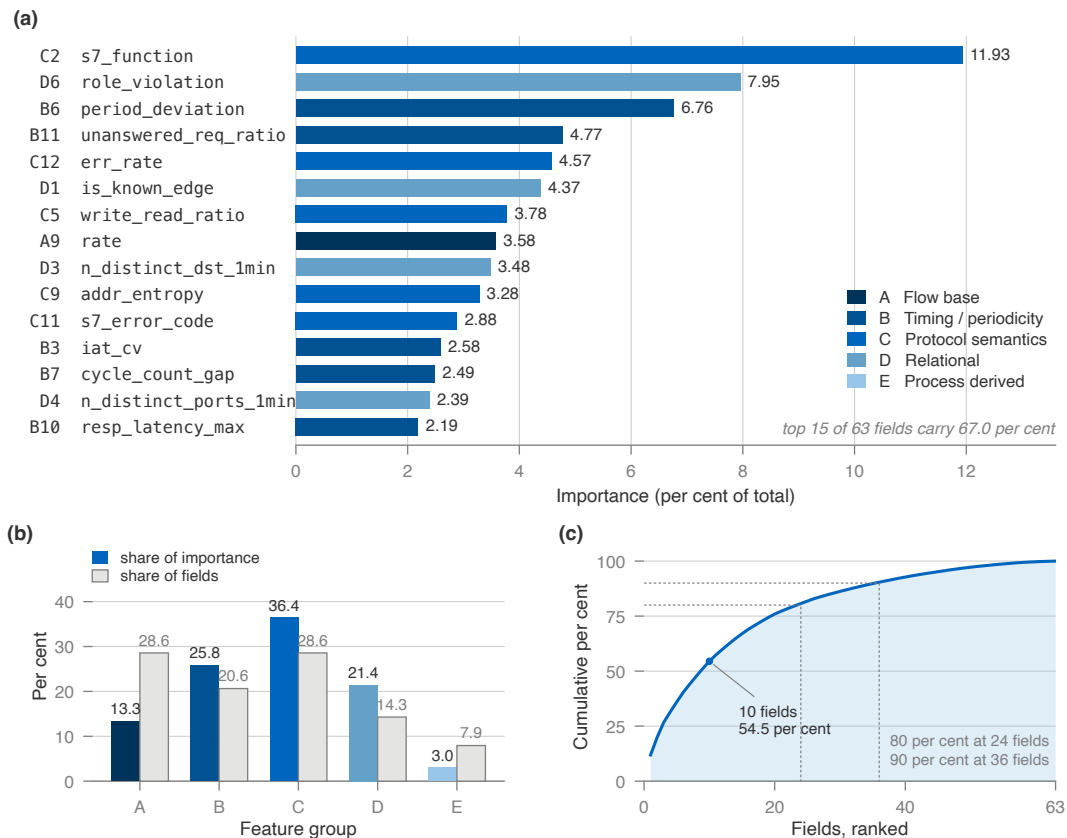


Figure 3.4. Random-forest importance of the 63 extraction fields. (a) The fifteen most important fields, coloured by group. (b) Each group's share of the total importance against its share of the field count. (c) Cumulative share over the ranked fields.

Figure 3.4 shows the importance of each field in the random forest of Section 3.4.2. The forest was fitted to the training partition; the impurity reductions from splits on each field were summed across its trees, normalised over all 63 fields, and ranked. Panel (a) shows the fifteen highest, and Appendix A lists all 63. `s7_function` leads at 11.9 per cent, because a stop or a start is an ordinary S7 operation whose function code is visible in the request, and `role_violation` follows at 8.0 per cent, because a write from a host whose role does not write is anomalous even over an established connection. Both signals have limits: legitimate maintenance issues the same function codes, and a compromised engineering station writes from a role that is permitted to write.

Panel (b) sets each group's share of the total importance against its share of the fields. Protocol semantics holds the largest share at 36.4 per cent, although it also holds the most fields. The comparison that carries information is the share per field: the nine relational fields average 2.38 per cent each, against 0.74 for the eighteen flow fields and 0.60 for the five process fields. Only the process fields receive less importance per field than the flow statistics. That is consistent with the position the schema takes: an industrial detector built on Argus and Bro flow features alone discards most of what separates an S7 attack from S7 polling. Panel (c) gives the cumulative share. Ten fields reach 54.5 per cent of the total importance, twenty-four reach 80 per cent and thirty-six reach 90 per cent, so the tail of the schema is thin and a reduced schema is plausible. NADICS exposes an importance ranking for exactly this purpose [33].

One record covers a non-overlapping ten-second window of an activity-bearing connection or channel, and a shorter final record is written when a session closes. PROFINET RT traffic is keyed at layer 2, since it carries no transport-layer connection. The windowing refines the per-connection unit of Figure 3.2 so that a change during a long-lived session can be scored while the session is still open. Channels that fall silent emit no record.

Three definitions were adjusted so that no record depends on information from after its own window closes. The duration field measures the exported segment and not the eventual lifetime of an open connection, while a separate field retains the time elapsed since the underlying connection opened. The ratio of unanswered requests counts only those requests whose one-second response deadline had already expired at window closure, and pending requests are excluded from both the numerator and the denominator. Trailing source and timing state is held over 60 seconds, and the period features are computed from past observations. Encoders, imputation and scaling were fitted on the training partition alone, and the classifier together with its preprocessing was then held fixed for validation and testing.

3.4.2 Detection Results

The classifier is a random forest of 300 trees with a maximum depth of 16 and a minimum leaf size of five. Class weighting was applied per bootstrap sample, and the training data were not resampled. Two decisions were read from the same fitted model: a binary verdict of normal against attack, and an assignment to one of six classes, the normal class and the five attack types. Every figure below comes from the held-out test session, which holds 18,608 normal and 1,233 attack records.

Table 3.4. Six-class confusion matrix on the 19,841-record test session, with rows giving the true class. Predicted classes are abbreviated: N normal, AB authentication bypass, CS CPU stop and start, RC reconnaissance, DS denial of service, BF brute force. The last three columns give precision, recall and F_1 for the class of that row.

True class	Predicted class						Score (%)		
	N	AB	CS	RC	DS	BF	Prec.	Rec.	F_1
Normal	18,518	21	5	31	18	15	99.74	99.52	99.63
Authentication bypass	22	292	8	12	1	13	88.75	83.91	86.26
CPU stop and start	3	1	134	2	3	1	90.54	93.06	91.78
Reconnaissance	9	7	0	337	6	8	84.25	91.83	87.87
Denial of service	5	0	0	4	178	0	85.99	95.19	90.36
Brute force	9	8	1	14	1	154	80.63	82.35	81.48

Table 3.4 reports the six-class confusion matrix together with the per-class scores. Collapsing the same predictions into the binary decision gives 18,518 true negatives, 90 false positives, 48 false negatives and 1,185 true positives. Binary accuracy is 99.30 per cent, attack precision 92.94 per cent and attack recall 96.11 per cent, for an attack F_1 of 94.50 per cent. The false-positive rate over normal records is 0.484 per cent and the balanced accuracy 97.81 per cent. On the six-class task, accuracy reaches 98.85 per cent, the macro- F_1 89.56 per cent and the weighted F_1 98.86 per cent.

Accuracy is the least informative of these numbers. Normal records make up 93.79 per cent of the test partition, so a detector that never fired would already score 93.79 per cent, and the gap between the weighted and the macro F_1 , 98.86 against 89.56 per cent, measures how much of the six-class figure the normal class carries. Per-class recall is highest for denial of service at 95.19 per cent and lowest for brute force at 82.35 per cent, with authentication bypass close behind at 83.91 per cent. Most errors of these two classes fall on other attack classes. Authentication bypass loses 22 of its 348 records to the normal class and 34 to the other attack types, while brute force loses nine of its 187 records to the normal class and 24 to the other attack types, 14 of them to reconnaissance. Both attacks reuse the protocol operations of legitimate engineering access, so a window holding only a few such operations offers little that separates them from ordinary traffic or from each other.

All of these scores are computed over records. They show that the extracted features separate the five executed attacks from the surrounding traffic of this plant under a session-disjoint split.

3.4.3 Benign Operation and Alarm Burden

A second exercise measured what the detector does when nothing malicious happens. It ran in the Volkswagen environment with four active PLCs and ordinary supervisory traffic. A separate two-hour baseline preceded 24 hours of benign monitoring. Over that day the collector recorded 8,472,913 frames, an average of 98.07 frames per second. The captured frames hold 2,186,039,471 bytes, about 2.322 GB in the classic PCAP format, and the extractor emitted 83,847 records. Engineering work and planned device restarts took place during the period and count as benign.

Table 3.5. Extraction workloads on the Raspberry Pi 3 prototype. Input frames are the frames presented to the extraction pipeline, and processed frames exclude those dropped before processing. Rates and CPU utilisation are averages over the replay, and CPU utilisation refers to the capacity of the whole device. The emission delay runs from the close of a window to the emission of its record.

Quantity	Light load	Moderate load	Overload
Replay duration (s)	600.14	599.87	600.09
Input frames	599,827	1,498,463	2,100,731
Dropped frames	4	361	245,897
Processed frames	599,823	1,498,102	1,854,834
Frame loss (%)	0.00067	0.02409	11.70531
Input rate (frames/s)	999.48	2,497.98	3,500.69
Processed rate (frames/s)	999.47	2,497.38	3,090.93
Feature records	5,842	5,887	5,781
Mean CPU utilisation (%)	22.6	43.1	50.4
Peak extractor memory (MiB)	302	381	436
95th-percentile emission delay (ms)	96	234	817

The detector classified 83,740 records as normal and 107 as attacks, all of them false, a record-level false-positive rate of 0.128 per cent. Positives were merged into alarm episodes before presentation to the control room: positives that share a source, a destination and a predicted attack class belong to one episode, and an episode closes once 60 seconds pass without a further positive. One notification opens an episode, and later positives update it. Nine alarm episodes were raised over the day, 0.375 per hour on average, and each was assigned a benign cause. Engineering workstation activity accounts for 43 positive records in three episodes and OPC UA reconnections for 28 records in two. Brief communication interruptions produced 23 records in two episodes, and planned device restarts 13 records in two.

No attack was executed during this period, so every alarm it raised was false. Its 0.128 per cent record rate lies below the 0.484 per cent of the test session, whose normal records also include the recovery intervals of executed attacks. The recovered causes share a structure that matters for deployment: each is a legitimate departure from routine polling, so the alarm burden of this design follows the rate at which such events occur in the plant.

3.4.4 Extractor Throughput on Constrained Hardware

The extractor was also exercised on a Raspberry Pi 3 Model B. The deliverable of June 2017 names Raspberry Pi boards as the prototype platform [41]. The board carries four cores at 1.2 GHz, 1 GB of RAM and 100 Mbit/s Ethernet. Only the extraction stage ran on it, comprising selected-field capture and dissection of S7 and PROFINET traffic, the bounded rolling feature computations and emission of CSV or JSON. Model fitting and inference remained on the server, and archival of full packet captures to the SD card was excluded. Each load replayed traffic over 100 configured communication channels with varying channel activity for about ten minutes.

Table 3.5 reports the three loads. At the moderate load, 1,498,463 frames were presented in 599.87 seconds and 361 were dropped, a loss of 0.024 per cent at a processed rate of 2,497.38

frames per second. This input rate is about 17 times the average frame rate of the corpus capture. The extractor used 43.1 per cent of the CPU capacity of the device on average and at most 381 MiB of memory. The 95th percentile of the delay from the close of a window to the emission of its record was 234 ms, a delay that excludes the ten seconds in which the window accumulates. The light load lost four of its 599,827 frames at 22.6 per cent CPU utilisation. The overload row does not describe a supported rate: 11.71 per cent of the input was dropped, and the processed rate stayed at 3,090.93 frames per second against an input rate of 3,500.69. Mean CPU utilisation reached 50.4 per cent there, and the 95th-percentile emission delay rose to 817 ms. The three runs emitted 5,842, 5,887 and 5,781 records, fewer than the 6,000 that 100 continuously active channels would give in ten-second windows over ten minutes, because channel activity varied.

3.5 A GAN-Based Multivariate Detector for SWaT

This section reports an empirical evaluation of a generative adversarial network (GAN) for detecting abnormal behaviour in an industrial water-treatment process. The study examines how the representation of process measurements and their temporal context influence the balance between detecting anomalous observations and generating false alarms. Its design combines the multivariate approach of MAD-GAN [127] with the encoder, cycle-consistency objective and Wasserstein critics described in TadGAN [128]. The reported results were measured on the SWaT evaluation cohort and compare five input representations.

The detector operates on process measurements from the Secure Water Treatment (SWaT) testbed. SWaT comprises six treatment stages, with programmable logic controllers coordinating sensors and actuators and a supervisory network connecting the controllers to SCADA (Figure 3.5). The process dataset contains 51 sensor and actuator variables recorded at a nominal frequency of one observation per second [123]. These variables describe the physical behaviour of the plant, allowing the detector to consider changes in individual measurements together with relationships among measurements across the treatment process.

The dataset documentation describes 36 attacks distributed across four categories (Figure 3.6): 26 single-stage, single-point attacks; four single-stage, multiple-point attacks; two multiple-stage, single-point attacks; and four multiple-stage, multiple-point attacks [123]. These categories describe the scope of the interventions. They do not imply that all attacks produce equally observable process changes. An intervention affecting one sensor or actuator may also influence measurements elsewhere in the plant, while an attack with little physical effect may remain difficult to identify from process data alone. The aggregate calculations in this section concern anomalous timestamps and do not establish the number of individual attacks detected.

The source collection campaign comprises seven days of normal operation followed by four days containing attacks [123]. Preprocessing removes the first 21,600 observations from the normal-operation trace, which cover the initial six-hour stabilisation period (Figure 3.7). The empirical study uses all 449,919 timestamps of the attack-period recording. Its first 56,240 timestamps are divided between the training and validation partitions, which also receive separate portions of the normal-operation recording. The later 393,679 attack-

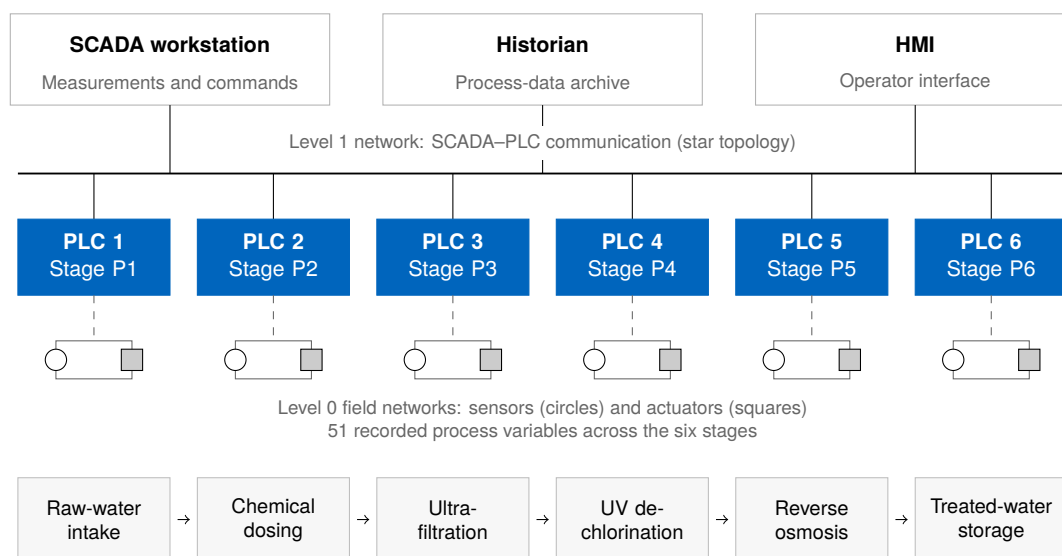
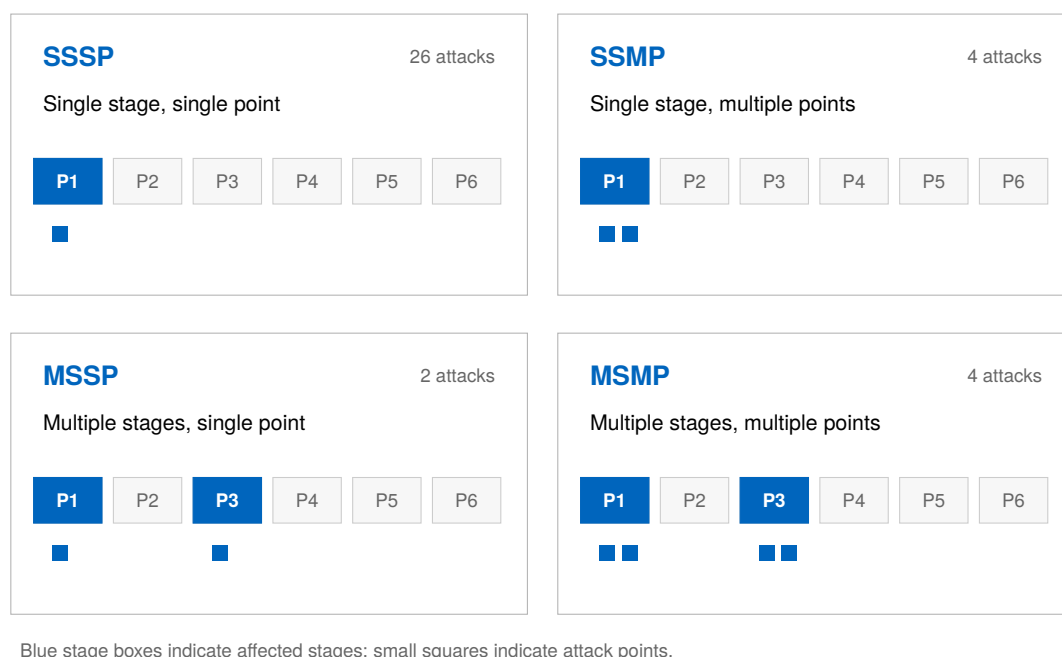


Figure 3.5. SWaT process and control architecture. Network levels follow the SWaT convention. The field-network symbols indicate sensors and actuators within each stage; the detector consumes their recorded values. Adapted from the supplied diagram and the dataset description [123].



Blue stage boxes indicate affected stages; small squares indicate attack points.

Figure 3.6. SWaT attack categories and their counts in the dataset paper [123]. Stage and point locations are schematic examples. The arrangement describes attack scope and carries no ranking of detection difficulty.

period timestamps form the test set. They comprise 51,453 anomalous and 342,226 normal timestamps, so anomalous observations account for 13.07% of the test cohort.

Collection campaign: 11 days, one sample per second



Figure 3.7. SWaT collection campaign and the partition used in the GAN study. The campaign follows the dataset paper [123]. After the initial normal-operation exclusion, the first 56,240 attack-period timestamps support training and validation; the later 393,679 timestamps form the test set.

All three partitions contain normal observations and labelled attack examples. Each attack episode is confined to one partition, and no window crosses a partition boundary or a discontinuity in the recordings. Partitioning precedes imputation, scaling, feature selection and window construction. Representations, model settings and detection thresholds are selected from the training and validation data. The model is therefore developed on earlier observations and evaluated on later observations that were not used for training or model selection.

The GAN objective does not require class annotations during gradient-based fitting. Nevertheless, the complete procedure includes supervised feature selection and model selection, and training includes attack observations. The resulting evaluation therefore concerns a detector developed with prior exposure to attacks. It does not establish detection performance for previously unseen attack types or a wholly unsupervised development process.

3.5.1 Model and Training

The model contains four jointly trained networks: an encoder, a generator, a data critic and a latent critic (Figure 3.8). For an input window X , the encoder E maps the observed process sequence to a latent representation $E(X)$. The generator G maps this representation back to the observation space, producing the reconstruction

$$\hat{X} = G(E(X)).$$

Latent samples z are drawn from a standard normal distribution. The data critic compares observed windows with generated windows $G(z)$, encouraging the generator to reproduce the distribution of the training sequences. The latent critic compares encoded observations $E(X)$



Figure 3.8. Multivariate GAN detection pipeline. (a) The four networks are trained together on earlier windows containing normal and attack observations; validation data select the representation, the settings and the threshold. (b) The later 393,679 test timestamps are scored with the trained networks; the latent critic has no part in the score a_t of Equation 3.1. The blue tag marks the input representation, the only factor varied between runs, and orange tags mark the steps that used attack ground truth. The window length is fixed at 60 samples. The scaler and PCA are fitted on training data and reused for validation and testing. The candidate implemented every processing step shown.

with samples from the latent prior. A cycle-consistency term penalises discrepancies between an observed window and its reconstruction,

$$\mathcal{L}_{cyc} = \mathbb{E}_X [\|X - G(E(X))\|_2].$$

Training combines the two Wasserstein objectives with this reconstruction constraint. The encoder provides a direct mapping from process observations to latent representations, while the cycle-consistency term encourages those representations to retain information required to reconstruct the input. The data critic supplies an additional measure of how closely an observed sequence resembles the learned training distribution. The latent critic contributes to training and is not included in the anomaly score at inference [128].

At inference, each test window is encoded and reconstructed using the fitted model. Discrepancies between the observed and reconstructed sequences provide a reconstruction score (Figure 3.9). The data critic supplies a second score for the observed window. Before combination, the scores require a common orientation, such that larger values indicate greater deviation from the reference behaviour. Their normalisation must also prevent a product of two negative standardised values from being interpreted as strong anomalous evidence. The reported score therefore uses non-negative, consistently oriented normalised components:

$$a_t = \tilde{r}_t \tilde{c}_t. \quad (3.1)$$

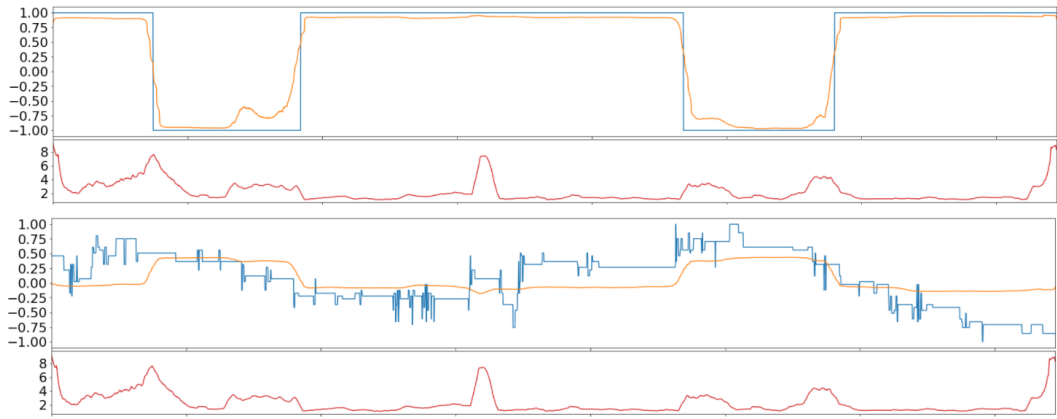


Figure 3.9. Observed signals (blue) and their reconstructions (orange) for two dimensions, reproduced at the original embedded resolution from Figure 4.15 of the internal report. The repeated red trace is the aggregate reconstruction discrepancy across dimensions, shared by the two displayed signals. It precedes the combination of Equation 3.1.

The transformation parameters are estimated from the development data and held fixed during evaluation. This scoring rule combines reconstruction discrepancy with the critic’s assessment of the observed sequence. A high reconstruction discrepancy alone does not establish that an attack occurred, because changes during normal operation may also be difficult to reconstruct. Similarly, a critic score is interpreted through the fitted scoring and thresholding procedure rather than as a calibrated probability of attack.

Overlapping windows produce several reconstructions and critic assessments for some timestamps. For the offline evaluation, reconstructed values are averaged at each corresponding timestamp and variable before calculating the multivariate reconstruction discrepancy. Critic outputs from windows covering the same timestamp are also averaged. Each observation then receives one combined anomaly score and one final prediction. Observations near a sequence boundary are aggregated over the windows available within their own partition. This convention ensures that overlapping windows do not cause a timestamp to be counted repeatedly in the confusion matrix.

An adaptive threshold converts the combined score into a binary anomaly prediction. The threshold rule, its historical context length and any interval-merging parameters are chosen using validation data and fixed before the test cohort is scored. Test ground truth is used only for the final performance calculation. The historical context used by the threshold is a separate parameter from the 60-second window used by the neural networks. Because the stated aggregation uses windows extending beyond the timestamp being scored, the evaluation measures offline detection. It does not measure causal online performance or real-time detection delay.

3.5.2 Input Representations and Settings

The representation comparison includes the complete set of 51 process variables, two principal component analysis (PCA) configurations, an autoencoder representation and a selected

feature subset. The full representation retains the original measurements and actuator states. It provides the broadest input coverage but also exposes the model to redundant relationships and normal operational variation across the process.

PCA represents the measurements through orthogonal combinations of the original variables. The five-component configuration imposes stronger compression, whereas the 24-component configuration retains 99% of the variance in the training data. Retained variance is a property of the fitted representation and does not measure the proportion of attack-relevant information preserved. A low-variance direction may still distinguish an attack from normal operation. The two PCA configurations therefore represent different balances between compression and retention of potentially informative changes.

The autoencoder provides a learned nonlinear representation before GAN processing. Its purpose is to retain relationships that support reconstruction while reducing the dimensionality of the input. The feature-selection branch combines correlation analysis with chi-squared selection, recursive feature elimination, LASSO and Random Forest importance. Supervised selectors use training ground truth only. Any non-negative transformation needed by chi-squared selection is fitted within the training partition, separately from the subsequent scaling used for GAN inputs. The identities of selected variables and the autoencoder bottleneck dimension remain implementation specifications; the aggregate performance results do not identify them.

Imputation and scaling are fitted on the training partition and applied consistently to validation and test observations. PCA and the preprocessing autoencoder are likewise fitted using training data. Features supplied to the GAN are scaled using a training-derived transformation with a reference range of $[-1, 1]$. An observation outside the training range need not fall inside this interval. Automatic clipping is not applied because it could suppress an informative deviation. These transformations preserve the separation between model development and final evaluation.

For a representation with d features, each input window contains 60 consecutive observations and has dimensions $60 \times d$. The window advances by one observation. At the nominal sampling frequency, this corresponds to a 60-second context with a one-second stride. An uninterrupted sequence of 393,679 observations would yield 393,620 complete windows, but its performance denominator remains 393,679 timestamps after aggregation. The 60-second setting is adopted for the reported evaluation; no claim of optimality across other window lengths is made.

The final configuration uses Adam with a learning rate of 0.0005, a latent dimension of 20, a batch size of 64 and 50 training epochs. All representation variants use the same evaluation cohort and counting convention. Their input dimensions vary, so comparison requires explicit control of the architecture and development procedure. A difference between variants cannot automatically be attributed to dimensionality reduction if network capacity or tuning effort also differs.

The reference computing platform comprises an AMD EPYC 7551P processor with 32 cores, 125 GB of RAM and a TITAN X GPU with 12 GB of memory. These specifications describe

Table 3.6. Measured timestamp-level performance of the input representations. Each timestamp is counted once and no point adjustment is applied.

Input representation	Precision (%)	Recall (%)	F_1 (%)	False-positive rate (%)
All 51 variables	70.00	94.50	80.43	6.09
PCA: 5 components	75.00	86.50	80.34	4.34
PCA: 24 components	73.00	92.00	81.41	5.12
Autoencoder representation	72.00	86.50	78.59	5.06
Selected feature subset	71.00	84.00	76.96	5.16

the reported execution environment. They do not establish training duration, inference throughput or memory consumption. Reducing the input dimension from 51 variables to 24 components constitutes a 52.94% reduction in input dimensionality; computational savings would require direct measurement.

3.5.3 Detection Results

Table 3.6 presents the measured performance of the five input configurations. Each row uses the same class totals: 51,453 anomalous timestamps and 342,226 normal timestamps. Precision describes the proportion of predicted anomalous timestamps that fall within recorded attack periods. Recall describes the proportion of actual anomalous timestamps that are detected. F_1 is the harmonic mean of precision and recall, while the false-positive rate is the proportion of normal timestamps incorrectly flagged. All table entries are calculated from integer confusion counts and rounded to two decimal places.

The complete representation has the highest recall, at 94.50%, together with the highest false-positive rate, at 6.09%. This profile represents a detector that identifies a larger proportion of anomalous observations while also flagging more normal observations. The five-component PCA configuration has the highest precision, at 75.00%, and the lowest false-positive rate, at 4.34%, accompanied by lower recall of 86.50%. In these results, stronger compression therefore corresponds to fewer false-positive timestamps and more missed anomalous timestamps.

The 24-component PCA configuration provides an intermediate balance, with precision of 73.00%, recall of 92.00% and an F_1 score of 81.41%. Relative to the complete representation, its precision is 3.00 percentage points higher and its recall is 2.50 percentage points lower. Its F_1 score is approximately 0.98 percentage points higher. It is consequently adopted as the final configuration on the basis of its numerical F_1 balance. This designation does not imply a statistically established advantage or preference under every operational cost model.

The autoencoder representation has a precision of 72.00% and recall of 86.50%, corresponding to an F_1 score of 78.59%. The selected feature subset has precision of 71.00%, recall of 84.00% and F_1 of 76.96%. These values show lower detection coverage for the two configurations. They do not identify a causal mechanism: loss of relevant input information, differences in representation learning and changes in model capacity would require separate investigation.

Table 3.7. Timestamp-level confusion matrix for the final 24-component PCA configuration. Counts refer to individual timestamps after aggregation of overlapping window outputs.

Actual class	Predicted anomalous	Predicted normal	Total
Anomalous	47,337 (TP)	4,116 (FN)	51,453
Normal	17,508 (FP)	324,718 (TN)	342,226
Total	64,845	328,834	393,679

The final 24-component PCA evaluation produced the confusion matrix in Table 3.7: 47,337 of the 51,453 anomalous timestamps are detected and 17,508 of the 342,226 normal timestamps are flagged as anomalous.

The performance measures follow directly from these counts and are calculated from the unrounded values:

$$\begin{aligned}
 \text{Precision} &= TP / (TP + FP) &= 47,337 / 64,845 &= 73.00\%, \\
 \text{Recall} &= TP / (TP + FN) &= 47,337 / 51,453 &= 92.00\%, \\
 F_1 &= 2TP / (2TP + FP + FN) &= 94,674 / 116,298 &= 81.41\%, \\
 \text{FPR} &= FP / (FP + TN) &= 17,508 / 342,226 &= 5.12\%, \\
 \text{Accuracy} &= (TP + TN) / (TP + FN + FP + TN) &= 372,055 / 393,679 &= 94.51\%.
 \end{aligned}$$

The recall of 92.00% means that the detector flags approximately 92 of every 100 anomalous timestamps. It does not mean that 92% of attack episodes are detected. Long attacks contribute more timestamps than short attacks, so the same timestamp-level recall could arise from different distributions of detection across attack episodes. Precision of 73.00% means that approximately 27% of flagged timestamps fall within recorded normal periods. This proportion differs from the false-positive rate of 5.12%, whose denominator includes all normal timestamps.

Accuracy is interpreted alongside the class distribution. Since normal observations account for 86.93% of the cohort, a rule assigning every observation to the normal class would achieve 86.93% accuracy while detecting no anomalous timestamps. The measured accuracy of 94.51% therefore provides supplementary information, whereas recall, precision and F_1 directly describe the balance relevant to identifying the anomalous class.

No point adjustment is applied. A timestamp inside an attack interval counts as a true positive only if that timestamp is predicted anomalous. Detecting one observation does not automatically convert all other observations in the interval into true positives. Contiguous positive predictions may be grouped into alarm intervals for display or operational analysis, but grouping does not retrospectively alter the timestamp-level classifications used in the matrix.

3.5.4 Interpretation and Evaluation Scope

The measured results describe a detector with high coverage of anomalous timestamps and a substantial false-positive burden. At the nominal one-second cadence, 17,508 false-positive observations correspond to approximately 4.86 hours of normal observations being flagged. This duration does not identify the number of alarms presented to an operator. A sustained interval and many isolated detections could contribute the same number of false-positive timestamps while requiring very different responses. Likewise, the 4,116 missed anomalous timestamps correspond to 68.6 minutes of anomalous observations, but their operational significance depends on which attacks and phases of process disruption they represent.

Attack-level assessment requires matching detected intervals to the retained attack episodes. Relevant outcomes include the number of episodes detected, the time between attack onset and the first qualifying detection, and the duration of detection within each episode. A minimum overlap or persistence rule must be specified before those outcomes are calculated. Attacks without a qualifying detection remain missed episodes and cannot be omitted from the interpretation of detection delay. The denominator is the number of attack episodes present in the evaluation cohort, which cannot be inferred solely from the 36 attacks described for the source dataset.

False alarm episodes similarly require an explicit grouping rule and a normal-operation exposure denominator. Their rate can be expressed as the number of distinct false alarm episodes per day of evaluated normal operation. Neither this rate nor the distribution of alarm durations follows from the confusion matrix alone. Timestamp-level precision also depends on the prevalence of anomalous observations: its value in a cohort containing 13.07% anomalous timestamps does not establish the precision expected during deployment with a different attack prevalence.

The comparison between representations specifies a modest difference in F_1 rather than evidence of a universal advantage for PCA. The complete input configuration retains the highest recall, while the five-component PCA representation produces the fewest false-positive timestamps relative to normal observations. The preferred configuration therefore depends on the relative consequences of missed anomalous behaviour and unnecessary alarms. Retention of 99% of training variance provides a description of the 24-component representation, not a guarantee of detection coverage or robustness.

The report gives no run-to-run variability from independently initialised training runs. A repeated-run analysis under the same partition and model-selection procedure should calculate precision, recall, F_1 and false-positive rate for each run, followed by their means and standard deviations. Resampling-based uncertainty estimates must account for temporal dependence and clustering within attack episodes because individual timestamps are not independent.

Further evaluation would separate the contributions of reconstruction scoring, critic scoring and their combination, and examine window length under a common validation procedure. Precision-recall curves would characterise changes across thresholds; their area cannot be derived from one confusion matrix. Training duration, inference throughput and peak memory use would establish computational cost on the stated hardware. The model topology, loss

weights, normalisation rule, threshold parameters, software versions, feature identities and partition records would form the implementation record needed for reproducibility.

Tables 3.6 and 3.7 report the empirical scope of this evaluation. The chronological partition prevents the later test observations from entering training or model selection. The report does not give the division of the first 56,240 attack-period timestamps between training and validation, and incomplete configuration records limit independent reproduction of the experiment.

3.6 Limitations, Validity, and Individual Contribution

Scope of the service-graph evidence. The artefact behind RQ1.1 is a two-page contribution to PAM 2018 on which the candidate is the fourth of four authors. The peer-reviewed anomaly-detection paper from the same group uses clustering rather than a graph, and the short paper that does use a graph is framed as access control [30, 31]. The portable service model was proposed but not implemented. RQ1.1 therefore carries no measured detection result.

Scope of the industrial evaluation. The S7-300 result rests on 25 attack episodes recorded in one plant over 24 hours, and the held-out partition holds a single episode of each class. Records taken from one episode are correlated, so the per-class scores of Table 3.4 summarise five attack events and not 1,233 independent trials. The benign monitoring day was a separate exercise with four active PLCs and executed no attack, so it bounds the false-alarm rate and says nothing about recall. No experiment reported here trains on one cell and tests for attacks on another, so transfer between installations is untested. The importances in Figure 3.4 are impurity decreases on the training partition, so they describe what this forest used and do not show which fields detection requires. Both exercises were labelled by the team that executed the attacks, and each two-hour baseline covers only the operating modes that occurred within it.

Sampling and selection. The GAN study develops the model on the first 56,240 attack-period timestamps and evaluates it on the later 393,679 timestamps. Training and validation also use separate normal-operation data, and attack episodes do not cross partition boundaries. This chronological split prevents test observations from entering training or model selection, but it remains within one SWaT campaign. The S7-300 evaluation holds out whole capture sessions. Neither study tests transfer to another installation or an operational class ratio. No repeated runs or uncertainty intervals are reported, so small differences between configurations cannot be interpreted as stable improvements.

Simple baselines. On SWaT, simple tests on process values have matched much more elaborate detectors under a common protocol [103]. Reimplementing published intrusion detectors in one framework has likewise left simple baselines competitive [121]. No such baseline is reported for the GAN study or for the S7-300 evaluation, so neither result shows that its model outperforms a simple rule on the same data.

Individual contribution. The candidate designed and implemented NADICS, its model interface, and the S7 and PROFINET extractor. He produced the labelled S7-300 attack corpus. For the process study, he designed and implemented the multivariate GAN pipeline, its preprocessing variants, dynamic architecture, training, scoring, and evaluation on SWaT. He analysed the results and wrote the internal report. His contribution to the service-graph study was the concept and the measurement design. The IUNO deliverables attribute authorship at document level [33, 34], so this account rests on the candidate’s attestation and on the surviving artefacts rather than on passage-level attribution.

3.7 Chapter Summary

RQ1.1 asks how service relationships can be represented as communication graphs. The reported design stores observed service relations as a node-local whitelist and permits a site owner to revise it as the deployment changes. Its memory and latency costs were measured. Detection quality was not.

RQ1.2 asks how a modular framework can analyse network and process features for anomaly detection in industrial systems. NADICS separates dataset layout, preprocessing, and learners behind fixed interfaces. On the S7-300 corpus recorded inside IUNO the framework reaches 94.50 per cent attack F_1 and 89.56 per cent macro- F_1 over six classes at a record false-positive rate of 0.484 per cent, and a benign day with four active PLCs raised nine alarm episodes, all of them false. For SWaT process windows, the GAN evaluation reports 92.00 per cent timestamp recall at a false-positive rate of 5.12 per cent with 24 PCA components. The full input gives the highest recall and five components the lowest false-positive rate, so the preferred representation depends on the relative cost of missed anomalies and false alarms.

Together, these mechanisms close gap G1 at the implementation level. No study shows transfer across time, sites, or adaptive attacks, which motivates Chapters 6 and 7.

Multimodal Android Malware Analysis with Hybroid

4.1 Research Problem and Contribution

This chapter addresses RQ2, stated in Section 1.3: To what extent does fusing static code-graph representations with dynamic network-flow features improve Android malware detection and category classification compared with either modality alone? It examines gap G2 through a comparison of program-code and network-flow representations on the same application cohort. The comparison measures the additional value of fusion for each prediction task and learner.

Hybroid combines program structure recovered from an Android application package with network traffic recorded during execution on a physical handset. Every sample enters the pipeline as such a pair (1), and the two observations are analysed in parallel (2). The static branch yields a 64-dimensional code vector, the dynamic branch 13 network-flow features, and the two are concatenated into one joint vector per sample (3). A supervised classifier is trained and cross-validated on that vector (4) and produces the predictions (5). Two tasks are evaluated separately: binary malware detection and five-class classification into Benign or one of four malware categories. Figure 4.1 presents the pipeline; Sections 4.5 to 4.8 define its stages.

The contribution comprises the multimodal pipeline and its within-system evaluation. The network branch aggregates bidirectional flow records into 13 features per application. The code branch uses learned instruction embeddings and program graphs to represent static structure. Their common application-level granularity permits a direct comparison of each branch with the joint representation. Section 4.11 uses these contrasts to answer RQ2.

The work was published at ISC 2021 [35]. The candidate is the first of four authors, with shared first authorship recorded for the first two. He designed the overall pipeline and its network-traffic analysis, together with the framework used for preprocessing and evaluation. He conducted the experiments and wrote the manuscript. His contribution to graph processing was partial, as stated in Section 1.5.

4.2 Related Work

Static Android detectors use features available in the application package. DREBIN combines manifest declarations with features extracted from bytecode, including API calls and network

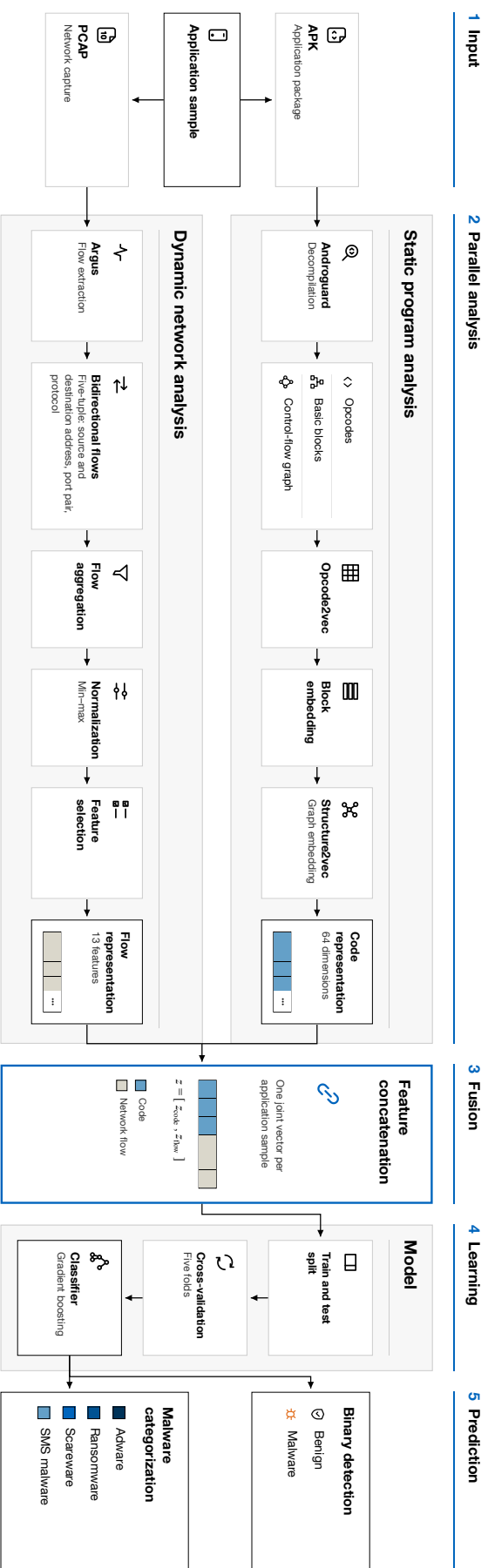


Figure 4.1. The Hybrid architecture, redrawn by the candidate from [35].

addresses, in a sparse representation classified by a linear model [1]. It is one of the baselines in Section 4.9. API-level detectors include DroidAPIMiner [129], while other approaches combine permissions and API calls [130]. Factorisation machines and static behaviour classifiers also use manifest and API feature vectors [131, 132].

Opcode-based methods describe instructions at a lower level. Canfora et al. evaluate opcode n -grams for malware detection [133]; McLaughlin et al. train convolutional networks on opcode sequences [134]. Hybroid uses opcode embeddings as features within a structured program representation. Basic blocks and their control-flow relations provide the intermediate units described in Sections 4.5 and 4.6.

Graph-based methods differ in both the relations they encode and their vertex features. Adagio maps neighbourhoods of a function-call graph to features for a linear classifier and permits the associated code structures to be inspected [64]. apk2vec learns application representations from several graph views [135]. MaMaDroid models transitions between abstracted API calls as Markov chains; package- or family-level abstraction reduces dependence on individual method identities [2]. Attributed control-flow graphs have also been used for binary-code similarity and malware classification [66, 136]. These methods operate at different program granularities, as defined in Section 2.2.1.

The choice of vertex features remains relevant in later work. Tran et al. augment function-call graphs with function metadata and code embeddings, and report improved classification under distribution shift relative to topology-based features [137]. Their MalNet-derived benchmarks provide evidence for that comparison, rather than a validation of Hybroid's opcode representation. A broader survey describes the extraction and learning choices used in graph-based malware detection [8].

Dynamic network analysis measures communication during execution. Arora et al. use traffic statistics with a rule-based classifier, and Zulkifli et al. evaluate a decision tree on network features [138, 139]. CREDROID examines network activity, including transmitted information [140]. In the broader malware-traffic literature, learned representations operate on raw packets or sessions [141, 142]. The observation unit determines which traffic is represented together and what a prediction label describes. Hybroid aggregates flow records at application level to align them with its static representation.

The CICAndMal2017 study classifies Android malware from flow features and provides the dynamic baseline used here [143]. A later extension combines network-flow and API features [144]. Hybrid methods cover a wider range of observation pairs and fusion mechanisms [145]. MARVIN compares static, dynamic and combined features, while the study by Onwuzurike et al. compares the modalities through a common behavioural model [57, 25]. Both report advantages for the combination in their evaluated settings. The benefit for a particular representation pair and task still requires a within-system comparison, which motivates RQ2.

HGDetector later combines static network-related call-graph features with graphs derived from network traffic [146]. Its experiments use CICAndMal2017 and a filtered CICMalDroid2020 corpus, with gains that depend on the available traffic. This supports examining fusion under

explicit observation conditions. Its reported gains are not directly comparable with Hybroid's because the representations and evaluation cohorts differ.

4.3 Observation and Threat Model

The static observation is the Dalvik bytecode recovered from the APK, including its methods and call sites. The dynamic observation is the packet capture associated with an application's execution on a physical handset. Argus converts that capture into bidirectional flow records. The classifier receives features derived from both observations after acquisition and preprocessing [35].

This design supports offline application analysis when an APK and a corresponding traffic capture are available. In the reported evaluation, those inputs were supplied by CICAndMal2017; the candidate's pipeline processed the recorded traffic. A new application would likewise require a capture before the fused classifier could be used. Chapter 5 examines a different observation requirement: analysis from the application package alone.

The static branch covers the bytecode visible to its analyser. Behaviour implemented in native libraries or code loaded during execution can be absent from that representation. The dynamic branch records network activity during the capture window, which may include consequences of such behaviour, but it supplies neither system-call traces nor decrypted payload contents. It therefore provides evidence about the observed execution rather than all possible executions.

The evaluation is non-adaptive: the malware samples were collected before the detector was developed, and no attacker modifies an application in response to its predictions. Attacker knowledge and perturbation budgets are not specified. The results consequently concern classification of the collected samples, with adversarial robustness remaining a separate evaluation question.

4.4 Dataset and Experimental Protocol

The evaluation uses CICAndMal2017, which supplies application packages and recorded network traffic [143]. Benign applications were collected from Google Play across 2015 to 2017; malicious applications came from VirusTotal and the Contagio archive. Traffic was captured on three physical handsets with scripted user behaviour and live accounts on messaging and social platforms. The capture scenarios cover installation and periods before and after a reboot [147]. These paired observations support the comparison of static and dynamic representations required by RQ2.

Table 4.1 gives the corrected source-corpus composition. The malware counts total 426 across 42 families; the category rows in the Hybroid publication instead sum to 446. The official dataset documentation records 5,065 installed benign applications, of which the Hybroid study uses 1,700 package/capture pairs.

Table 4.1. CICAndMal2017 source-corpus composition before extraction failures are excluded. Malware-category counts are recomputed from the official family listing [147]; the 1,700 benign samples are the package/capture pairs used in [35].

Class	Families	Samples	Share
Benign		1,700	80.0 %
Adware	10	104	4.9 %
Ransomware	10	101	4.8 %
Scareware	11	112	5.3 %
SMS malware	11	109	5.1 %
Malware	42	426	20.0 %
Total		2,126	100.0 %

Both analysis branches excluded the 47 graph-extraction failures from the 2,126 source samples and used the same remaining 2,079 applications. This corresponds to processing coverage of 97.8 per cent. All modality comparisons concern that common cohort; excluded samples are not counted as classification successes or errors. Their class labels are unavailable, so the post-exclusion class counts cannot be reconstructed.

The paper reports five-fold cross-validation with random shuffling and scores averaged over the folds [35]. Implementation used Python with scikit-learn, TensorFlow and Keras, on a server with 32 cores, 128 GB of memory and a 16 GB GPU. Accuracy and F_1 are reported alongside precision, recall and ROC AUC. The class-averaging convention is unspecified; Section 4.9 explains the implications for interpreting these values.

A supplementary static-only experiment used 45,592 malicious and 90,313 benign applications from AndroZoo, VirusShare and VirusTotal. Its 135,905 samples were divided at random into 80 per cent for training and 20 per cent for testing [35]. This experiment evaluates the code representation on a larger corpus; it provides no additional modality comparison. Its detection results are reported in Section 4.9.

4.5 Opcode and Basic-Block Representation

Dalvik instructions encode an operation and its operands. Operand fields may identify registers, contain literal values or reference other DEX items, such as methods and strings [45]. Hybroid retains the opcode and omits these operand fields. This abstraction limits the vocabulary to the 216 opcode types used in the study, at the cost of information about the values and objects on which each operation acts.

The opcode vectors are learned with the skip-gram formulation of word2vec [148]. The training corpus contains 18,240,542 opcode occurrences extracted from AndroZoo applications

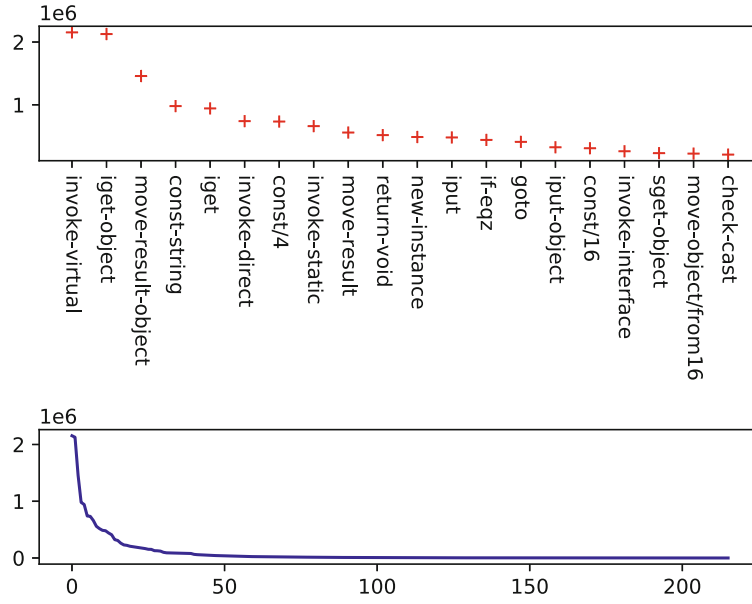


Figure 4.2. Opcode frequency across the pre-training corpus. The upper panel names the twenty most common operations, and the lower panel plots all 216 in rank order. The figure describes empirical frequencies, without a statistical test of a power-law fit. Reproduced from [35].

[35]. For a sequence $o_1, \dots, o_L$ over a vocabulary $\mathcal{O}$, let $\mathcal{C}_t$ contain the positions within a context window of o_t , excluding t itself. The canonical objective is

$$\max_{\theta} \frac{1}{L} \sum_{t=1}^L \sum_{j \in \mathcal{C}_t} \log p_{\theta}(o_j | o_t), \quad (4.1)$$

$$p_{\theta}(o' | o) = \frac{\exp(u_{o'}^T v_o)}{\sum_{r \in \mathcal{O}} \exp(u_r^T v_o)}.$$

Here $v_o, u_o \in \mathbb{R}^l$ are the input and output embeddings, and θ contains their parameters. The objective rewards prediction of local opcode contexts. It defines the learning aim; the Hybroid publication does not specify which training approximation was used.

Figure 4.2 shows that opcode frequencies are uneven. Skip-gram uses contextual co-occurrence and does not require a power-law distribution. A learned lookup table can be reused for applications whose opcode types it covers, although vocabulary coverage alone does not establish generalisation to new applications or independence of the evaluation data.

Operand removal also makes calls with the same opcode share a token embedding regardless of their callee [149]. Call targets can still contribute to graph construction through their operand references. Thus, the token features and graph edges retain different information from the same bytecode; omitting operands from the former does not imply independence of the two.

Basic-block features provide the input to the method-level CFG encoder. The publication describes them as weighted means of opcode vectors [35]. For a block b with opcode positions I_b , this design can be written as $x_b = \sum_{j \in I_b} w_{bj} v_{o_j} / \sum_{j \in I_b} w_{bj}$, with $w_{bj} \geq 0$ and

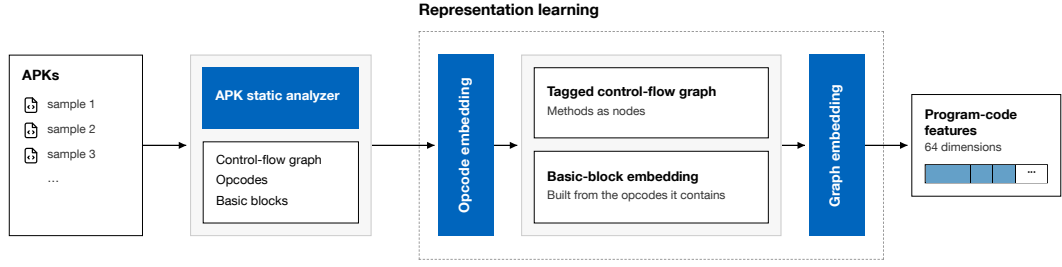


Figure 4.3. Published design of the program-code branch, redrawn by the candidate from [35].

$\sum_{j \in I_b} w_{bj} > 0$. The result lies in $\mathbb{R}^l$. This notation expresses the stated block-level design: the source uses function-level indexing in its formula and does not specify the weighting rule, so the exact aggregation procedure remains under-specified.

4.6 Program-Graph Representation

The static branch uses Androguard to extract bytecode and program graphs [51, 35]. A method’s control-flow graph describes relations between its basic blocks. An application-level function-call graph instead relates methods through their call sites. The published design places opcode-derived features within this hierarchy and produces a 64-dimensional application vector (Figure 4.3). The equations below define its documented method-level encoder; the application-level composition is discussed separately.

Static graph recovery depends on the analyser and its treatment of Android execution (Section 2.2.1). It can include infeasible calls and omit calls whose targets are unresolved, for example through reflection or runtime-loaded code. The representation therefore describes recovered structure, with coverage bounded by the analysis.

Hybrid adapts structure2vec to propagate information between neighbouring basic blocks [65, 35]. Let $G_f = (V_f, E_f)$ be the CFG of method f , with block features $x_v \in \mathbb{R}^l$. Each vertex has a hidden state $\mu_v^{(t)} \in \mathbb{R}^p$, and $N_f(v)$ denotes its neighbour set for propagation. The Hybrid algorithm initialises these states at random and updates them for T rounds:

$$\mu_v^{(t+1)} = F \left(x_v, \sum_{u \in N_f(v)} \mu_u^{(t)} \right), \quad v \in V_f, \quad (4.2)$$

The update function is

$$F \left(x_v, \sum_{u \in N_f(v)} \mu_u^{(t)} \right) = \tanh \left(W_1 x_v + \sigma \left(\sum_{u \in N_f(v)} \mu_u^{(t)} \right) \right), \quad (4.3)$$

where $W_1 \in \mathbb{R}^{p \times l}$ is trainable and $\sigma : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is a fully connected network. For k compatible trainable weight matrices, its stated form is

$$\sigma(h) = P_1 \text{ReLU}(P_2 \cdots \text{ReLU}(P_k h)). \quad (4.4)$$

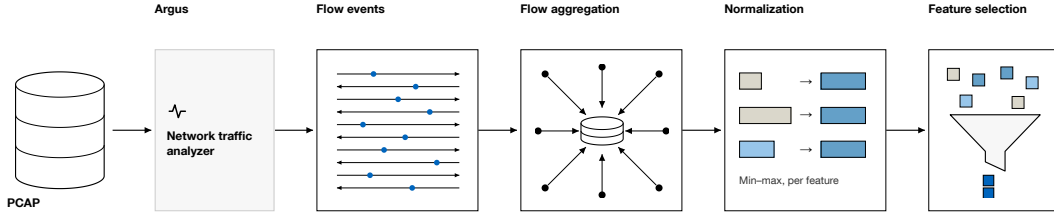


Figure 4.4. The network branch, redrawn by the candidate from [35].

After propagation, the readout in the publication’s Algorithm 1 averages the block states and applies a linear transformation [35]:

$$h_f = W_2 \left(\frac{1}{|V_f|} \sum_{v \in V_f} \mu_v^{(T)} \right), \quad W_2 \in \mathbb{R}^{d_m \times p}, \quad |V_f| > 0. \quad (4.5)$$

The resulting $h_f \in \mathbb{R}^{d_m}$ represents one method. Its hidden states depend on information propagated through at most T edges of the chosen neighbour relation. For $T \geq 1$, $\tanh$ bounds each state coordinate between -1 and 1 , the coordinatewise mean remains in that interval regardless of the number of basic blocks. This controls the direct size dependence of the pooled values without guaranteeing that different graphs remain distinguishable.

The published design subsequently forms a 64-dimensional application representation from the method information and call structure. Its composition operator is not specified, so Equation 4.5 should not be read as an application-level pooling rule. The symbol d_m denotes the unreported method-vector dimension. The encoder can be applied to new graphs with compatible inputs; that architectural property alone does not establish how training and test applications were separated during representation learning.

4.7 Network-Flow Representation

Argus produces bidirectional flow records from each capture [150]. Records are identified through source and destination addresses and ports, together with the transport protocol, and contain roughly forty statistics. Transaction direction and state flags are categorical and are one-hot encoded; the remaining fields are numeric [35].

Let $r_{ij} \in \mathbb{R}^q$ denote the encoded feature vector for flow record j in the capture associated with application i , where q is the dimension after categorical encoding. For m_i records, the application-level mean is

$$\bar{r}_i = \frac{1}{m_i} \sum_{j=1}^{m_i} r_{ij}, \quad m_i > 0. \quad (4.6)$$

This expresses the aggregation described in the publication [35]. The label applies to the application and its capture; individual flows are not assigned the application’s label as separate training samples. For a one-hot indicator, the mean gives the fraction of records with that category. For a byte-count field, it gives mean bytes per record, not the total volume of the capture.

Table 4.2. The 13 retained network-flow fields, grouped by the quantities they describe [35]. Their encoded values are averaged at application level as in Equation 4.6.

Kind	Argus fields before application-level averaging
Volume and duration	mean record duration, total bytes, source bytes
Transport state	destination TCP window, source loss and retransmission flag, destination out-of-order flag, transaction direction
Path and service	source and destination TTL, source and destination TOS byte
TCP sequence state	source and destination TCP base sequence number

Table 4.3. Reported HTTP and TLS flow shares by malware category [35]. The remainder is calculated as 100 per cent minus those shares; its protocol composition is unspecified.

Category	HTTP	TLS	Unspecified
Adware	52.00 %	8.00 %	40.00 %
Ransomware	29.22 %	0.00 %	70.78 %
Scareware	61.38 %	10.89 %	27.73 %
SMS malware	52.20 %	10.28 %	37.52 %

Reordering the extracted records leaves Equation 4.6 unchanged because addition is commutative. Record order and deviations from the mean are not encoded separately, although means of temporal fields such as duration are retained. Traffic associated with malicious activity may also be averaged together with background traffic in the same capture.

The averaged features are then min-max scaled [35]. For feature k , let $\mathcal{I}_{\text{fit}}$ denote the applications used to fit the scaling bounds:

$$\begin{aligned} \tilde{r}_{ik} &= \frac{\bar{r}_{ik} - a_k}{b_k - a_k}, & b_k &> a_k, \\ a_k &= \min_{i \in \mathcal{I}_{\text{fit}}} \bar{r}_{ik}, & b_k &= \max_{i \in \mathcal{I}_{\text{fit}}} \bar{r}_{ik}. \end{aligned} \quad (4.7)$$

The fitted range maps to $[0, 1]$; values outside that range can map outside the interval. Leakage-controlled cross-validation requires fitting the bounds and data-dependent feature selection within each training fold. The publication describes these operations but does not locate their fitting within the cross-validation procedure.

Feature selection retained the 13 fields in Table 4.2. The publication reports Pearson correlation, an extremely randomised trees estimator and univariate scoring as selection methods [35]. Kendall correlations were then examined for strong pairwise associations (Figure 4.5). This checks one form of redundancy; low pairwise correlation does not establish statistical independence or exclude nonlinear dependence.

Table 4.3 reports the observed protocol shares. HTTP accounts for 29.22–61.38 per cent of flows across the malware categories, and TLS for 0.00–10.89 per cent. The publication does not characterise the remaining protocols.

The network branch uses flow statistics to avoid processing payload contents in the classifier [35]. Application-layer encryption can occur within HTTP or other non-TLS traffic [151], so

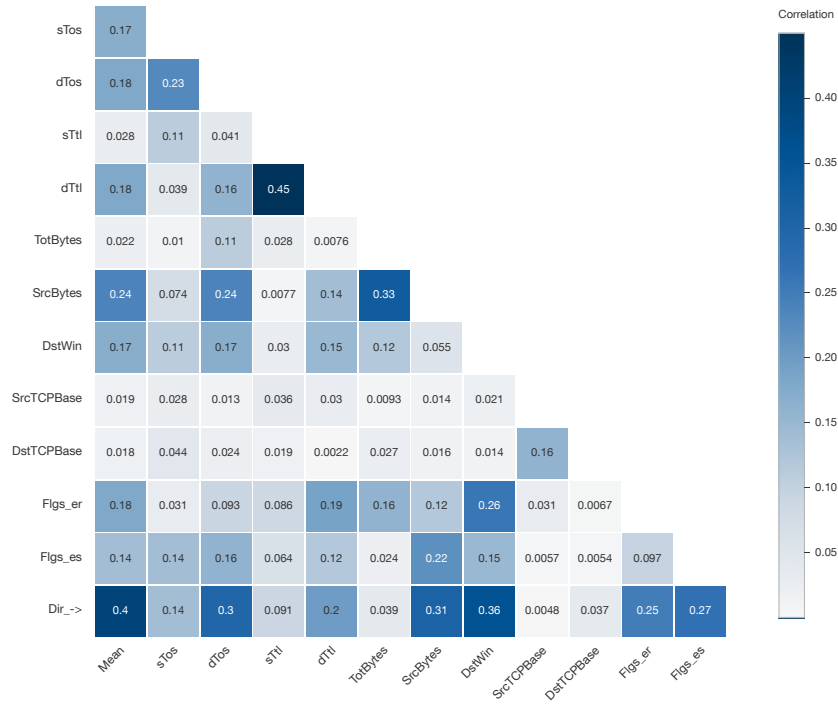


Figure 4.5. Kendall correlations among the retained flow features. Reproduced from [35].

the TLS share is not a measure of all encrypted content. The evaluation does not separate classification performance by encryption status.

4.8 Multimodal Fusion and Classification

Representation learning and final classification have distinct roles. Within the static branch, a two-layer fully connected network is trained for binary detection, with labels +1 for malicious applications and -1 for benign applications, and a decision threshold of zero [35]. This supervised stage fits the code representation. Its reported output for subsequent classification is a 64-dimensional application vector.

For application i , denote the static vector by $s_i \in \mathbb{R}^{64}$ and the selected flow features by $d_i \in \mathbb{R}^{13}$. Both feature groups form the joint input to the final classifier. This is feature-level fusion: the learner receives both observations, rather than predictions made by separate modality-specific classifiers. The 77 input features need not contribute equally, and their counts do not establish which modality influences the decision.

The final learner is trained separately for binary detection and five-class categorisation. The binary-trained static representation is used for both tasks; a category-specific refitting of that representation is not reported. The experiment therefore evaluates the available representation on each task, without isolating the effect of its binary training objective.

Table 4.4. Published binary-detection comparison with four baselines. Performance values are read from Figures 6 and 7 of [35]; their metric-averaging conventions are not fully specified.

System	Accuracy	F_1	Recall	Precision	AUC
SVM [152]	0.939	0.860	0.856	0.865	0.876
DREBIN [1]	0.954	0.867	0.873	0.861	0.961
Adagio [64]	0.893	0.932	0.913	0.953	0.867
CIC 2017 [143]	0.876	0.872	0.878	0.871	0.914
Hybroid	0.970	0.969	0.970	0.970	0.996

The reported comparisons cover a decision tree, random forest and gradient boosting [35]. Support vector machines and naive Bayes were excluded after preliminary tests. A decision tree partitions the feature space through axis-aligned splits; a random forest combines predictions from multiple trees. Gradient boosting constructs an additive model through sequential tree fits. These are different learning procedures, not a controlled ordering of model capacity. Each is evaluated with static features, flow features and their joint input in Sections 4.9 to 4.11.

4.9 Binary Detection Results

Table 4.4 reports the binary-detection comparison [35]. Hybroid has the highest reported value in each column, with accuracy of 0.970 and AUC of 0.996. The possible post-exclusion class counts imply a calculated accuracy of 79.5–81.8 per cent for an always-benign rule, which provides a majority-class reference. The baseline table compares complete detection pipelines; its interpretation also depends on preprocessing and tuning, as well as on common metric definitions.

The missing averaging convention can be illustrated through a confusion-matrix identity. For one binary test set of N applications, let π be the malware proportion, with precision P and recall R defined for malware as the positive class. The false-negative and false-positive counts satisfy $\text{FN}/N = \pi(1 - R)$ and $\text{FP}/N = \pi R(1 - P)/P$, so accuracy satisfies

$$\text{Acc} = 1 - \pi \left[(1 - R) + \frac{R(1 - P)}{P} \right], \quad P > 0. \quad (4.8)$$

Treating Hybroid’s reported accuracy, precision and recall of 0.970 as values from one such matrix would imply $\pi = 0.50$. The evaluated malware proportion is instead between 18.2 and 20.5 per cent, depending on the labels of the excluded applications. This illustrates why the reported row cannot be interpreted as a pooled malware-class confusion matrix. Class-averaged metrics have different meanings, and separately averaged fold metrics need not satisfy the nonlinear identity in Equation 4.8. The publication supplies neither the class-averaging definition nor raw predictions for a reconstruction. The table therefore retains the published scores without assigning an unreported convention to them.

Figure 4.6 shows the corresponding ROC curves. DREBIN has the second-highest reported AUC at 0.961. The dynamic CIC 2017 system reaches 0.914, above the SVM and Adagio entries despite its lower accuracy. Baseline ordering therefore depends on the metric. AUC

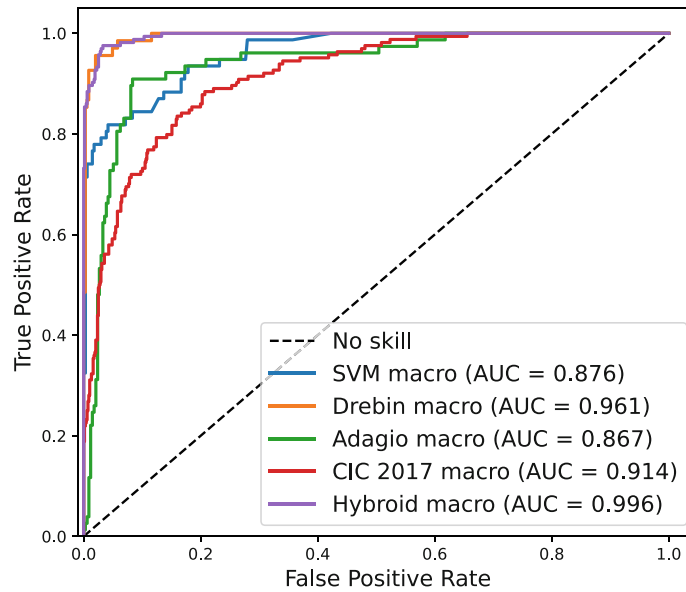


Figure 4.6. Receiver operating characteristic curves for binary malware detection, reproduced from the published vector content of Figure 7 in [35]. The source labels the curves as macro averages and reports their AUC values in the legend.

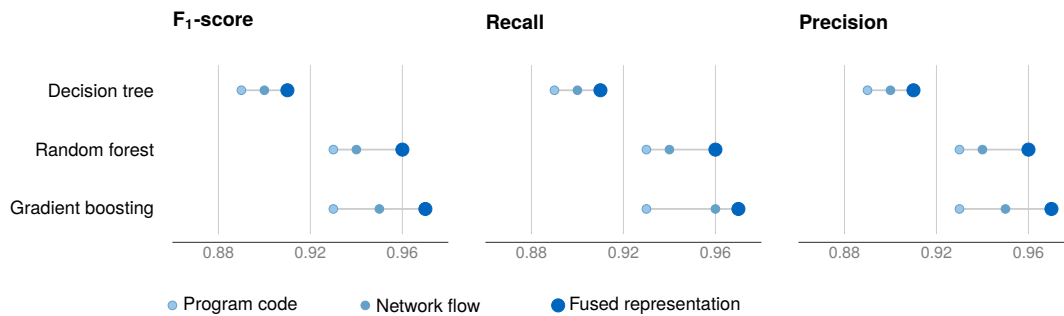


Figure 4.7. Binary detection by learner and observation modality, with a panel for each metric. Values are the five-fold means reported to two decimal places in Figure 8 of [35]. The score axis is truncated to 0.86–0.98. Fold-level scores are unavailable, so no error bars are shown. Redrawn by the candidate.

summarises ranking across thresholds; the source does not specify an operational decision threshold [35].

The external comparison includes three static systems and one dynamic system. Their different representations and implementations prevent attributing the score differences to observation modality alone. The within-system comparison below holds the application cohort and learner type fixed while changing the feature groups.

Gradient boosting with the fused representation reaches 0.97 for F_1 , recall and precision (Figure 4.7). Random forest reaches 0.96 and the decision tree 0.91 on those metrics. Flow features alone have higher point values than code features alone for each learner. With code features, random forest and gradient boosting both report 0.93 for all three metrics, so their results tie at the published two-decimal precision [35].

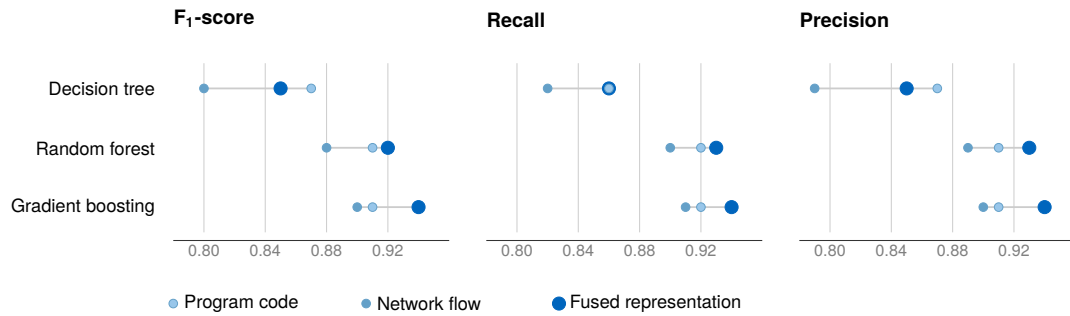


Figure 4.8. Five-class classification by learner and observation modality. Values are the five-fold means reported to two decimal places in Figure 9 of [35]. The score axis is truncated to 0.78–0.96. Decision-tree recall is 0.86 for code and fused inputs, so their markers coincide. Fold-level scores are unavailable, so no error bars are shown. Redrawn by the candidate.

Table 4.5. Area under the ROC curve per class, gradient boosting on the combined representation, with the pre-exclusion counts of Table 4.1. Values are read from Figure 10 of [35].

Class	Source samples	AUC
Benign	1,700	0.995
Ransomware	101	0.991
Scareware	112	0.979
Adware	104	0.963
SMS malware	109	0.946
Macro average		0.976

For the supplementary static-only corpus described in Section 4.4, the paper reports 95.0 per cent accuracy and 96.0 per cent F_1 [35]. These values come from the separate 80/20 evaluation on 135,905 applications.

4.10 Malware-Category Classification

The second task assigns each application one of five mutually exclusive labels: Benign or one of the four malware categories in Table 4.1. It is a single-label multiclass task, evaluated separately from binary detection [35].

Figure 4.8 gives the category-classification results. Gradient boosting on the fused representation reaches 0.94 on all three metrics. Random forest follows at 0.92 for F_1 and 0.93 for recall and precision. The decision-tree result differs: the fused representation reaches 0.85 for F_1 and precision, below the program-code values of 0.87, while recall is equal at 0.86. The effect of fusion therefore depends on the learner [35].

Table 4.5 gives the per-class ROC AUC for gradient boosting on the joint representation. The source figure reports a macro-average AUC of 0.976 [35]. This is a ranking measure, distinct from five-class accuracy.

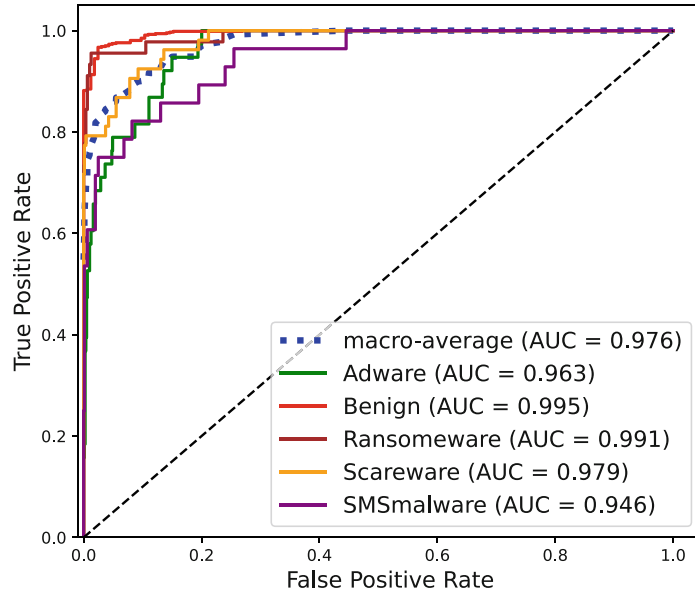


Figure 4.9. Per-class receiver operating characteristic curves for gradient boosting on the fused representation, reproduced from the published vector content of Figure 10 in [35]. The legend reports the AUC of each class and a macro-average AUC of 0.976.

Figure 4.9 preserves the published curves. Benign and ransomware have the largest reported AUCs, 0.995 and 0.991; SMS malware has the smallest at 0.946. The five displayed areas have an arithmetic mean of 0.9748, which does not reproduce the reported macro value through rounding alone. The construction of the macro curve is not specified in enough detail to resolve that difference, so both the class values and the reported macro AUC are retained.

Per-class precision and recall at the five-class decision rule are not reported [35]. Benign applications remain the majority after the exclusions; the aggregate score of 0.94 therefore does not establish equal performance across the classes.

4.11 Modality Comparison

RQ2 is evaluated through the within-system contrasts in Figures 4.7 and 4.8 [35]. For a fixed learner and task, let $\overline{M}_s$, $\overline{M}_d$ and $\overline{M}_f$ be the reported mean scores for static, dynamic and fused inputs. The contrast against the better single modality is

$$\Delta_M = \overline{M}_f - \max(\overline{M}_s, \overline{M}_d). \quad (4.9)$$

Positive values favour fusion. Table 4.6 applies this definition to the published two-decimal means, separately for each metric.

The single-modality ordering changes with the task. Network flow has the higher reported F_1 for detection under all three learners: 0.90 against 0.89, 0.94 against 0.93, and 0.95 against 0.93. Program code has the higher value for categorisation: 0.87 against 0.80, 0.91 against

Table 4.6. Difference between the fused score and the better of the two single-modality scores. Positive values favour fusion. The differences are computed from Figures 8 and 9 of [35], whose values are printed to two decimal places.

Task	Learner	ΔF_1	Δ recall	Δ precision
Detection	Decision tree	+0.01	+0.01	+0.01
Detection	Random forest	+0.02	+0.02	+0.02
Detection	Gradient boosting	+0.02	+0.01	+0.02
Categorisation	Decision tree	-0.02	0.00	-0.02
Categorisation	Random forest	+0.01	+0.01	+0.02
Categorisation	Gradient boosting	+0.03	+0.02	+0.03

0.88, and 0.91 against 0.90. This reversal is consistent across the three learners reported in the paper. The experiment does not identify a causal mechanism for it [35].

Fusion adds one or two percentage points of F_1 in binary detection. For categorisation, the contrast is minus two points for the decision tree, plus one for random forest and plus three for gradient boosting. For example, gradient boosting improves from the better single-modality score of 0.91 to 0.94 on that task. These comparisons support a modest reported fusion benefit for detection and a learner-dependent result for category classification [35].

Fold-level scores and raw predictions are unavailable, so uncertainty and paired statistical tests cannot be reconstructed. The contrasts are differences between rounded reported means, not estimates of a statistically established improvement [35].

4.12 Limitations and Threats to Validity

Observation conditions and temporal validity. The traffic describes the configured execution windows, so behaviour that was not triggered may be absent. Its age also limits transfer to current applications. For applications targeting Android 9 (API level 28) or higher, cleartext traffic is disabled by default, although an application can opt in [153]. Flow statistics remain observable with encrypted payloads, but their distribution may change with the protocol mix. Neither the main cross-validation experiment nor the larger static-only experiment enforces temporal ordering. Their results therefore do not establish prospective performance on later applications [12].

Capture-related feature associations. Some flow fields may reflect endpoints or capture conditions as well as application activity. An audit of UNSW-NB15 found that TTL features could distinguish operating-system populations associated with benign and malicious traffic [101]. That finding concerns a different corpus, but it identifies a mechanism relevant to the fields retained here. An ablation of TTL, TOS and TCP sequence features would assess reliance on those fields and whether their associations persist under changed capture conditions. No such test is reported for Hybroid.

Processing coverage and obfuscation. The common-cohort comparison is conditional on successful extraction of both representations. The excluded graph-extraction failures are

attributed to obfuscated applications in the source [35]. Excluding them from both branches permits comparison on the same samples, but does not measure how a dynamic-only fallback would perform on those applications. Identifier renaming can preserve call relations, while packing or structural rewriting can affect recovery. The study does not establish resistance across these transformations.

Reproducibility and interpretation. The block aggregation and application-level composition remain under-specified (Sections 4.5 and 4.6). The recorded protocol also does not fully document the fitting boundaries for feature preparation and model selection. A common cohort does not by itself establish identical fold assignments or independence of every fitted stage from test data. These are limits of the available specification, not evidence that leakage occurred. Interpretation of the scores remains subject to the metric definitions discussed in Section 4.9 and the uncertainty limits in Section 4.11.

Adversarial evaluation. A relevant robustness test would apply functionality-preserving APK transformations to the complete pipeline. Published attacks provide operators for modifying function-call graphs [112, 154]. Inserted unreachable calls need not add network communication, but functionally equivalent applications may still differ in timing-dependent flow features. Hybroid’s resilience therefore requires measurement of both branches and their joint classifier under a specified attacker model.

4.13 Chapter Summary

Hybroid combines a learned representation of static program structure with 13 features of recorded network flows. The application-level representation supports separate binary and five-class prediction tasks. On CICAndMal2017, gradient boosting with the joint input achieves a reported detection accuracy of 97.0 per cent and category classification precision, recall and F_1 of 0.94 [35].

The within-system comparisons answer RQ2 for the evaluated cohort. Fusion improves reported detection F_1 by one or two percentage points over the better single modality. Its category classification contrast ranges from minus two to plus three points across learners. Flow features lead code features in detection, while code features lead in categorisation for each reported learner. These are descriptive point differences under the published protocol [35].

Chapter 5 examines contextual representation within the application package alone. Its hypergraphs encode relations among groups of methods, with no runtime traffic capture required by the analysis pipeline.

Higher-Order Android Malware Analysis with HGANN-Mal

5.1 Research Problem and Relation to Hybrid

This chapter answers RQ3, stated in Section 1.3: To what extent do attention-weighted hypergraph representations of static Android program structure improve binary malware detection and malware classification over pairwise graph models and non-attentive hypergraph models? It takes up gap G3. What that gap records is an asymmetry in the evidence. The cost of projecting a group relation onto pairwise edges has been measured [70, 71], and the conditions under which the projection loses nothing are known [68, 69], yet the benefit of avoiding it in Android malware analysis rests on a single published study [155]. The weighting usually applied alongside the higher-order encoding carries its own unsettled evidence [77].

HGANN-Mal represents an application as a hypergraph whose vertices are the methods of its function-call graph. Three families of hyperedge are laid over those vertices. The first records proximity in the call graph, the second groups methods whose learned representations are close, and the third follows a backward program slice from a call site that reaches a security-sensitive API. The three are merged into one incidence structure, a hypergraph operator with learned attention over hyperedges reduces the application to a single vector, and a classifier with two heads decides malicious against benign and, separately, the class within a labelled taxonomy. The source publication numbers the pipeline in eleven stages [36], and the chapter keeps those numbers. Stages 0 to 2 catalogue and unpack the packages and recover the call graph and the slices, and Stage 3 computes the node features. Stages 4 to 6 build and assemble the hypergraph, Stage 7 forms mini-batches, Stage 8 is the attention operator, and Stages 9 and 10 are the read-out and the two heads. The construction is described in Sections 5.4 to 5.8 and the evaluation in Sections 5.9 to 5.13.

Three claims are made for the work, and each is stated here in the form the evidence supports. The first is the construction: a method-level function-call graph is lifted to a hypergraph by three rules of different kinds, one topological, one semantic and one derived from a program slice, with a weight on the third that rises with the number of data-transmission calls the slice contains. The second is the operator: membership in a hyperedge is weighted by a learned score rather than by a binary indicator, so the aggregation is trained rather than fixed by the incidence structure. The third is the comparison, which measures the resulting system against two pairwise graph networks and two hypergraph operators without attention, on two corpora and three tasks, under one protocol.

Two limits belong here rather than in the limitations section, because they qualify what is being claimed rather than what was found. The comparison changes the hyperedge vocabulary

and the aggregation together, so a difference in the totals cannot be assigned to either change on its own; Section 5.13 states what experiment would separate them and Section 5.14 carries the consequence. And the field this work enters is narrow. A sweep of the published record finds one other hypergraph-based Android malware detector [155], so the comparison against hypergraph baselines is a comparison against operators taken from other domains rather than against competing Android systems.

Relation to Hybroid

Chapter 4 and this chapter answer one question about context in two ways, and they are not stages of a single development. Hybroid widens the evidence base: it observes the application twice, once as a program and once as a source of network traffic, and asks what the combination adds. HGANN-Mal keeps to one source of evidence and widens the relation instead: it asks what is gained when a group of methods can be recorded as a group rather than as a set of pairs. The first buys context by paying for a second observation, with the acquisition cost that Chapter 4 reports. The second buys context inside an observation already made. Chapter 7 takes up that trade directly, and it is the reason the two chapters sit next to each other.

The comparison between them is conceptual. Their corpora, class definitions and protocols differ, so their numerical results are not ranked against one another in this dissertation. Where the source publication of this chapter cites Hybroid as third-party prior work on hybrid analysis, it is naming the candidate's own earlier work, and this chapter states the relationship rather than repeating that framing.

The work was published at ISC 2025 [36], the candidate first of two authors with his supervisor as the second. The method, its implementation, the experiments and the manuscript are the candidate's own. Section 1.5 states the same in the form the affidavit covers.

5.2 Related Work

Section 2.2 defined the representations and Section 2.3 the operators. What this section adds is the systems: which of them have been built for Android, what they represent, and where the work of this chapter sits among them.

Graph representations of Android applications

Detection over program graphs predates graph neural networks. Adagio embedded call graphs with a neighbourhood hash kernel and classified the result [64], and MaMaDroid abstracted call sequences to package or family level before modelling transitions between them [2]. Their learned successors keep the graph and train the representation over it. The survey that organises this line covers control-flow graphs, call graphs, dependency graphs, heterogeneous application-and-API graphs, system-call graphs and network-flow graphs, and no hypergraph operator appears among them [8].

One of those successors is close enough to this chapter to be worth stating in detail. MsDroid gives up the whole program and works on local snippets around security-sensitive APIs, one subgraph per snippet, classifying the subgraphs and combining their verdicts; it locates malicious snippets under application-level labels alone, and it returns an explanation naming the control flows responsible [156]. The localisation is the same instinct that produces the slice-aware hyperedges of Section 5.7, reached through a different construct. MsDroid cuts subgraphs out of the call graph, while this chapter keeps one graph and lays a hyperedge over the methods a slice touches.

What every member of the line shares is the restriction Section 2.2 describes. An edge relates two methods. A behaviour realised by several methods acting together is representable only as a set of pairs, and the grouping is not recoverable from that set.

Hypergraph learning

Learning over hypergraphs supplies the missing construct. HGNN defined a spectral convolution over the incidence structure [27], its successor generalised the propagation with a random-walk transition matrix and is the operator this chapter calls HGNN+ and the source publication calls HGNNP [28], and attention was carried across by making membership itself real-valued and learned [80]. The line has since moved again: a stability-oriented successor from the same group supersedes the operator used as a baseline here [82], which is where a comparison run today would start.

Hypergraphs for Android malware, and how narrow that field is

A sweep of the published record finds exactly one prior application of hypergraph learning to Android malware, and it is worth describing precisely, both because it is the only true competitor and because the source publication of this chapter misattributes it once in its own text [155]. That work builds a function-call graph at method granularity, describes each vertex by six centrality measures together with a one-hot encoding of what the function does, and forms hyperedges by two rules: the K -order call neighbours of a function, and the set of functions sharing a permission. It evaluates HGNN, HGNN+, HyperGCN and UniGNN against GCN, GIN and GraphSAGE, on Malnet-Tiny and on Drebin, and it reports its best Drebin result as 97.1 per cent accuracy with HGNN+. Its hyperedge weights are fixed in advance and no learned weighting is evaluated anywhere in it, so every member of a hyperedge contributes equally to the aggregation.

HGANN-Mal differs from that work in three respects. It adds a third construction rule that neither topology nor permissions supply, taking a hyperedge from the backward slice of a security-sensitive call site, so that a group of methods jointly implicated in reaching a sensitive API becomes one object. It weights that family by the transmission and access calls the slice contains, which is the only place in the construction where a hyperedge carries a weight other than one. And it replaces fixed membership with a learned score, which is the change RQ3 asks about. Being the second entrant in a two-member field is itself worth stating plainly:

the hypergraph baselines in Section 5.10 are general operators applied to this chapter's own representation, because there is almost nothing else to compare against.

The comparison that the published record does not support

A reader will want the two Android hypergraph systems placed side by side, and the published figures do not allow it. Zhang et al. report 97.1 per cent accuracy for HGNN+ on the twenty-family Drebin task over 4,664 samples under an 80/20 split [155], while Section 5.11 reports 95.0 per cent for HGANN-Mal and 93.3 per cent for the same operator on the same task over the same samples under a 70/20/10 split. Setting those numbers against each other would compare implementations, feature sets, splits and training regimes at once.

The clearest evidence that such a comparison misleads is contained in the pair itself. One operator, HGNN+, on one corpus and one task, differs by 3.8 points of accuracy between the two papers, and that difference is larger than three of the four margins this chapter reports over its own baselines. The problem is not peculiar to these two publications. A recent study that re-implemented twelve Android detectors under one toolchain names unfair comparison as a field-wide defect, on the ground that published approaches are evaluated on datasets of different sizes, different ratios of benign to malicious samples and different training and test proportions [157], which is the same objection in general form [13, 122]. This chapter therefore reports what its own protocol measures and makes no claim against the published figures of the other work.

What the evaluation literature requires

Three further lines bear on the chapter and are taken up where they apply rather than here. Methodological work on machine learning in security names the pitfalls that this chapter's evaluation has to be checked against [13], and studies of detector behaviour under obfuscation, drift and adversarial perturbation give the conditions under which published accuracy figures fail to hold [61, 158, 157]. Attacks aimed specifically at graph-based Android detection show what an adversary can do to the representation itself [159, 112, 154]. Section 5.14 states what each of them implies for the results reported here.

5.3 Observation and Threat Model

Everything this chapter measures follows from one decision about what is observed. HGANN-Mal reads the application package and does not run it. The manifest and the Dalvik bytecode are the whole of the evidence, which fixes both the cost of the analysis and the limits on what it can see.

Static analysis removes the need for a device or emulator and for scripted interaction. Packages can be analysed independently across a corpus; Section 5.13 treats the computational cost of representation construction and learning. The second modality of Chapter 4 depends on traffic

that the CICAndMal2017 authors captured on physical handsets, and Section 7.2 compares the two observation requirements directly.

The limits are the standard limits of static analysis on Android, stated in Section 2.1.2, and three of them bear on the representation this chapter builds. Code loaded at runtime through the class loader is absent from the recovered bytecode, so no vertex exists for it. Reflective invocation resolves a target from a string at runtime, so the call edge is missing from a statically recovered call graph unless the analyser models the reflection explicitly, which the pipeline described here does not. Packing and native payloads move the interesting behaviour outside the DEX files that are parsed at all. Each of the three removes vertices or edges from the object the model consumes, and a representation that is built on relations between methods is affected by a missing method more than a representation built on a bag of permissions.

The adversary assumed by the evaluation is not adaptive. The corpora are historical collections, their samples were written without knowledge of this detector, and no sample is modified in response to what the detector learns. That assumption is standard for the reported task and it is also the assumption Chapter 6 argues should be stated rather than left implicit, which is why it is stated here. It has a consequence the chapter does not evade: an adversary who rewrites the application changes the call graph the extraction recovers, and everything downstream of that graph inherits the change. Section 5.14 works that exposure out against the published attacks on function-call-graph detectors.

5.4 APK Processing and Function-Call Graph Extraction

Stage 0 builds a catalogue. Every application package under a root directory is hashed with SHA-256, and the hash is joined against the label list distributed with the corpus, so each sample carries a benign label or a family label before any analysis runs. Working from the hash rather than from the file name matters for a corpus assembled from several sources, since it makes duplicate files visible; Section 5.9 returns to what was and was not done with that information.

In Stage 1 each package is decompressed with `apktool` and the Python `zipfile` module, which yields the manifest and every `classes*.dex` file the package contains. Applications built with multi-dex are therefore covered, and the bytecode of all of them enters the next stage. The reported runs used `apktool`, `MobSF`, `Androguard` with its DAD decompiler and a ready-made `code2vec` model.

`Androguard` [51], which the source runs beside `MobSF` in Stage 2a, disassembles the DEX files and produces the object the rest of the chapter is built on: a function-call graph $G = (V, E)$ at method granularity, in which a vertex is a method identified by its class, its name and its descriptor, and a directed edge records that one method may invoke another. The same pass emits a flat list of pairs that records, for each method, which framework APIs it calls. That list is what allows a flagged API to be located at its call sites rather than merely counted, and Section 5.5 uses it for exactly that.

Two properties of this object were fixed in Section 2.2.1 and are not restated here beyond their consequences. A statically recovered call graph over-approximates, so an edge in G records that an invocation is possible and not that it occurs. And the recovered graph depends on the analyser: a different toolkit, or a different treatment of the component lifecycle and of callbacks registered with the framework, yields a different vertex and edge set for the same application. Both properties travel through every construction in Section 5.7, because each of the three hyperedge rules is defined over G or over an artefact derived from it.

5.5 Security-Sensitive API Identification and Program Slicing

Stage 2 decides which parts of an application are worth isolating. In Stage 2a two tools run over each package. MobSF parses the manifest, disassembles the bytecode and reports the permissions an application requests together with the framework APIs it uses. Of the permissions reported, only those whose status is *dangerous* in the platform’s own classification are retained, which narrows attention to the capabilities that guard user data and device functions [48].

The APIs reached from those capabilities are then sorted into two classes by a hand-written rule. An *access* API reads something the user would consider private, and the examples the source publication names are contacts, location, short messages and storage. A *transmission* API moves data off the device, with HTTP clients, raw sockets and `sendTextMessage` as its examples. The pairing is deliberate. Data exfiltration is a sequence in which something is read and then sent, and a rule that marks both ends of that sequence gives the construction in Section 5.7 something to join into one group. The source publication describes the classification as a heuristic.

Stage 2b then computes a backward slice for every call site of a marked API. The variables that reach the call site are identified first, and the control-flow graph of the enclosing code is walked backwards, keeping only the blocks and statements that influence those variables. What the stage stores is a compact control-flow graph per slice. The purpose is localisation rather than verification: the slice does not establish that data flows from a source to a sink, in the sense that a taint analysis would [53], and it is not offered as evidence of exfiltration. It marks a neighbourhood of code that participates in reaching a sensitive call, and that neighbourhood becomes one hyperedge.

5.6 Node Features

In Stage 3 each vertex of the call graph receives a feature vector assembled from four groups, listed in Table 5.1 and drawn in Figure 5.1. The groups are concatenated and the result is standardised to zero mean and unit variance across the corpus, giving the feature matrix X with one row per method. The figure labels the blocks h_{op} , s_{struct} , a_{api} and c_{2v} , and the last of

Table 5.1. The four groups of node feature. The width of the first group is the size of the opcode vocabulary.

Group	Width	What it records
Opcode histogram	vocabulary	the distribution of Dalvik instructions in the method body
Structural metrics	4	in-degree, out-degree, PageRank and betweenness centrality of the method in G
API indicators	4	whether the method reaches an access API and whether it reaches a transmission API, with the counts of each within the slices attached to it
Semantic embedding	320	a code2vec representation of the decompiled method body

them is the embedding written c_v in Section 5.7. With O the opcode vocabulary, the width is $F = |O| + 328$.

The first group describes what a method does at instruction level, and it is the feature the program-code branch of Chapter 4 also rests on, there as a learned embedding of the opcode sequence rather than as a histogram. Operand values are not used, which removes the constants and register numbers that vary between builds of the same code. The histogram can be taken as raw counts or weighted by inverse document frequency across the corpus.

The second group places the method in the call graph rather than describing its body, and it is the only group that changes when the graph changes. The third connects the vertex to the analysis of Section 5.5, so that a method reaching a sensitive API is distinguishable before any hyperedge is built. Both are small, and together they contribute eight of the dimensions.

The fourth group is the largest. code2vec represents a method as a bag of paths through its abstract syntax tree and learns a fixed-width vector from that bag [67]. The published method consumes Java source, and the released model was trained on a corpus of Java source, whereas the input here is Dalvik bytecode recovered from a distributed package. The candidate confirms that the two were bridged by decompilation: each method was decompiled to Java with DAD, the decompiler built into Androguard, and embedded with a ready-made, pre-trained code2vec model. The source publication records neither step, and the width it reports differs from the released model's.

The embedding of a method therefore depends on how completely the decompiler recovers its source, as well as on the model. That dependence matters for the construction too. The semantic hyperedges of Section 5.7 are built from cosine similarities between these embeddings, so the decompiler and the model shape one of the three construction rules as well.

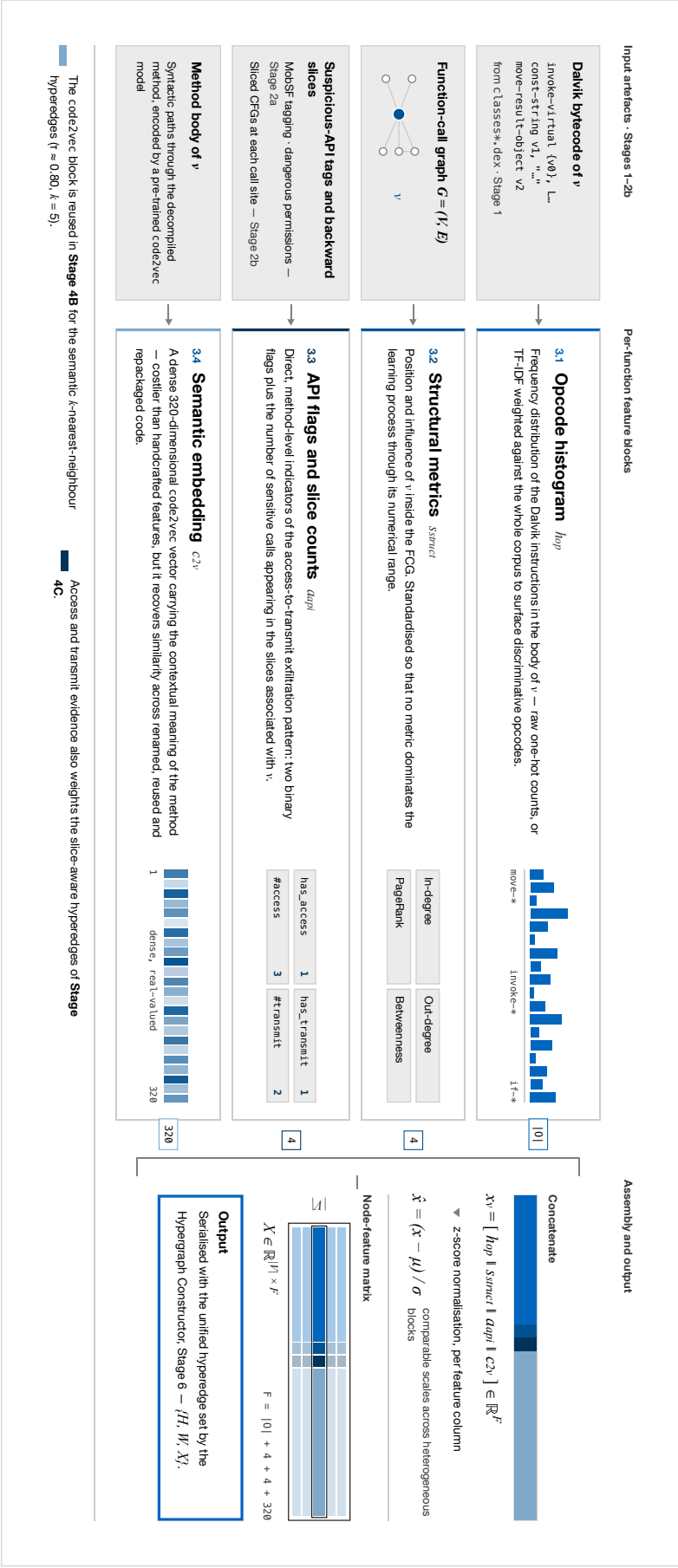


Figure 5.1.

Construction of the node feature vector in Stage 3. For each method v of the function-call graph, an opcode histogram, four structural centrality metrics, four suspicious-API flags and counts, and a 320-dimensional code2vec embedding are computed from the artefacts of Stages 1 to 2b. The four blocks are concatenated into x_v and z-score normalised, which yields the node-feature matrix $X \in \mathbb{R}^{|V| \times F}$ consumed by the hypergraph constructor in Stage 6.

5.7 Hyperedge Construction

The lifting from a graph to a hypergraph is where the representation is decided. Stages 4A to 4C apply three rules of different kinds to the same vertex set, and Stages 5 and 6 merge and assemble what they produce. Write $\mathcal{H} = (\mathcal{V}, \mathcal{E}, w)$ for the result, with $\mathcal{V} = V$ the methods of the call graph. Figure 5.2 draws one hyperedge of each kind together with the parameters of its rule. Its panels keep the names H_K , H_{SEM} and H_{API} that the source gives the three lists, which Equation 5.5 writes as the candidate sets $\mathcal{C}_{\text{topo}}$, $\mathcal{C}_{\text{sem}}$ and $\mathcal{C}_{\text{sus}}$.

Topological hyperedges. The source collects, for each method v , its callers and callees up to order K . Let $\vec{d}_G(v, u)$ be the number of call edges on a shortest directed path from v to u , with $\vec{d}_G(v, u) = \infty$ when there is none, and let $\text{dist}_G(v, u) = \min\{\vec{d}_G(v, u), \vec{d}_G(u, v)\}$. Then

$$e_{\text{topo}}(v) = \{u \in V : \text{dist}_G(v, u) \leq K\}, \quad K = 3, \quad (5.1)$$

holds v together with its callers and callees up to K steps away. This convention follows the wording of the source. A distance measured with call directions ignored would also admit the other callees of a caller and would give larger hyperedges, and the source does not record which convention the implementation used. This is the rule the earlier hypergraph work on Android uses, there in combination with a permission-based rule [155]. It records a neighbourhood as one object rather than as a set of pairs, which is the whole of the higher-order claim at this point: a vertex and its K -hop neighbourhood enter the operator together.

Semantic hyperedges. A nearest-neighbour search over the code2vec embeddings, computed with the FAISS library, groups methods whose representations are close. Write $c_v \in \mathbb{R}^{320}$ for the embedding of method v , the fourth group of Table 5.1; the full feature vector x_v also carries the other three groups, and the search uses the embedding alone. The candidate confirms that the search compares cosine similarities, $\text{sim}(a, b) = a^\top b / (\|a\| \|b\|)$, and that it runs within one application. With $\mathcal{N}_k(v)$ the k methods of the same application, other than v , that score highest under sim ,

$$e_{\text{sem}}(v) = \{v\} \cup \{u \in \mathcal{N}_k(v) : \text{sim}(c_v, c_u) \geq \tau\}, \quad k = 5, \quad \tau \approx 0.80. \quad (5.2)$$

The original algorithm, whose test panel 4B of Figure 5.2 reproduces, names the search output D , a letter that suggests a distance, and keeps a neighbour when $D \geq \tau$. The test agrees with the cosine similarity, since for a distance the inequality would run the other way. The query method is included and is not counted among its k neighbours, so $|e_{\text{sem}}(v)| \leq k + 1$; the source does not say whether its search counted the query itself. Nothing in this rule refers to the call graph, which is the point of including it: two methods that never call one another, and that sit in different parts of the application, are joined when they do the same kind of work. The source names polymorphic malware and code reuse as the intended cases. Because the search stays inside one application, a semantic hyperedge joins methods of that application only, and code shared across a family can make the hyperedges of different applications alike without joining them.

Inputs
from Stages 2 and 3

- $G = (V, E)$ – method-level function call graph, from Androguard
- E – 320-dimensional code2vec embedding per function
- S – backward slices around suspicious API call sites

Hyper-parameters

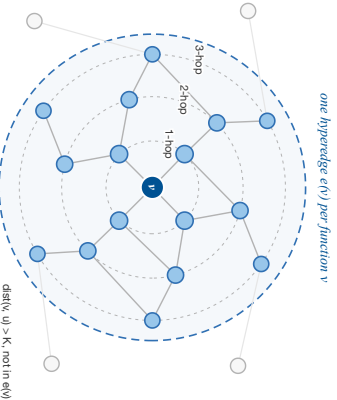
- K call-graph hop radius, 3
- k semantic neighbours, 5
- τ similarity floor, 0.80
- L_p maximum path in-fill length
- c_{max} hyperedge cardinality cap, 64

Every construction emits the same triple – (e, w, l) – a node set, a scalar weight and a one-byte type tag, so the three views stay separable downstream.

4 Hyperedge generation Structure, semantics and security-critical data flow over one vertex set

4A K-hop topological

hyperedge set H_K , type tag **topo**



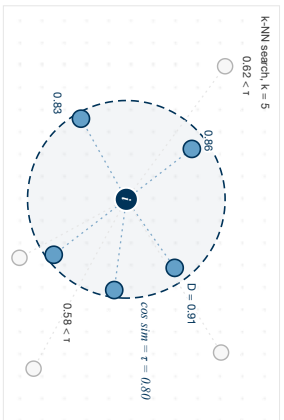
$$e(v) = \{u \in V : \text{dist}_G(v, u) \leq K, 2 \leq |e| \leq c_{max}\}$$

One hyperedge per function: its **K-order callers and callees** collapsed into a single higher-order relation. Uniform weight $w = 1$, because topology asserts adjacency, not importance.

- anchor v
- member, $\text{dist} \leq K$
- excluded

4B Semantic k-nearest neighbour

hyperedge set H_{kNN} , type tag **sem**



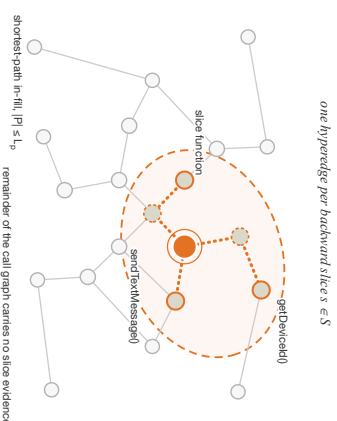
$$e(l) = \{l_{ij} : D_{ij} \geq \tau, k = 5, \tau \approx 0.80\}$$

A k-nearest neighbour query over the code2vec embeddings, gated by the similarity floor τ . This groups **functions that behave alike but never call one another**, the signal for code reuse, repackaging and polymorphic variants that a call graph cannot express.

- query function
- neighbour, $\text{sim} \geq \tau$
- candidate dropped

4C Slice-aware suspicious API

hyperedge set H_{API} , type tag **sus**



$$w = 1 + 0.2 \times (\#transmit) + 0.1 \times (\#access)$$

Every function a backward slice touches forms one hyperedge; shortest paths between slice members of length at most L_p are folded in so the exfiltration chain stays connected. The weight **prices transmission above access**: data leaving the device outranks data merely being read.

- suspicious call site
- function in slice
- path in-fill

Figure 5.2.

Hyperedge generation in Stage 4: three complementary constructions over one vertex set of methods. (4A) Topological hyperedges take the K -hop ball around every method in the call graph, with $K = 3$ and uniform weight $w = 1$. (4B) Semantic hyperedges are k -nearest-neighbour groups in the 320-dimensional code2vec space, with $k = 5$, admitted only above the similarity floor $\tau \approx 0.80$. (4C) Slice-aware hyperedges are built from the backward slices of Stage 2b, extended with shortest paths of length at most L_p between slice members, and weighted $w = 1 + 0.2 \times \#transmit + 0.1 \times \#access$. A candidate with fewer than 2 or more than 64 members is rejected. Node colour marks hyperedge membership only.

Slice-aware hyperedges. Each backward slice s from Section 5.5 yields one hyperedge over the methods the slice contains, extended by the methods on short call paths between them. Let F_s be that member set, fixed before any path is added. For a directed path P in the call graph, let $V(P)$ be its methods and $\ell(P)$ its number of call edges, and let P_{uv} be a shortest path from u to v . Then

$$e_{\text{sus}}(s) = F_s \cup \bigcup_{\substack{u,v \in F_s, u \neq v \\ P_{uv} \text{ exists, } \ell(P_{uv}) \leq L_p}} V(P_{uv}). \quad (5.3)$$

Only pairs of original members are joined, so a method that one path adds does not start further paths. The original algorithm loops over the pairs of the set it is enlarging, and freezing F_s is the convention adopted here to make the result independent of the loop order. A pair with no directed path contributes nothing. Where several shortest paths exist, the one chosen decides which methods join. This family carries the only non-unit weight in the hypergraph:

$$w(e_{\text{sus}}(s)) = 1 + 0.2 \# \text{transmit}(s) + 0.1 \# \text{access}(s), \quad (5.4)$$

so a slice that both reads private data and sends it weighs more than one that only reads, and the transmission end weighs twice the access end. The weight is a construction heuristic, so it cannot be read as a probability of exfiltration.

Merging. Each rule proposes candidate triples (e, w, t) of a member set, a weight and a type tag. Collected as a multiset and passed through the post-processing of the source, they give

$$\mathcal{C} = \mathcal{C}_{\text{topo}} \uplus \mathcal{C}_{\text{sem}} \uplus \mathcal{C}_{\text{sus}}, \quad \mathcal{E} = \text{Post}_\eta(\{(e, w, t) \in \mathcal{C} : 2 \leq |e| \leq c_{\text{max}}\}), \quad c_{\text{max}} = 64. \quad (5.5)$$

The lower bound removes singletons, which carry no relation, and with them every slice that has a single member, since such a slice has no pair to join. The upper bound is described in the source publication as necessary for stable message passing. Its prose says that hyperedges are capped, whereas its algorithm rejects a candidate outside the interval. This chapter follows the algorithm: truncation would need a rule for choosing which members to keep, and none is given. Rejection has a consequence of its own. A method whose K -step neighbourhood holds more than 64 methods loses its topological hyperedge entirely, so the topological family omits exactly the methods with the largest neighbourhoods.

The operator Post_η removes duplicates. Two candidates with the same members can carry different weights or tags, as a slice hyperedge and a topological one over the same methods would, and the policy η that decides which weight and which tag survive is not recorded; this chapter does not supply one. Filtering can also leave a method in no hyperedge, for instance an isolated method that belongs to no slice and has no semantic neighbour above τ . Its degree is then zero and the normalisation of Section 5.8 is undefined for it, and the source does not say whether such methods were dropped or handled in another way.

Each surviving hyperedge is tagged with its family in one byte, and the incidence matrix H and the diagonal weight matrix W are assembled from the result and serialised per application. Algorithm 5.1 restores the construction procedure of the source publication with the conventions above written into it. It also adds the post-processing and the sparse assembly, which the output line of the original announces and its body omits. The type tag records

Algorithm 5.1 Construction and assembly of the HGANN-Mal hypergraph. Stages 4A to 4C restore the algorithm of the source publication [36] with the conventions of Equations 5.1 to 5.3 written in; stages 5 and 6 add the post-processing and assembly that its prose describes.

Input: call graph $G = (V, E)$; embeddings c_v ; slices $s \in \mathcal{S}$ with member sets F_s ; parameters $K, k, \tau, L_p, c_{\max}$; duplicate policy η

Output: sparse incidence matrix H , weight matrix W , type tags t

Stage 4A: topological hyperedges

```

1:  $\mathcal{C} \leftarrow \emptyset$  ▷ multiset of candidates
2: for  $v \in V$  do
3:    $e \leftarrow \{u \in V : \text{dist}_G(v, u) \leq K\}$ 
4:   if  $2 \leq |e| \leq c_{\max}$  then
5:      $\mathcal{C} \leftarrow \mathcal{C} \uplus \{(e, 1, \text{topo})\}$ 

Stage 4B: semantic ( $k$ -NN) hyperedges
6:  $(I, S) \leftarrow \text{FAISS}(\{c_v\}_{v \in V}, k)$  ▷  $k$  others per method;  $S$  holds cosine similarities
7: for  $v \in V$  do
8:    $e \leftarrow \{v\} \cup \{I_{v,j} : S_{v,j} \geq \tau\}$ 
9:   if  $2 \leq |e| \leq c_{\max}$  then
10:     $\mathcal{C} \leftarrow \mathcal{C} \uplus \{(e, 1, \text{sem})\}$ 

Stage 4C: slice-aware hyperedges
11: for  $s \in \mathcal{S}$  do
12:    $e \leftarrow F_s$ ;  $w \leftarrow 1 + 0.2 \# \text{transmit}(s) + 0.1 \# \text{access}(s)$ 
13:   for  $(u, v) \in F_s \times F_s$  with  $u \neq v$  do ▷  $F_s$  is not enlarged
14:     if  $P_{uv}$  exists and  $\ell(P_{uv}) \leq L_p$  then
15:        $e \leftarrow e \cup V(P_{uv})$ 
16:   if  $2 \leq |e| \leq c_{\max}$  then
17:      $\mathcal{C} \leftarrow \mathcal{C} \uplus \{(e, w, \text{sus})\}$ 

Stage 5: post-processing
18:  $\mathcal{E} \leftarrow \text{Post}_\eta(\mathcal{C})$  ▷ policy  $\eta$  not recorded

Stage 6: sparse assembly
19: for  $j = 1, \dots, |\mathcal{E}|$ , with  $(e_j, w_j, t_j)$  the  $j$ -th element of  $\mathcal{E}$  do
20:    $H_{vj} \leftarrow 1$  for every  $v \in e_j$ ;  $W_{jj} \leftarrow w_j$ ;  $t(j) \leftarrow t_j$ 
21: return  $H, W, t$ 

```

something the operator never reads. The layer described in Section 5.8 consumes H , W and X , so a hyperedge's family influences the model only through its weight, and since two of the three families carry weight one, the topological and semantic hyperedges are indistinguishable to the operator once they are built. That is a design gap rather than an error, and Section 5.13 names it as the cleanest available extension.

5.8 Hypergraph Attention Model

The operator is defined over the structure fixed in Section 2.2.2, whose notation is used here without restatement: the incidence matrix H , the weighted vertex degree $d(v)$ and the hyperedge degree $\delta(e)$ with their diagonal matrices D_v and D_e , and the diagonal weight matrix W . Write $n = |\mathcal{V}|$, and assume for now that $d(v) > 0$ for every method; Section 5.7 records that filtering can violate this. The hypergraph Laplacian is

$$L = I - P_0, \quad P_0 = D_v^{-1/2} H W D_e^{-1} H^\top D_v^{-1/2}, \quad (5.6)$$

and for a signal $f \in \mathbb{R}^n$ over the methods it has the energy

$$f^\top L f = \frac{1}{2} \sum_{e \in \mathcal{E}} \frac{w(e)}{\delta(e)} \sum_{u,v \in e} \left(\frac{f_u}{\sqrt{d(u)}} - \frac{f_v}{\sqrt{d(v)}} \right)^2, \quad (5.7)$$

where the inner sum runs over ordered pairs [26]. The energy is small when the degree-normalised signal varies little inside the constructed groups, and each hyperedge counts in proportion to its weight and in inverse proportion to its size. It describes the operator and is not a term of the training loss. One layer of hypergraph convolution propagates through P_0 :

$$X^{(l+1)} = \sigma \left(D_v^{-1/2} H W D_e^{-1} H^\top D_v^{-1/2} X^{(l)} \Theta^{(l)} \right), \quad (5.8)$$

with $\Theta^{(l)}$ the learned projection of layer l and σ a non-linearity. Read from right to left, the operator gathers vertex states into the hyperedges that contain them, averages within each hyperedge through D_e^{-1} , scales by the hyperedge weight, and returns the result to the member vertices with the symmetric normalisation of $D_v^{-1/2}$. One layer therefore couples every pair of methods that share a hyperedge. The matrix P_0 is positive semidefinite and similar to the row-stochastic matrix $D_v^{-1} H W D_e^{-1} H^\top$, so its eigenvalues lie in $[0, 1]$. Equation 5.8 thus multiplies each eigencomponent of L with eigenvalue λ by $1 - \lambda$, and the components with the highest energy are damped most. This is the smoothing behind the low-pass bias recorded in Section 2.3.3.

Attention over hyperedges

Equation 5.8 weights a vertex's hyperedges by W , which is fixed when the hypergraph is built. Attention replaces the binary membership indicator by a learned real value, so that the strength with which a vertex participates in a hyperedge is trained [80]. The layer index is dropped from here on. Let $X \in \mathbb{R}^{n \times F}$ hold the input states as rows, let $\Theta \in \mathbb{R}^{F \times F'}$ be the projection, and set $Y = X\Theta$, with $y_v \in \mathbb{R}^{F'}$ the row of method v read as a column vector. A hyperedge is first represented by the mean of the projected states of its members,

$$z_e = \frac{1}{\delta(e)} \sum_{u \in e} y_u, \quad (5.9)$$

and the coefficient joining a vertex to one of its hyperedges is the normalised score

$$\alpha(v, e) = \frac{\exp(\sigma(a^\top [y_v \parallel z_e]))}{\sum_{e' \ni v} \exp(\sigma(a^\top [y_v \parallel z_{e'}]))}, \quad (5.10)$$

with $a \in \mathbb{R}^{2F'}$ a learned vector and $\parallel$ concatenation. Collecting the coefficients into H_{att} , whose entry at (v, e) is $\alpha(v, e)$ where H has a one and zero elsewhere, gives the layer actually used:

$$X' = \sigma \left(D_v^{-1/2} H_{\text{att}} W D_e^{-1} H_{\text{att}}^\top D_v^{-1/2} X \Theta \right). \quad (5.11)$$

The degree matrices are those of the binary H , as in the source publication; degrees recomputed from H_{att} would define a different layer. Gradients reach H_{att} as well as X and Θ , so the membership weights are learned. The support is not. $H_{ve} = 0$ implies $\alpha(v, e) = 0$, and a finite score gives every present membership a positive coefficient, so training reweights the

memberships the construction produced and cannot add a method to a hyperedge or remove one.

Equations 5.9 and 5.10 state the additive form of the score, which the source publication leaves as an uninstantiated similarity function and which the candidate confirms is what was implemented. The published formulation also writes the score between two vertices while indexing it by a hyperedge; the form above resolves that by scoring a vertex against a representation of the hyperedge, which is what the implementation computes.

The messages between methods

Expanding Equation 5.11 shows what one method receives from another. The layer runs in two stages. Each hyperedge first forms a message, and each method then combines the messages of its hyperedges:

$$\mu_e = \frac{1}{\delta(e)} \sum_{u \in e} \frac{\alpha(u, e)}{\sqrt{d(u)}} y_u, \quad x'_v = \sigma \left(\frac{1}{\sqrt{d(v)}} \sum_{e \ni v} w(e) \alpha(v, e) \mu_e \right). \quad (5.12)$$

Substituting the first expression into the second gives one matrix of couplings between methods,

$$X' = \sigma(P_{\text{att}} X \Theta), \quad P_{\text{att}}(v, u) = \frac{1}{\sqrt{d(v)} d(u)} \sum_{e \ni v, u} \frac{w(e)}{\delta(e)} \alpha(v, e) \alpha(u, e). \quad (5.13)$$

Two methods interact only through the hyperedges they share, and each shared hyperedge adds one term. In that term the coefficient of the sender and the coefficient of the receiver appear as a product. The construction weight $w(e)$ is a separate factor, so the slice heuristic of Equation 5.4 and the learned coefficients act on a message independently. The coefficients depend on the layer input through Equations 5.9 and 5.10, which makes P_{att} a function of X ; for a fixed input it is a symmetric matrix.

The key z_e and the message μ_e are different aggregations over the same members. The key is an unweighted mean, whereas the message weights each member u by $\alpha(u, e)/\sqrt{d(u)}$, so attention changes the average inside a hyperedge as well as the weighting across the hyperedges of a method. It does not change the sign of any term. Every member enters μ_e with a positive coefficient, so before the non-linearity each output is the projection of a non-negative combination of input states, and one layer cannot form a difference between two members of a hyperedge. The high-frequency component that standard hypergraph convolution discards has been characterised [81]. Attention changes the proportions of that average and keeps every coefficient positive, so the propagation of one layer remains an averaging of member states.

Grouping the sum in Equation 5.13 by hyperedge gives a structural statement:

$$P_{\text{att}} = \sum_{e \in \mathcal{E}} \frac{w(e)}{\delta(e)} \psi_e \psi_e^\top, \quad (\psi_e)_v = \frac{\alpha(v, e)}{\sqrt{d(v)}}, \quad (5.14)$$

with $\alpha(v, e) = 0$ for $v \notin e$. Each hyperedge adds a positive semidefinite matrix of rank one, and because every weight is positive, P_{att} is positive semidefinite for every input. The statement concerns the propagation matrix at a fixed input. It carries no guarantee about the convergence of training.

Because $\alpha(v, e)$ depends on both the method and its hyperedge, two different hyperedge groupings can produce the same fixed matrix P_0 yet different attentive propagation matrices. The reported experiments do not determine whether the learned model used this capacity or whether it caused the measured accuracy differences.

What the mechanism can and cannot be claimed to do

Two further properties bound what the chapter claims for the operator.

The first concerns the ranking that the score induces. Splitting $a = (a_1, a_2)$ across the concatenation writes the argument of σ in Equation 5.10 as $q_v + b_e$, with a query term and a hyperedge key

$$q_v = a_1^\top y_v, \quad b_e = a_2^\top z_e. \quad (5.15)$$

If σ is strictly increasing, then for a fixed layer input and any two hyperedges e_1 and e_2 that contain v ,

$$\alpha(v, e_1) > \alpha(v, e_2) \iff b_{e_1} > b_{e_2}. \quad (5.16)$$

The proof is short. Because σ and $\exp$ are strictly increasing, the numerators of Equation 5.10 are ordered as the keys are, and the two coefficients share one positive denominator. Any two methods therefore rank the hyperedges they share in the same order, and the keys alone fix that order. Methods can still attend to different hyperedges, since each normalises over its own. A non-decreasing activation such as ReLU can merge two hyperedges into a tie but cannot reverse their order between methods. The argument is the one Brody et al. give for the additive score of pairwise graph attention [29], applied to the score of Equation 5.10. It holds for each attention head separately and says nothing about hypergraph attention with other scoring functions. Under either kind of activation, the restriction on wording that the chapter keeps is thus a result for this operator: within one layer, HGANN-Mal can learn which hyperedges matter, and it cannot reverse their order from one method to another.

The query term does not cancel, however. With σ a leaky ReLU of slope 0.2 and keys -1 and 1 , the coefficients are approximately 0.23 and 0.77 for $q_v = 0$, and 0.12 and 0.88 for $q_v = 2$. The order agrees and the values differ. Cancellation is exact when σ is affine on the arguments of one method, as a leaky ReLU is when they share a sign.

The second concerns normalisation. Uniform coefficients do not reproduce Equation 5.8, because the degrees in Equation 5.11 are those of the binary H . Let r_v be the number of hyperedges containing v and $R = \text{diag}(r_v)$. Equal scores give $H_{\text{att}} = R^{-1}H$ and hence

$$P_{\text{att}} = R^{-1}P_0R^{-1}, \quad (5.17)$$

which reduces to P_0/r^2 when every method lies in r hyperedges, and to $P_0/4$ in the example above. An ablation that replaced the learned coefficients by uniform ones would therefore

change the scale of propagation as well as its weighting. A comparison with the non-attentive operator needs a stated normalisation convention before a difference can be read as an effect of attention.

Read-out and classification

Applications differ in size, so the vertex states are collapsed into one vector per application. Two variants are named in the source publication, an unweighted mean over the vertices and a learned attention pooling.

Training and inference run on batches. The source assembles a batch of B applications by placing their incidence and weight matrices on a block diagonal and stacking their feature matrices,

$$H_{\mathcal{B}} = \text{diag}(H_1, \dots, H_B), \quad W_{\mathcal{B}} = \text{diag}(W_1, \dots, W_B), \quad X_{\mathcal{B}} = \begin{bmatrix} X_1 \\ \vdots \\ X_B \end{bmatrix}, \quad (5.18)$$

together with an index that maps each row of $X_{\mathcal{B}}$ to its application. No hyperedge spans two blocks, so degrees, keys, coefficients and messages are all computed inside one application. The output for an application therefore does not depend on the other applications in its batch, and the read-out uses the index to pool each block on its own.

The layer is also equivariant under relabelling. Let Π reorder the methods of an application and Q its hyperedges, and write $\Phi(X, H, W)$ for the output of Equation 5.11. Then

$$\Phi(\Pi X, \Pi H Q^{\top}, Q W Q^{\top}) = \Pi \Phi(X, H, W), \quad g(\Pi X, \Pi H Q^{\top}, Q W Q^{\top}) = g(X, H, W), \quad (5.19)$$

where g is the application vector. The first identity holds because the degree matrices, the keys and the coefficients all permute with H , and the second because both read-outs are symmetric in the vertex states. The prediction therefore does not depend on the order in which method rows or hyperedge columns are stored. This concerns the hypergraph once it is built. It gives no resistance to obfuscation, and order-dependent choices earlier in the pipeline, such as tie-breaking among nearest neighbours or among shortest paths, can change H itself.

That vector passes through a shared multi-layer perceptron with two hidden layers, rectified linear activations and dropout, and then into two heads. The first emits a single logit whose sigmoid is the probability that the application is malicious. The second emits one logit per class of the labelled taxonomy, normalised by a softmax. Training minimises the sum of the two objectives,

$$\mathcal{L} = \lambda_{\text{bin}} \text{BCE}(z_{\text{bin}}, y_{\text{bin}}) + \lambda_{\text{fam}} \text{CE}(z_{\text{fam}}, y_{\text{fam}}), \quad (5.20)$$

weighted by two constants.

Equation 5.20 applies both terms unconditionally. On the corpus of Section 5.12, benign applications form one of the labelled classes and the second term is defined for them. On the corpus of Section 5.11 the classes are twenty malware families. A benign application has no

family among them, and neither has a malicious application from one of the smaller families, so the family term has no target for either. One well-defined specification averages each term over the training samples that carry its label,

$$\mathcal{L} = \frac{\lambda_{\text{bin}}}{|\mathcal{D}_{\text{bin}}|} \sum_{i \in \mathcal{D}_{\text{bin}}} \text{BCE}(z_{\text{bin},i}, y_{\text{bin},i}) + \frac{\lambda_{\text{fam}}}{|\mathcal{D}_{\text{fam}}|} \sum_{i \in \mathcal{D}_{\text{fam}}} \text{CE}(z_{\text{fam},i}, y_{\text{fam},i}), \quad (5.21)$$

with $\mathcal{D}_{\text{bin}}$ and $\mathcal{D}_{\text{fam}}$ the non-empty sets of samples that have a binary label and a class label.

5.9 Datasets, Tasks, and Split Policy

Two corpora are used, and they answer to different needs. Drebin is the reference collection of the field, which makes results on it comparable with a long line of published work, and its family labels support a classification task that a category taxonomy cannot. CICMalDroid is newer, it carries benign applications alongside malicious ones, and its labels are categories of behaviour rather than families. Using both guards against a conclusion that holds only for one collection, which is the reason the source publication gives for the second corpus.

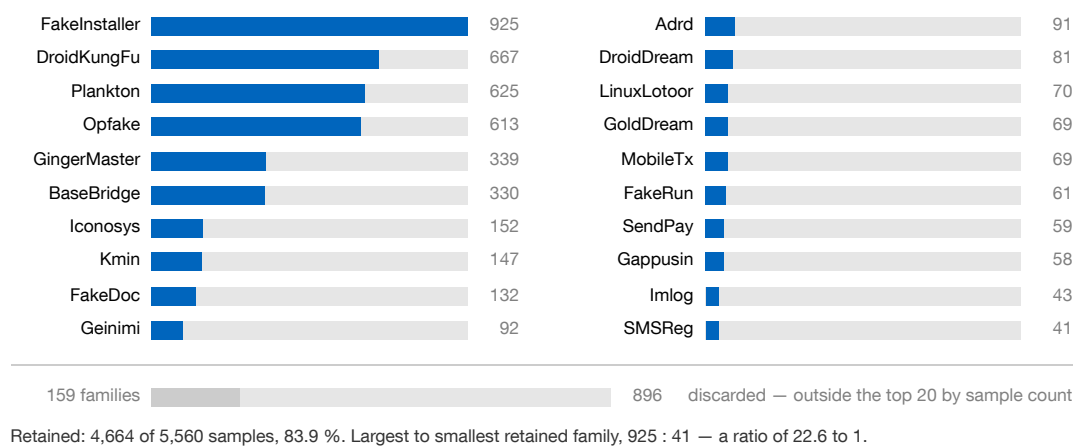
Drebin

Drebin is the malicious part of a larger collection gathered between August 2010 and October 2012 from the Google Play Store, from Chinese and Russian alternative markets, from Android websites and malware forums, and from the whole of the Malgenome project [1]. Every sample in that collection was submitted to VirusTotal and judged by ten antivirus scanners; an application was counted as malicious when at least two of the ten flagged it, applications labelled adware were then removed, and 5,560 malicious samples remained. The family names are taken from the labels of one of the ten scanners. The source publication of this chapter counts 179 families in the set [36], a figure the dataset paper does not state. Family sizes are heavily skewed, so the classification task uses the twenty largest, which cover 4,664 of the samples and are drawn with their sizes in Figure 5.3(a). The 159 smaller families hold the other 896 samples, 16.1 per cent of the set, and they are left out of the family task only, since binary detection uses all 5,560 malicious samples.

Three properties of that construction bear on the results. The malicious label is a vote of two scanners out of ten, which is a lower bar than a majority and admits samples the other eight consider harmless. The ground truth for malware-family classification is one vendor's naming, which is a convention rather than a taxonomy of behaviour. And the 123,453 benign packages the original DREBIN work evaluated against are not part of what is distributed, which is why a negative class had to be found elsewhere.

Binary detection needs a negative class, and Drebin supplies none, so the benign side was drawn at random from AndroZoo [161]. AndroZoo collects Android applications by crawling Google Play, the Chinese markets Anzhi and AppChina, and several smaller repositories; the crawl began in late 2011 and has continued since, and the collection held more than three million applications when it was first described. Every application in it has been submitted to

(a) Drebin — the twenty retained families



(b) CICMalDroid2020 — five categories

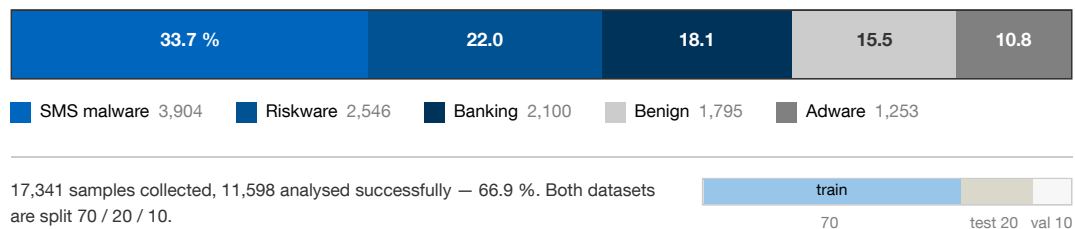


Figure 5.3. Class distribution of the two corpora. (a) The twenty largest Drebin families, the classes of the malware-family classification task. The 159 smaller families, with 896 samples, take no part in that task and remain in binary detection. (b) The five CICMalDroid 2020 categories of the published distribution [160]; Benign is the negative class of binary detection on this corpus. The long tail of Drebin is the reason the source publication reports an F1 score, which it labels Macro-F1, alongside accuracy.

VirusTotal, and what the collection stores per application is the number of antivirus products that flagged it, together with the compilation date of its bytecode, its size and the market it came from. About 9,470 benign applications were used against the full set of 5,560 malicious ones, giving a corpus of roughly 15,030 in which malware is 37.0 per cent.

Turning that stored count into a benign label requires a threshold, and the threshold is a decision rather than a property of the data. The AndroZoo authors report the flagged share at one antivirus product and again at ten, and state that antivirus results have to be treated with care; the two thresholds give very different populations. Section 5.14 states what follows from assembling the two classes from different collections.

CICMalDroid

CICMalDroid 2020 was collected between December 2017 and December 2018 and holds five categories, four malicious and one benign. Its benign class is not a separate acquisition: applications that fell into none of the four malicious categories were taken as benign and then checked against VirusTotal. Its published distribution, drawn in Figure 5.3(b), holds 11,598 applications, which is not the number collected: 17,341 applications were gathered, 13,077 executed successfully in the dataset authors' sandbox, and the labelled distribution retains 11,598 after the analysis records of the remainder failed to parse [160]. This chapter uses the published distribution, so it inherits a selection made by dynamic analysis, and Section 5.14 records what that implies for a static detector. The categories are far more even than the Drebin families, since the largest holds 3.1 times as many samples as the smallest.

Three tasks, and they are not interchangeable

Binary detection separates malicious from benign, and it is reported on both corpora. On Drebin the negative class is the AndroZoo sample described above; on CICMalDroid it is the 1,795 benign applications of Figure 5.3(b) against the remaining 9,803.

The malware-family classification task assigns a malicious application to one of the twenty Drebin families. No benign application takes part, and the task presupposes that the sample is malicious.

The malware-category classification task assigns an application to one of the five CICMalDroid categories, one of which is Benign. It therefore includes the detection decision within a wider one.

The distinction is kept throughout this chapter and throughout the dissertation, because the three tasks have different priors, different error costs and different trivial baselines, and reporting them under one name would make the numbers look comparable when they are not.

The two corpora do not even agree on what counts as malicious. Drebin removed every application its scanners labelled adware, while CICMalDroid keeps Adware as one of its four

Table 5.2. Binary detection, in per cent, as published. The source labels the first column Macro-F1 and states no averaging convention. Section 5.14 shows that the HGANN-Mal values on Drebin are positive-class values for the malicious class, and that several rows cannot follow one convention. Best value in each column in bold.

Method	Drebin and AndroZoo				CICMalDroid			
	F1	Prec.	Rec.	Acc.	F1	Prec.	Rec.	Acc.
GCN	92.2	92.2	91.5	92.8	90.6	90.6	90.6	90.9
GraphSAGE	92.1	92.1	92.5	93.3	91.0	91.1	91.1	91.5
HGNN	96.2	95.8	96.3	96.8	94.2	94.8	94.3	94.6
HGNN+	96.1	95.9	96.1	97.1	95.9	96.1	96.5	97.7
HGANN-Mal	97.8	97.8	97.6	98.3	96.2	96.2	97.6	97.6
Majority class	63.0				84.5			

malicious categories. A sample of the kind Drebin discarded is therefore a positive case on one corpus and absent from the other, which is one more reason the figures in Sections 5.11 and 5.12 are not read against each other.

Split policy

Each corpus is divided at random into 70 per cent for training, 20 per cent for testing and 10 per cent for validation, and the same division is used for every method in the comparison, so the baselines and HGANN-Mal see the same samples. At that test fraction the smallest Drebin family, with 41 samples, places about eight samples in the test partition in expectation, so a per-family score for the smallest classes would rest on a handful of decisions. Two properties of that policy are stated here and their consequences are worked out in Section 5.14: the division is random rather than ordered by time, so training and test samples come from the same period, and no deduplication or hash-aware grouping was applied, although Drebin is known to contain repackaged near-duplicates.

5.10 Binary Detection Results

Binary detection asks one question of an application: malicious or not. It is reported on both corpora, and the two experiments differ in a way that matters for reading the numbers. On CICMalDroid the benign applications belong to the corpus, and the task uses the 1,795 benign samples against the 9,803 malicious ones. On Drebin they do not, because Drebin holds malware only, so the benign side was assembled separately as Section 5.9 describes. Table 5.2 reports both. Figures 5.4 and 5.5 plot its F1 and accuracy columns together with those of the multiclass table in Section 5.11, for all four task and corpus combinations.

Three readings follow, and they are of unequal strength.

The clearest is the gap between the pairwise and the higher-order representations. On Drebin every hypergraph operator sits above every pairwise one, by between 3.5 and 5.5 points of

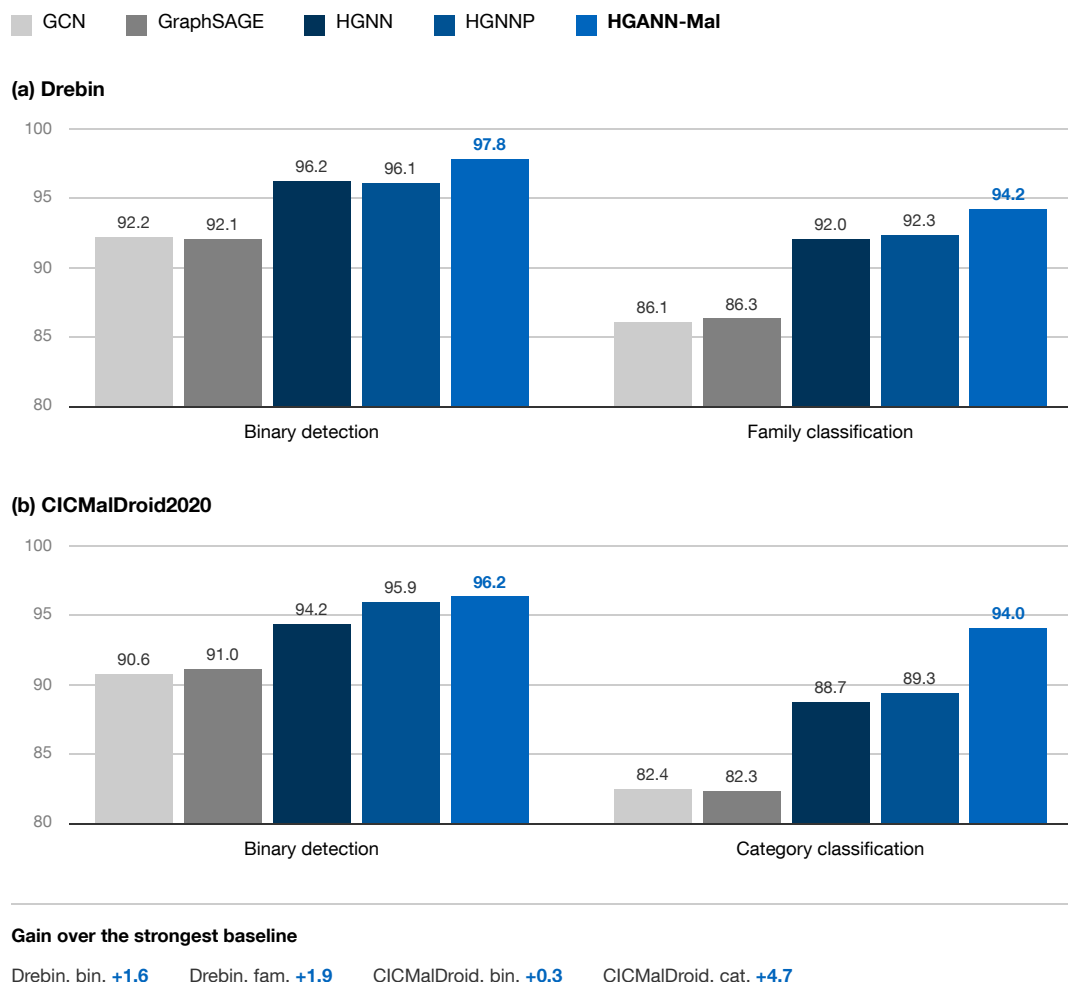


Figure 5.4. F1 on both corpora and both tasks, as published in Tables 5.2 and 5.3. The source labels these values Macro-F1, and Section 5.14 shows that several rows cannot follow one averaging convention. The vertical axis is truncated at 80 so that differences among the hypergraph methods remain visible. HGNNP is the source publication’s name for HGNN+.

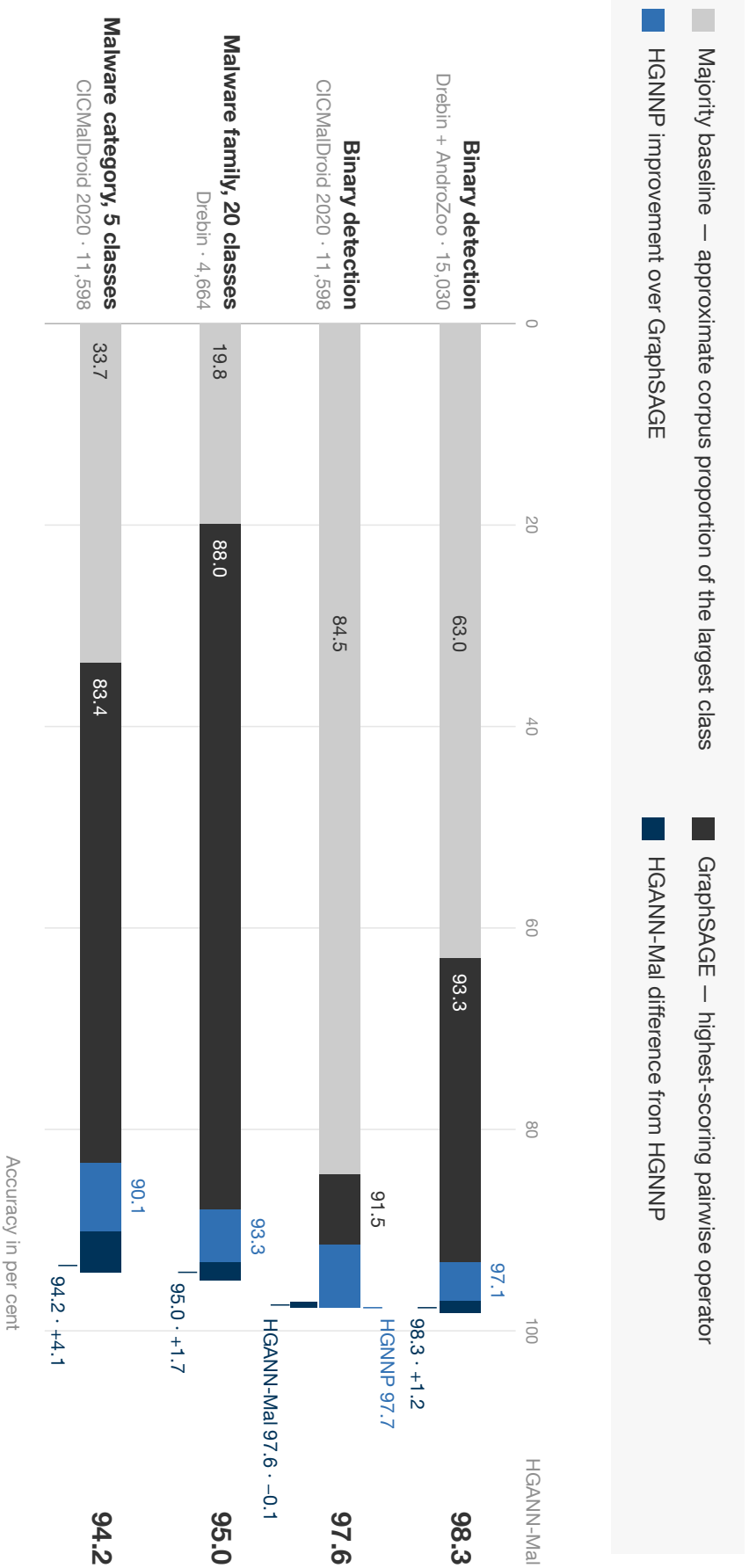


Figure 5.5. Accuracy reported in Tables 5.2 and 5.3, divided at the majority baseline, at GraphSAGE and at HGNNP, which the figure labels HGNNP after the source publication. Variability across runs and the isolated contribution of attention are not established by these reported results.

accuracy and between 3.9 and 5.7 points of F1, and the ordering holds on CICMalDroid as well. Figure 5.5 shows the same step on all four combinations: HGNN+ exceeds the best pairwise operator, GraphSAGE in every case, by 3.8 to 6.7 points of accuracy, and in each combination this step is larger than the difference between HGANN-Mal and HGNN+. All four baselines were re-implemented on this pipeline and consumed the same call graph and the same node features, so the difference belongs to the representation rather than to the input. That is the comparison RQ3 asks for first, and it is the one the evidence supports best.

The second is the gap between HGANN-Mal and the two hypergraph operators without attention. On Drebin it is 1.2 points of accuracy against the stronger of them, and on CICMalDroid it is not there at all: HGNN+ holds the best accuracy at 97.7 against 97.6, while HGANN-Mal holds the better recall at 97.6 against 96.5 and the better F1 at 96.2 against 95.9. A system that leads on three columns of four, and trails by one tenth of a point on the fourth, is level with its baseline on that corpus rather than ahead of it. The source publication reads the same pattern as a precision-recall trade-off, which is one explanation among several; single runs without dispersion cannot separate a trade-off from noise, and Section 5.14 returns to that.

The third concerns the base rate, and it is the reason the last row of the table is there. A constant predictor scores 63.0 per cent on the Drebin corpus, where benign applications are the majority, and 84.5 per cent on CICMalDroid, where they are 15.5 per cent of the samples. Reported accuracy stands 35.3 points above the trivial baseline in the first case and 13.1 points above it in the second. Neither figure is diminished by this, but the second is much less impressive than 97.6 per cent sounds, and the source publication reports no such baseline.

Drebin confusion matrix and class balance

The Drebin binary experiment evaluated HGANN-Mal on 1,112 malicious and 1,894 benign applications. The classifier identified 1,085 malware samples and missed 27. It raised 24 false alarms among the benign applications and classified 1,870 as benign. These 51 errors across 3,006 decisions give a false-positive rate of 1.3 per cent.

The experiment used a corpus with 37.0 per cent malware, whereas deployments may contain a lower share. A separate sensitivity calculation holds recall and the false-positive rate fixed while reducing the malware share to 10 per cent. Under this assumption, precision falls from 97.8 to about 89.5 per cent because benign applications outnumber malicious applications nine to one. This calculation illustrates the effect of class balance under fixed rates, as discussed in Section 2.6.

5.11 Malware-Family Classification on Drebin

The second task assigns a malicious application to the family it belongs to. It runs on the 4,664 Drebin samples of the twenty largest families, shown in Figure 5.3(a), and no benign application takes part. Results are in the left panel of Table 5.3.

Table 5.3. Multiclass classification, in per cent, as published. The Drebin task assigns a malicious application to one of twenty families; the CICMalDroid task assigns an application to one of five categories, of which Benign is one. The two tasks are not comparable with each other. Section 5.14 identifies two rows whose three columns cannot follow one averaging convention.

Method	Drebin, 20 families				CICMalDroid, 5 categories			
	F1	Prec.	Rec.	Acc.	F1	Prec.	Rec.	Acc.
GCN	86.1	85.9	86.2	87.8	82.4	82.0	82.4	82.7
GraphSAGE	86.3	86.7	86.2	88.0	82.3	82.2	82.6	83.4
HGNN	92.0	92.1	92.4	92.0	88.7	88.8	88.8	89.0
HGNN+	92.3	92.5	92.5	93.3	89.3	89.5	89.8	90.1
HGANN-Mal	94.2	94.1	93.8	95.0	94.0	94.1	94.0	94.2
Majority class	19.8				33.7			

The ordering repeats: pairwise operators lowest, non-attentive hypergraph operators four to five and a half points of accuracy above them, and HGANN-Mal 1.7 points above the better of those. Against the trivial baseline the margin is wide here, since the largest family covers 19.8 per cent of the samples.

One property of the task deserves more caution than the totals invite. The twenty families range from 925 samples down to 41, a ratio above twenty to one. Read as a macro average, which is how the source labels it, an F1 of 94.2 per cent would require the smallest families to be classified nearly as well as the largest, since every class contributes equally to that average. Section 5.14 shows that the printed precision and recall of the same row exclude any reading in which the three columns share one convention. The classes are also one vendor’s labels rather than an agreed taxonomy, as Section 5.9 records, so a disagreement between the model and the label is not always an error of the model. The families are further related to one another: several of the twenty are known to contain repackaged applications built from shared code. Shared code gives the applications of one family similar semantic hyperedges, each built inside its own application under Section 5.7, and it is also the condition under which duplication inflates a measured score. Section 5.14 returns to that threat.

5.12 Malware-Category Classification on CICMalDroid

The third task assigns an application to one of the five CICMalDroid categories. Benign is one of them, so this task subsumes the detection question rather than sitting beside it, and an application misclassified between two malicious categories is a different kind of error from one misclassified as benign. Results are in the right panel of Table 5.3.

The margin over the non-attentive operators is 4.7 points of F1 and 4.1 of accuracy, the widest of the four task and corpus combinations reported anywhere in this chapter. The narrowest, on the same corpus, is binary detection, where Section 5.10 found no margin at all. The ranking of the four combinations does not depend on the measure. Over the strongest baseline, the F1 margins in Figure 5.4 are 4.7 points here, 1.9 and 1.6 on the Drebin family and binary tasks

and 0.3 on CICMalDroid binary detection, which is also the order of the accuracy margins. That pattern is worth stating plainly because it constrains what may be concluded: whatever the model adds, it adds most where the decision is between several classes and least where it is between two. A representation that groups methods by shared behaviour would plausibly help most in exactly that setting, since categories such as adware and riskware are separated by what an application does with what it collects rather than by whether it collects anything. The observation is offered as a reading of the pattern and not as a finding, since one run per configuration cannot support more.

5.13 Ablation, Efficiency, and Interpretability Analysis

The ablation that was not run

HGANN-Mal differs from the hypergraph baselines in two ways at once. It builds a different set of hyperedges, fusing a topological, a semantic and a slice-aware family where the baselines take k -hop and permission-based ones [155], and it weights membership by a learned score instead of a fixed indicator. Both changes are present in every number in Tables 5.2 and 5.3, and no experiment in this chapter varies one while holding the other. The measured differences are therefore differences between two systems, and they do not establish that attention is what produces them.

Four experiments would settle it, and they are recorded here so that the gap is specified rather than merely admitted. First, the baseline operator without attention, run over the identical fused hyperedge set, which isolates the aggregation. Second, HGANN-Mal run over topological hyperedges alone, which isolates the vocabulary. Third, a leave-one-out over the three hyperedge families, which apportions the vocabulary’s contribution among them. Fourth, a sweep over K , k , τ , $c_{\max}$, L_p and the two weight constants, none of which is varied anywhere in the source publication. The first two are the ones that bear on RQ3; the second two would tell a reader how sensitive the construction is to choices made once and never revisited. Equation 5.17 adds a condition to the first: uniform coefficients also rescale propagation, so that comparison has to fix a normalisation convention before its difference can be attributed to attention.

The question is not idle. On pairwise graphs, over node-property benchmarks and against non-attentive baselines, no unmodified attentive model has been shown to outperform all of those baselines on any dataset or task tested [77], and the scoring functions of both the original and the revised formulations realise only finitely many neighbour rankings at bounded width [29, 77]. Neither result was stated for hyperedges, as Section 2.3.2 records. The static-ranking argument carries over to the score used here, as Equation 5.16 shows, while the finite-ranking bound has not been examined for hyperedge scores. What the two results establish is that the attribution cannot be assumed, and the evidence needed to make it was not gathered.

Efficiency

The stated procedure admits bounds. Let n be the number of methods, m the number of retained hyperedges, $|\mathcal{S}|$ the number of slices and $\nu = \text{nnz}(H) = \sum_{e \in \mathcal{E}} |e|$ the number of stored memberships. Algorithm 5.1 proposes at most one topological and one semantic candidate per method and one candidate per slice, and post-processing only removes candidates. Semantic hyperedges hold at most $k + 1$ methods under the convention of Equation 5.2, and the others at most $c_{\max}$. Hence

$$m \leq 2n + |\mathcal{S}|, \quad \nu \leq (c_{\max} + k + 1)n + c_{\max}|\mathcal{S}| = 70n + 64|\mathcal{S}|, \quad (5.22)$$

and the representation grows linearly in the numbers of methods and slices. For one attention head with input width F and output width F' , a sparse evaluation of Equation 5.11 computes the projection, the keys, the scores and the two incidence products in

$$T_{\text{layer}} = O(nFF' + \nu F'), \quad M_{\text{layer}} = O(nF + (n + m)F' + \nu + FF') \quad (5.23)$$

time and memory, where $m \leq \nu/2$ because every retained hyperedge has at least two members. The bound follows the two stages of Equation 5.12 and never forms P_{att} . That matrix can be far denser than H , since two methods are coupled whenever they share a hyperedge: it can hold up to $\sum_e |e|^2 \leq c_{\max}\nu$ non-zero entries, and the cap bounds this density. Further heads and layers add their own terms.

Construction costs more. With $|E|$ call edges, breadth-first searches from every method build the topological hyperedges in $O(n(n + |E|))$. An exhaustive search within one application evaluates the similarity of $O(n^2)$ pairs of 320-dimensional embeddings, and since the FAISS index is unspecified, no sub-quadratic bound can be claimed for it. The slice paths need one breadth-first search from each member of each slice, $O(\sum_s |F_s|(n + |E|))$ in total. The cap is applied after a candidate is built, so it does not limit the cost of candidates that are then rejected, and centrality computation and the semantic embedding add costs outside these bounds. None of these quantities is a measurement. Static analysis removes the need for runtime capture, while graph searches and semantic-neighbour construction retain the computational costs specified above.

Interpretability

The slice-aware hyperedges are the part of the construction that could carry an explanation. Each one is anchored at a call site of a security-sensitive API and covers the methods a backward slice passes through, so an attention weight on such a hyperedge points at a concrete region of code and at the API that made it interesting. Reporting the highest-weighted hyperedges for a sample, with the call sites they contain, would turn a score into something an analyst can check. Equation 5.16 makes that report compact: within one layer and head every method ranks its hyperedges by the same keys b_e , so one ranked list of keys per application summarises what the layer attends to.

That was not done. The source publication states interpretability as future work, and no attention weight is inspected, visualised or reported anywhere in it. A single retrained model

on one corpus would be enough: the weights of one application, ranked, with the API call at the anchor of each hyperedge, would turn this section from a description of what the design permits into a result about what the model attends to. It is named here because it is the cheapest substantial improvement available to the design, and because the construction was built in a way that makes it reachable.

One design gap, and it is the cleanest extension available

Section 5.7 records that every hyperedge is tagged with the family that produced it, and that the operator never reads the tag. A hyperedge reaches the layer as a column of H and an entry of W , so a topological hyperedge and a semantic one over the same methods are the same object to the model, and only the slice-aware family is distinguished at all, through a weight it happens to carry. The type is available, it is stored, and it is discarded.

Two ways of spending it suggest themselves. The weight matrix could carry a learned scalar per family, so that the training decides how much a topological grouping is worth against a semantic one, which costs three parameters. Or the attention score of Equation 5.10 could take the type as an input. Entered additively, the type would only shift the keys b_e , and Equation 5.16 shows that every method would still rank its hyperedges in one shared order; a weighting that depends on the family and the vertex together needs a score that is not additive, such as the revised form of Brody et al. [29]. Neither was tried here, and the second is the more interesting because it is the version of the mechanism that the confound in Section 5.14 leaves unexamined.

5.14 Limitations and Threats to Validity

The results above are measurements on two historical corpora under one protocol. This section states the limits of what they show and closes with a brief account of threats to validity.

The gain cannot be attributed to attention. HGANN-Mal changes the hyperedge vocabulary and the aggregation together, and no experiment separates the two (Section 5.13). The evidence supports a claim about the complete system: it performs better than four baselines on three of the four task and corpus combinations. Whether learned weighting of hyperedges produces that gain is an open question.

Every figure is a single run. No configuration was repeated, and the tables therefore carry no standard deviation, confidence interval or significance test. Small margins are affected most. A tenth of a point separates HGANN-Mal from HGNN+ on CICMalDroid accuracy, and that difference cannot be interpreted. Even the 1.2 points on Drebin would need repeated runs before they could be called reliable.

The evaluation covers favourable conditions only. Both splits are random, so training and test samples come from the same collection period, and the figures do not describe performance on applications released later [12]. Neither corpus was obfuscated, and no adversary was present. Under obfuscation, concept drift and adversarial examples, none of

twelve representative Android detectors kept the performance it reached in the ideal setting, and one call-graph-based detector scored 0.912 of F1 on one year's test set and 0.463 on another's [61]. Sections 5.10 to 5.12 thus report the conditions that a comparison between representations needs. A deployment would meet harder ones.

Some behaviour lies outside the representation. Code fetched at runtime and targets resolved by reflection are absent from the recovered call graph, and so are payloads held in packed archives or native libraries (Section 5.3). A representation built from relations among methods loses more from a missing method than one built from a bag of permissions.

The baselines are general operators. GCN, GraphSAGE, HGNN and HGNN+ were re-implemented on this pipeline's own representation. They test whether the higher-order encoding and attention help within one design, but they do not place HGANN-Mal against the Android detectors that the field uses as reference points. The hypergraph baseline has since been superseded by a stability-oriented successor from the same group [82], so the comparison reflects that line of work as it stood in 2022.

The threat model is non-adaptive. No adversary in this evaluation knows that the detector exists, although published attacks act on the object this pipeline consumes. The first structural attack on graph-based Android detection reports over 90 per cent success in feature space and up to 100 per cent in problem space against function-call-graph detectors under white-box access [159]. FCGHunter rewrites an application with semantics-preserving operators and reaches a white-box success rate of 87.9 per cent against forty detectors built from graph-summary features [112]. A4M reports an average of 85.17 per cent against seven detectors built with MaMaDroid, APIGraph and a graph neural network, under a perturbation budget of five per cent [154]. None of these attacks was run against HGANN-Mal. Every hyperedge of Section 5.7 is derived from the call graph or from quantities computed over it, so methods and edges injected into that graph also change the hyperedges. Attention operates downstream of an extraction that the adversary controls. HGANN-Mal therefore inherits the exposure of its representation, and in line with Chapter 6 this chapter makes no robustness claim.

Threats to validity. The result tables head their first column Macro-F1 without stating a convention. For the HGANN-Mal Drebin binary row a macro reading is impossible: over 1,112 malicious and 1,894 benign test applications, macro precision 97.8 and macro recall 97.6 at an accuracy that rounds to 98.3 would require at least 1,897 true negatives. That row is read as positive-class values. Under positive-class, macro and support-weighted averaging alike, $F_1 \leq (P + R)/2$, and four rows exceed that mean by more than the 0.1 point that rounding allows: GCN and HGNN in the Drebin binary panel, HGANN-Mal in the malware-family classification panel, and GCN in the malware-category classification panel. No single convention fits these rows, so the multiclass columns are not read as macro averages; accuracy is unaffected. On the data side, the benign Drebin applications come from AndroZoo, which began crawling in late 2011, while Drebin covers August 2010 to October 2012, so a classifier may separate the two collections by build tools or API levels that have nothing to do with malice [13, 161]. The source publication records that about 49 per cent of Drebin samples are repackaged applications, and none was removed. Near-duplicates across the partitions may inflate the family scores of Section 5.11 more than the binary ones [15, 14]. CICMalDroid, for

its part, keeps only applications that ran in its authors' sandbox [160], which may remove evasive samples that a static detector most needs to catch.

5.15 Chapter Summary

This chapter answered RQ3 by lifting a method-level function-call graph to a hypergraph, joining its vertices by three rules of different kinds, and learning how strongly each method participates in each group rather than fixing that strength when the structure was built. The system was measured against two pairwise graph operators and two hypergraph operators without attention, on Drebin and on CICMalDroid, for binary detection and for classification into families and into categories.

Two findings hold. The higher-order representation is better than the pairwise one on every task and corpus reported, by between three and eleven and a half points of accuracy depending on the task and on which pair is compared, with the input to both held identical. The attention-weighted system is ahead of the non-attentive hypergraph operators on three of four combinations and level on the fourth, and its widest margin appears on the multiclass task rather than on detection. The analysis of Section 5.8 sharpens what the second finding can mean without adding evidence for it. Attention reweights the memberships that the construction fixed, and within one layer it ranks the hyperedges of every method in one shared order. A constructed example shows that it can still separate structures that the fixed operator identifies, although nothing shows that training exploits this.

One thing does not hold, and the chapter states it rather than working around it. Because the hyperedge vocabulary and the weighting were changed together, the improvement over the hypergraph baselines cannot be attributed to attention, and the experiment that would separate them was not run. Chapter 7 takes the representational question up against Chapter 4, and Chapter 8 records the gap as the first item of future work.

Evaluating Adversarial Robustness: Benchmark Design and Metric Analysis

6.1 Research Problem and Contribution

This chapter addresses RQ4: How can an explicit adversary model be translated into a reproducible benchmark for adversarial attacks and defences, and what does such a benchmark reveal about the relation between clean predictive performance and adversarial robustness? The question concerns gap G4, which links adversary assumptions to the validity of the resulting measurements.

Adversarial machine learning studies how deliberate changes to training data or inference inputs can cause a learning system to fail, together with methods for resisting such changes [18]. A poisoning attack intervenes during training, whereas an evasion attack modifies an input presented to an already trained model. Defences may expose the model to adversarial examples during training, transform its inputs or internal representations, or detect and respond to suspected adversarial inputs. This chapter concerns evasion attacks and the evaluation of these three defence strategies. The broader attack taxonomy and terminology are defined in Section 2.5.

An empirical robustness estimate depends on the permitted perturbations and the attacks used to search for them [104, 21]. An unsuccessful search may reflect resistance to the tested attack, or a failure of the attack to exploit the model. For example, Athalye et al. identified obstructed gradients in seven of nine non-certified defences accepted at ICLR 2018 and circumvented six completely and one partially [19]. Evaluators therefore need to examine the measurement procedure alongside the defended classifier.

The SPARTA SAFAIR work produced two complementary evaluation mechanisms: an adversarial machine-learning competition and a reusable benchmark framework [162, 39]. The competition evaluated submitted defences on hidden test samples. The benchmark provided common interfaces for evaluating trained models against reference attack implementations, with task-specific components for different prediction problems. Its versioned source code is publicly available on GitHub [163].

Using the project records, this chapter reconstructs both evaluation mechanisms and assesses whether the reported measurements are consistent with the documented attack objectives, attacker access and perturbation constraints. The empirical evidence comprises the contest submissions and the application of the benchmark to partner defences on histopathology

images. The chapter then derives properties of the accuracy-loss score and the binary detector response policy. These analytical results establish what the published aggregates support and what additional observations would be required to distinguish competing explanations. A short comparison with the project's PDF-malware study examines how feature constraints alter the evaluation problem.

SPARTA was funded under Horizon 2020 grant agreement No. 830892. Its seventh work package, SAFAIR, concerned the security of artificial-intelligence systems. The candidate wrote the whole of D7.6, which reports the contest and benchmark, and Chapter 5 of D7.5 [39, 38]. Section 6.10 distinguishes this contribution from the partner methods and experiments discussed below.

6.2 Threat Analysis and Evaluation Scope

SPARTA's threat analysis links attack techniques to affected ML assets and the adversary's knowledge [37, Sections 3.3.2–3.3.6]. Applied to the evaluations in this chapter, these distinctions locate the attack within the ML life cycle. Poisoning concerns training data or the learning process; evasion concerns inputs supplied to a trained model.

The evaluations examined here concern evasion. The contest specified attacks that induce a chosen identity or errors in selected facial attributes [162, Section 3.3]. The histopathology benchmark reports classification accuracy on clean and perturbed images [39, Section 3.1.3]. The affected asset is therefore the inference input, and the measured outcome is prediction correctness under attack. Sections 6.3 and 6.4 specify attacker access and permitted perturbations, which determine the conditions under which these measurements can be interpreted.

6.3 Evaluation Requirements and Threat Models

The threat model specifies the attack objective, the information available to the attacker and the operations permitted on the input. Section 2.5 defines the white-box, black-box and grey-box terminology used throughout the dissertation. Under its grey-box convention, some model information is known and the training process cannot be influenced. Knowledge of training data and the ability to alter those data are separate assumptions.

The contest granted knowledge of the training corpus, withheld the defence technique and model architecture, and allowed gradient-based updates against the deployed model [162, 164]. Access to gradients is an oracle capability. The term grey box alone therefore leaves the experimental interface underspecified. The conditions stated in Section 6.4, including the query interface and input budget, define the contest adversary used here. This is an academic evaluation setting; a survey of 271 practitioners found that academic knowledge assumptions often grant more information than attackers have in deployment [107].

D7.3 adopted the evaluation checklist of Carlini et al. It called for attacks adapted to the complete defence, explicit attack hyperparameters and comparisons with transfer or gradient-

free attacks. It also requested perturbation-budget curves and ablations [162, 21]. These requirements provide a basis for assessing the delivered results. Participation by independent attack teams can broaden the search, but the scoring rule and the validity of each generated input require separate checks.

6.4 Tasks and Evaluation Protocol

6.4.1 Prediction Tasks and Permitted Perturbations

The contest defined two tasks, each with an attack track and a defence track: targeted face re-identification and attribute alteration. In targeted face re-identification, the attacker modifies a face image so that the classifier assigns the image to a specified target identity. In attribute alteration, the attacker modifies an image so that the classifier misclassifies a nominated subset of its forty binary facial attributes. The corresponding defence tracks evaluated model robustness against the same perturbations [162]. A participant could attack one task and defend the other, but could not enter both sides of the same task.

Let f_θ denote the model scores and h the resulting class decision. For an image x in a specified input coordinate system, define the feasible perturbations by

$$\mathcal{S}_\varepsilon(x) = \{x' \in \mathcal{X}_{\text{valid}} : \|x' - x\|_\infty \leq \varepsilon\}. \quad (6.1)$$

Here $\mathcal{X}_{\text{valid}}$ includes the permitted pixel range and quantisation. For pixel values scaled to $[0, 1]$, the contest budget was $\varepsilon = 8/255$, corresponding to a maximum change of eight intensity levels per channel in the discrete 0–255 encoding. The rules required discrete output images, deterministic and stateless submitted models, and an upper limit of four hours for generation of 1,000 perturbed examples [39]. The report identifies two 12 GB NVIDIA GPUs, a Titan X Pascal and a GeForce GTX, with a 32-core AMD CPU and 128 GB RAM. Container execution used Ubuntu 18.04 and CUDA 10.1. These are reported resource limits, not measured per-example runtimes.

The budget must be interpreted before any model-specific normalisation. For example, a linear rescaling from $[0, 1]$ to $[-1, 1]$ doubles the numerical distance. A reproducible evaluation therefore records both the input coordinate system and preprocessing.

6.4.2 SAFAIR Contest Scoring and Evaluation Protocol

The SAFAIR contest ranked submissions by the change between clean and perturbed accuracy,

$$\Delta = A_{\text{initial}} - A_{\text{final}}. \quad (6.2)$$

Attacks sought the largest Δ and defences the smallest. Unless stated otherwise, accuracies and Δ are proportions in this chapter; 100Δ is a difference in percentage points. D7.5 specifies clean accuracy and runtime as tie-breakers [38]. These do not prevent a low-utility model with a smaller Δ from outranking a more accurate model.

Table 6.1. Planned and reported SAFAIR evaluation procedures.

	Plan [162]	Reported delivery [38, 39]
Participants	attack and defence tracks for both tasks	defence submissions only
Attack selection	organiser baselines, then five selected participant attacks	four named organiser baselines
Defence score	weighted opening Δ ; mean final Δ over opponents	aggregate best and worst perturbed accuracies; Δ from the best value
Final comparison	five attacks against five defences	no participant-attack round
Reproduction	hidden evaluation set and submitted containers	sampling code and tool descriptions; incomplete run-level records

The planned opening round evaluated defences against FGSM, BIM and PGD, with respective weights 0.2, 0.4 and 0.4 [162]. FGSM makes one update in the sign of the input-loss gradient [114]. BIM repeats smaller FGSM-style updates and clips each result to the permitted input range and perturbation neighbourhood [165]. PGD also applies iterative gradient updates, projects each result onto the feasible perturbation set, and can use multiple starting points [115]. Carlini–Wagner instead formulates adversarial-example generation as an optimisation problem that balances the required classification outcome against perturbation size [166]. The five selected attacks would then face the five selected defences, with the final score averaged over opponents [162]. No attack submissions arrived. The organisers used fixed baseline attacks instead; D7.5 lists Carlini–Wagner in addition to FGSM, BIM and PGD [38]. Table 6.1 records the change in protocol.

The lack of independent attacks reduced the diversity of the evaluation. It does not explain all weaknesses of the resulting measurement. As derived in Section 6.8.4, a constant classifier has $\Delta = 0$ against any set of label-preserving attacks, including the opponents in the planned final round. The scoring defect and the attack-coverage defect are therefore distinct.

The contest drew inspiration from the NIPS 2017 competition, whose defence score was the proportion of correctly classified attacked images averaged across attacks [167]. That score differs from Equation 6.2. The competitions also differed in task and participation, so their results do not isolate the effect of independent contestants. Details of the schedule are retained in the cited project records; the scientific comparison here concerns the scoring and attack interfaces.

6.5 Data and Evaluation Measures

6.5.1 Corpora and Hidden Evaluation Samples

The contest used CelebA for model training, a corpus described by Liu et al. as approximately 200,000 images of 10,000 identities, with forty binary attributes per image [168]. For hidden attribute evaluation the organisers used LFWA, the forty-attribute annotation of Labeled Faces in the Wild, which contains 13,233 images of 5,749 identities [169, 168]. A separate corpus

Table 6.2. Evaluation units and data provenance. Stated sample sizes describe the printed procedures, not recovered per-example result files.

Evaluation	Input and prediction	Source and sampling
Contest identity	one image; one identity label	augmented CelebA; 1,000 selected test records [39]
Contest attributes	one image; forty binary labels	LFWA; 1,000 selected images [39]
Partner benchmark	one 50×50 histopathology patch; IDC/non-IDC	training, validation and test subsets described; evaluated subset size and patient separation unreported [170]
PDF comparison	21 tag counts; benign/malicious	20,000 files; 45/45/10 per cent defender/proxy/crafting split [38]

reduces direct image reuse, but does not establish equality of image distributions or attribute prevalences.

Attribute imbalance can give a constant predictor high mean accuracy, as reported for SD [39]. Attribute prevalences and per-attribute outcomes are therefore needed alongside the aggregate score; the available results do not establish demographic performance differences.

Re-identification required identity labels compatible with the trained classifier. The organisers therefore used CelebA images under a random choice of colour jitter, greyscale conversion, horizontal reflection or rotation of at most five degrees [39]. The printed procedure applies a face-detection check and retries failed transformations before excluding an image. The resulting selection may differ from an unfiltered sample. The report recognises that participants could have used similar augmentations during training. Augmentation alone does not demonstrate independence from training images.

6.5.2 Accuracy and Task-Specific Attack Success

For N examples with identity labels y_i , the decision is $h(x_i) = \arg \max_k f_{\theta,k}(x_i)$. Attribute decisions are $h_j(x_i) = \mathbf{1}[f_{\theta,j}(x_i) \geq 0]$ when the outputs are logits, with equivalent threshold 0.5 for sigmoid probabilities. The corresponding accuracies are

$$A_{\text{id}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[h(x_i) = y_i], \quad A_{\text{attr}} = \frac{1}{40N} \sum_{i=1}^N \sum_{j=1}^{40} \mathbf{1}[h_j(x_i) = y_{ij}]. \quad (6.3)$$

These conventions follow the task-specific scoring code printed in D7.6. Attribute accuracy counts individual attribute decisions; it does not require all forty predictions for an image to be correct [39].

Accuracy loss does not identify success against a specified target. For identity targets $t_i \neq y_i$, let $I_t = \{i : h(x_i) = y_i, t_i \neq y_i\}$ contain eligible, initially correct examples. Given valid attacked inputs x'_i , conditional targeted success is

$$\text{ASR}_{\text{target}} = \frac{1}{|I_t|} \sum_{i \in I_t} \mathbf{1}[h(x'_i) = t_i]. \quad (6.4)$$

A prediction of a third identity reduces accuracy without satisfying the targeted objective. An empty eligible set makes this rate undefined and its size must be reported.

For a nonempty nominated attribute subset J_i , define $I_J = \{i : h_j(x_i) = y_{ij} \text{ for all } j \in J_i\}$. One explicit success criterion is

$$\text{ASR}_{\text{subset}} = \frac{1}{|I_J|} \sum_{i \in I_J} \prod_{j \in J_i} \mathbf{1}[h_j(x'_i) \neq y_{ij}]. \quad (6.5)$$

This requires errors on every nominated attribute of an initially correct subset. Reporting $|I_J|$ separates induced errors from those already present; the rate is undefined if $|I_J| = 0$. Equations 6.4 and 6.5 specify measurements needed to evaluate the stated objectives; they are not additional scores recovered from the contest. If an attack flips all k initially correct bits in the nominated subset and leaves all other decisions unchanged, the per-image contribution to attribute accuracy falls by only $k/40$. For $k = 1$ this is 2.5 percentage points, although the specified attack succeeds.

6.5.3 Aggregation over Attacks

An attack suite can fail on different examples under each method. Selecting the lowest aggregate accuracy does not account for that complementarity. Following per-example evaluation guidance [21], let $\mathcal{A}_\varepsilon$ be a finite set of attacks whose outputs satisfy Equation 6.1. For a single-label task, define

$$\hat{A}_{\mathcal{A}}(\varepsilon) = \frac{1}{N} \sum_{i=1}^N \min_{a \in \mathcal{A}_\varepsilon \cup \{\text{id}\}} \mathbf{1}[h(a(x_i)) = y_i]. \quad (6.6)$$

The identity transformation includes clean errors in the score. The suite must respect the same threat model for each defence. Because it searches only a subset of valid perturbations, its surviving accuracy is an upper bound on worst-case accuracy on these examples, not a certificate of robustness. Moreover,

$$\hat{A}_{\mathcal{A}}(\varepsilon) \leq \min_{a \in \mathcal{A}_\varepsilon \cup \{\text{id}\}} \frac{1}{N} \sum_i \mathbf{1}[h(a(x_i)) = y_i], \quad (6.7)$$

since each per-example minimum is no greater than its value for any fixed attack. The aggregate tables do not identify which examples fail under which attack, so this suite score cannot be calculated retrospectively from them.

6.6 Benchmark Architecture and Reproducibility

The benchmark implements a common software procedure for adversarial evaluation. It separates attack execution from the task-specific model and data interfaces. D7.3 proposed five modules, which D7.6 reports as implemented: attacks, datasets and models, together with use cases and a wrapper [162, 39]. Figure 6.1 redraws the published architecture.

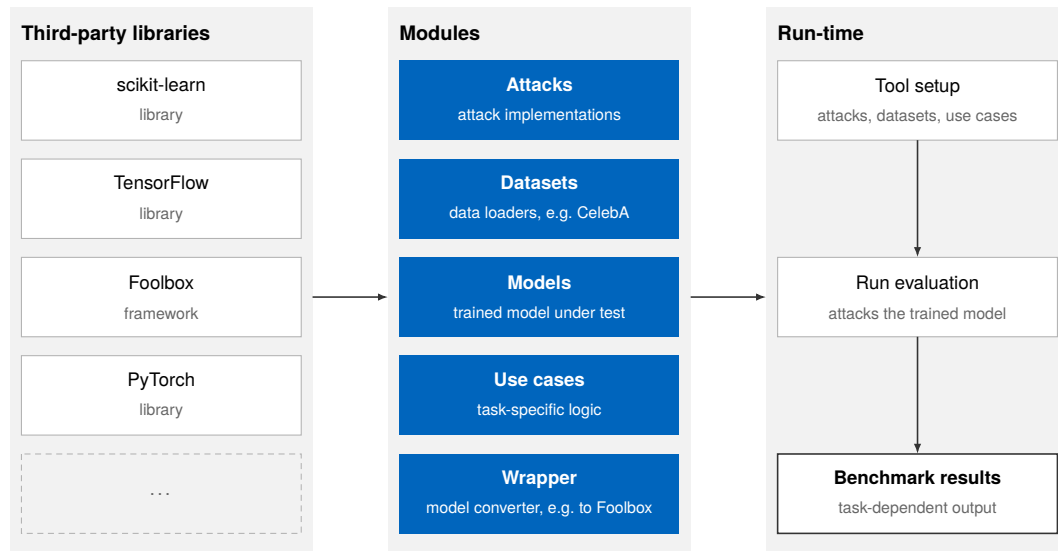


Figure 6.1. Benchmark architecture, redrawn after D7.6 Figure 2, p. 22 [39]. Box labels follow the original. The five modules separate task definitions and data loading from attack execution.

Foolbox supplies reference attack implementations [171]. The SAFAIR framework integrates them into a shared evaluation procedure with task-specific adapters. Identity classification requires a single-label loss, whereas facial attributes require a multi-label loss. The implementation selects cross-entropy or binary cross-entropy for these tasks and passes the selected loss to supported attacks [163]. Researchers can thus change the prediction task while retaining the evaluation loop.

A run loads trained model weights and evaluation data, converts the model to the attack-library interface, and records clean and attacked accuracies in JSON format. New attack classes extend a base class and are discovered by the loader; a configuration list excludes named attacks. PyTorch is the native model interface, and D7.6 describes conversion of TensorFlow or Keras models [39, 163]. The public benchmark source is separate from the contest development toolkit [40].

Reuse requires compatible prediction criteria and data processing. The multi-label adapter handles fixed-budget attacks but omits some minimum-distance interfaces. Vicomtech’s histopathology evaluation in Section 6.8.3 documents use on a different dataset and prediction task [39]. Standardisation here concerns the component interfaces and execution procedure.

Perturbation validation. The public revision identified by commit 2fac62b947ed pins Foolbox 3.3.1. Its attack call returns a raw candidate and a budget-clipped candidate, with a success indicator. However, lines 134–135 of `main.py` pass the raw candidate to the accuracy function [163]. In Foolbox’s minimisation interface the clipped candidate is obtained by restricting the raw perturbation to the requested distance budget; the raw output need not satisfy that budget [172].

A further distinction concerns targeted runs. The same implementation replaces the reference labels with the nominated target before computing its fields labelled accuracy. Those fields

then measure target-label agreement, with a different interpretation from true-label accuracy [163]. Each output therefore needs an explicit task, reference label and denominator.

6.7 Defence Mechanisms and Integration

Vicomtech used the benchmark described in Section 6.6 to evaluate five defence configurations documented in D7.2 [170, 39]. They instantiate the three defence strategies introduced in Section 6.1: adversarial training, transformations of inputs or internal representations, and adversarial-input detection.

The underlying histopathology classifier combines a pretrained VGG16 feature extractor with a dense prediction network [170, 39]. Write its scores as $f_\theta(x) = c_\psi(\phi_\omega(x))$, where ϕ_ω is the convolutional feature extractor and c_ψ the prediction head. The following notation makes the intervention of each defence explicit without supplying missing layer configurations.

6.7.1 Training and Representation Transformations

The adversarial-training configuration retrains the classifier with clean and perturbed examples. D7.6 identifies FGSM, BIM and PGD as the attacks used to generate its adversarial training examples [39]. This method exposes the model to those attacks during training; resistance to other attacks remains an empirical question.

For the dimensionality-reduction variants, let r_η denote an autoencoder. The two configurations have score functions

$$f_{\text{top}}(x) = c_\psi(\phi_\omega(r_{\eta_x}(x))), \quad f_{\text{mid}}(x) = c_\psi(r_{\eta_z}(\phi_\omega(x))). \quad (6.8)$$

The first transforms pixels and the second transforms extracted features. D7.2 states that the original convolutional network and prediction head retain their weights in these autoencoder variants. Its separate encoder-only variant trains a new prediction head, but that variant is not a column of the D7.6 benchmark table [170]. Figure 6.2 maps the initial and middle autoencoder designs of D7.2 to the Top AE and Middle AE configurations compared in Table 6.4.

A smaller reconstruction error alone does not establish robustness of either composition. The relevant attack must optimise against the complete transformed classifier, including its preprocessing. Reusing examples generated against the original classifier tests a different condition from attacking the defended composition [21].

6.7.2 History and Activation Detectors

Figure 6.3 shows the two partner detectors and their response paths.

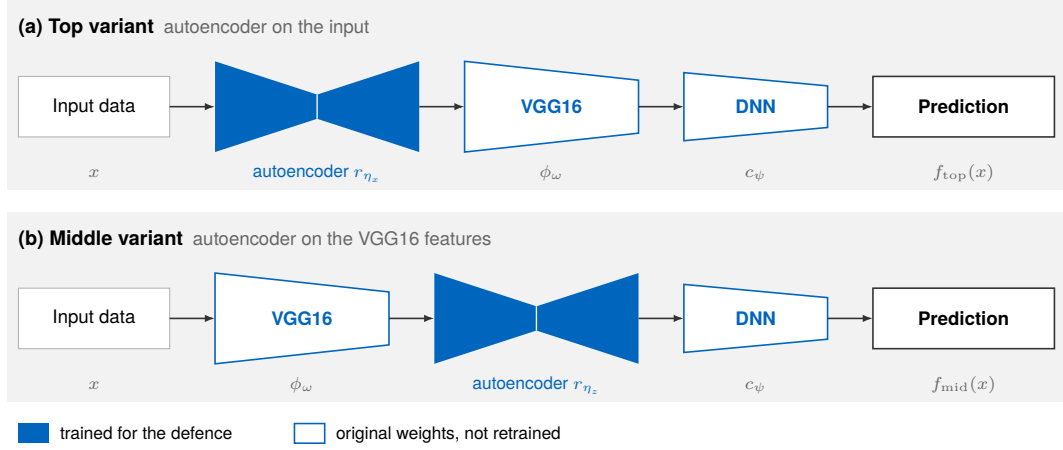


Figure 6.2. Autoencoder placement in the partner classifier, redrawn after D7.2 Figures 13 and 11, p. 14 [170]. In the top variant (a), the autoencoder r_{η_x} transforms the input before feature extraction. In the middle variant (b), r_{η_z} acts between the feature extractor ϕ_ω and the prediction head c_ψ , as in Equation 6.8.

Prediction similarity adds a history of queries and their predictions. The partner study records user information with image similarity and alarm statistics, and reports SSIM as the similarity measure [170]. Its alarm can be written $d_t = d(x_t, H_{t-1})$, where H_{t-1} contains the preceding queries associated with the monitored interaction. A sequence of related optimisation queries may then trigger an alarm. A transferred adversarial example submitted once presents a different observation to this stateful detector.

The activation detector instead characterises the response of the underlying classifier to an input. For each of nineteen hidden layers it calculates nine statistics: mean and median; skewness and kurtosis; standard deviation and interquartile range; minimum and maximum; and the number of nonzero activations [170]. With layer activations $a_\ell(x)$ and the statistic vector s , the representation is

$$z(x) = \text{concat}_{\ell=1}^{19} s(a_\ell(x)) \in \mathbb{R}^{171}. \quad (6.9)$$

After standardisation and feature selection, two support-vector classifiers are trained, one for each class predicted by the original model. The selected detector therefore depends on that prediction. The 94% detector accuracy reported in D7.2 concerns detection of adversarial inputs; it is not the defended task accuracy of the D7.6 table.

6.7.3 From an Alarm to a Classification

To place the detectors in an accuracy table, D7.6 uses the binary response policy of returning the opposite class whenever the detector raises an alarm [39]. Let $h(x) \in \{0, 1\}$ denote the base prediction, $d(x) \in \{0, 1\}$ the alarm and $g(x)$ the returned class. For a fixed history, where applicable,

$$g(x) = h(x) \oplus d(x), \quad (6.10)$$

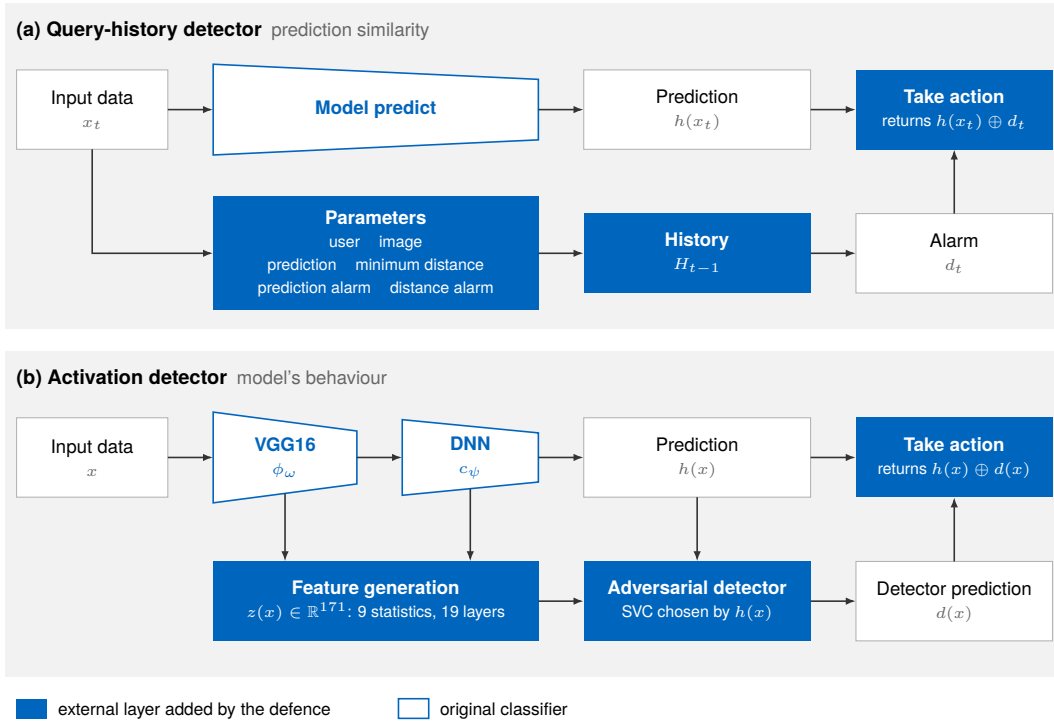


Figure 6.3. Partner detection mechanisms, redrawn after D7.2 Figures 15 and 16, p. 16 [170]. Panel (a) uses preceding queries, whereas panel (b) uses internal activations of the classifier. The recorded parameters, the activation features of Equation 6.9 and the per-class detector follow D7.2. The response is the label reversal of Equation 6.10, which D7.6 applies in the benchmark [39]. The final action is part of the defended system and must be included when task accuracy is measured.

Table 6.3. Reported contest scores, including the worst-perturbed column, from D7.6 Table 1, p. 16 [39]. Task names follow that table. The SD task label conflicts with its detailed attribute-model description; best and worst values are not identified as paired outcomes for one checkpoint under a common attack suite.

Rank	Team	Task	Clean	Best	Worst	Δ
1	SD	re-identification	0.87	0.85	0.67	0.02
2	Vicomtech	re-identification	0.85	0.72	0.72	0.13
	Vicomtech	attribute alteration	0.80	0.65	0.65	0.15
3	BPI	re-identification	0.84	0.68	0.68	0.16

where $\oplus$ denotes exclusive OR. This rule corrects an erroneous binary prediction when the detector raises an alarm, but turns a correct prediction into an error under the same alarm. Consequently, on a common labelled evaluation set,

$$A(g) - A(h) = \Pr(d = 1, h \neq y) - \Pr(d = 1, h = y). \quad (6.11)$$

The probabilities denote empirical fractions on that set. To obtain the identity, partition the observations by the alarm and the correctness of h : unflagged predictions are unchanged, flagged errors are corrected, and flagged correct predictions are reversed. If $d(x) = 1$ for every observation, then $A(g) = 1 - A(h)$, while g can still depend on the input.

Detection accuracy alone does not determine either term in Equation 6.11. A useful evaluation records alarms jointly with base and returned predictions, and measures false alarms on clean inputs. Abstention would be a different response policy, for which coverage and error among accepted predictions must be reported [21]. The historical benchmark reports the reversal policy; its aggregate accuracies do not identify the two joint fractions.

6.8 Results and Metric Analysis

6.8.1 Contest Evidence

Table 6.3 preserves the numerical columns of D7.6 Table 1. The source reports one row per ranked team and task, rather than one row per submitted checkpoint. Its surrounding text describes best and worst scores across submissions, but does not link each value to an attack and checkpoint [39].

The SD row reports 0.85 as best perturbed accuracy and 0.67 as worst, a spread of eighteen percentage points. The printed $\Delta = 0.02$ uses the best value. Subtracting 0.67 from 0.87 would give 0.20, but it would not establish a worst-case loss for the same model without the missing checkpoint association. The report’s prose also substitutes 87% for attacked accuracy and gives 61% and 88% values that do not match its table. The numerical record used here is the table, with those inconsistencies retained as limits on interpretation.

6.8.2 SD Contest Submissions

The contest permitted each participant up to three submissions, and D7.6 describes the three models submitted by SD [162, 39]. Although the report introduces them as re-identification models, their technical descriptions specify forty attribute outputs. The analysis therefore follows the attribute-model description; their mapping to the official SD row in Table 6.3 remains unresolved.

The first submission used a convolutional network trained on PGD examples with $\varepsilon = 0.3$, approximately 9.6 times the contest's 8/255 budget if both use $[0, 1]$ pixels. The team reports that training produced the same output for every input, with approximately 80% clean attribute accuracy and zero accuracy loss. It attributes that accuracy to the prevalence of absent attributes and recognises a weakness in the dataset and contest rules. A constant attribute vector cannot be interpreted as a demonstrated identity-classification result, and neither the checkpoint nor individual predictions accompany the reported collapse.

The second submission combines six EfficientNet and three pruned ECA-ResNet models. Their averaged outputs are thresholded to obscure gradients. For a nonzero scalar attribute logit u , write the transformation as

$$q(u) = \begin{cases} 10, & u > 0, \\ -10, & u < 0. \end{cases} \quad \mathbf{1}[q(u) \geq 0] = \mathbf{1}[u \geq 0], \quad q'(u) = 0 \ (u \neq 0). \quad (6.12)$$

For $u \neq 0$, thresholding preserves the binary decision while suppressing its direct derivative. Adaptive or transfer attacks are therefore needed to assess this construction [19, 20].

The second submission achieved approximately 92% clean accuracy and 79% under transferred attacks in the team's development account. The third submission used one pruned ECA-ResNet, added adversarial training, and retained the threshold operation. Its reported clean accuracy was approximately 87%, with 84–85% under transferred attacks [39]. Figure 6.4 compares the reported values for these two submissions.

The third submission trades five percentage points of reported clean accuracy for five to six points of transferred-attack accuracy relative to the second submission. This is a descriptive comparison between systems: ensemble size and training changed together, and the attacks' individual settings are not given. It does not isolate the effect of adversarial training.

6.8.3 Histopathology Benchmark

D7.6 reports that Vicomtech used the benchmark tool to evaluate five defence configurations on a histopathology classification task [39, Section 3.1.3]. This application documents use of the framework for a prediction problem distinct from the competition's facial image tasks. Table 6.4 reproduces the reported accuracies for binary IDC classification. For the two detectors, these include the response policy in Equation 6.10. The clean row records the prediction cost associated with each configuration.

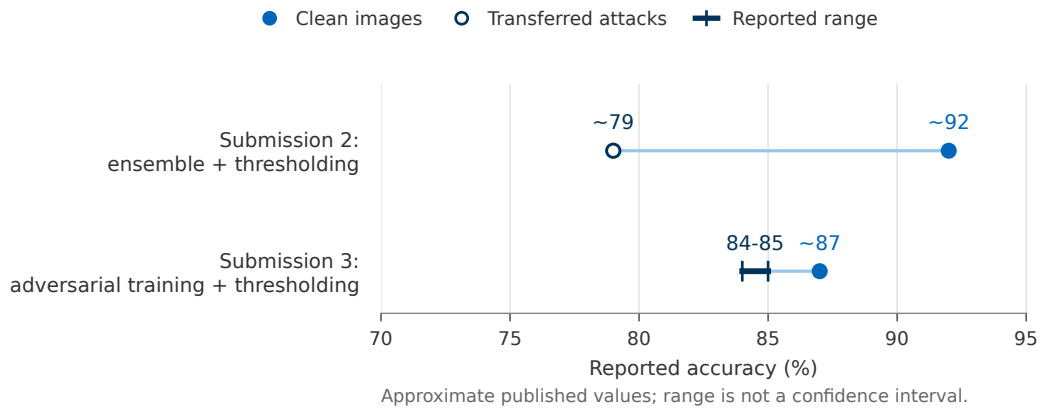


Figure 6.4. Reported clean and transferred-attack accuracies for SD submissions 2 and 3, from D7.6 Sections 2.8.1.2–2.8.1.3, pp. 18–19 [39]. Submission 1 is omitted because D7.6 reports no comparable transferred-attack accuracy for its collapsed model. The 84–85% interval is an approximate reported range, not a confidence interval; these development values are separate from the official contest scores.

Table 6.4. Published accuracies for the five partner defence configurations, from D7.6 Table 2, pp. 28–29 [39]. Attack hyperparameters and the evaluated subset are unspecified. Each entry is a reported aggregate, with no dispersion estimate.

Attack	Adv. training	Top AE	Middle AE	Pred. similarity	Activations
none (clean)	0.844	0.726	0.817	0.844	0.708
FGSM	0.837	0.716	0.812	0.837	0.708
L_2 basic iterative	0.844	0.725	0.817	0.842	0.708
BIM	0.837	0.716	0.812	0.837	0.708
DeepFool	0.003	0	0.001	0.015	0.708
additive noise	0.844	0.725	0.817	0.844	0.708
NewtonFool	0.464	0	0	0.460	0.708
PGD	0.837	0.717	0.813	0.837	0.708

DeepFool iteratively linearises the classifier around the current input and moves towards the closest estimated decision boundary [173]. NewtonFool applies gradient steps that reduce the original-class softmax probability while bounding the perturbation at each iteration [174].

The middle autoencoder exceeds the top autoencoder by 9.1 percentage points on clean data and 9.6 points under the reported PGD condition. Its PGD loss is 0.4 points, compared with 0.9 for the top variant. These values favour the middle configuration within the reported evaluation. They do not establish that placement alone caused the difference, because the complete training configurations and repeated-run outcomes are unavailable.

Adversarial training and prediction similarity have equal reported clean accuracy and equal FGSM, BIM and PGD values. Prediction similarity is higher under DeepFool by 1.2 percentage points, but lower under NewtonFool by 0.4 points. It therefore does not dominate adversarial training across the table. The source’s explanation that similarity detects gradient-based construction is insufficient to explain the results, since DeepFool and NewtonFool also use gradient information. Query sequences and detector alarms would be needed to test that mechanism.

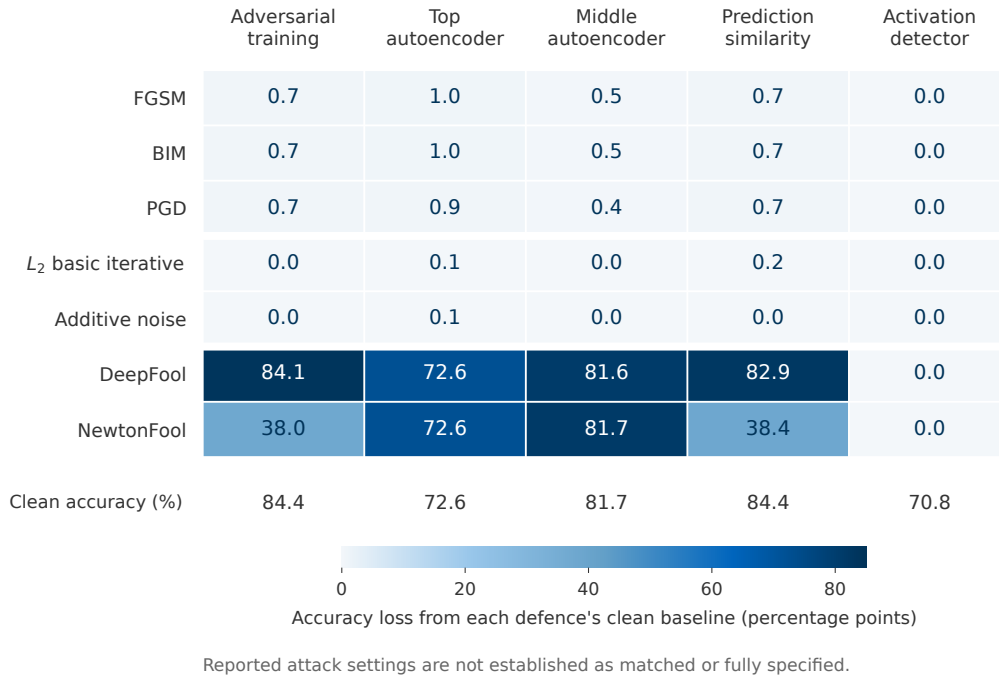


Figure 6.5. Accuracy losses computed from D7.6 Table 2 [39]. Each cell is $100(A_{\text{clean}} - A_{\text{attack}})$ in percentage points; the original accuracies remain in Table 6.4. The separation of attack rows does not imply matched budgets. In particular, zero loss for the activation detector describes its aggregate score, not a demonstrated mechanism or robustness certificate.

The near-zero DeepFool results for four configurations require separate interpretation. DeepFool searches for an adversarial input with low distortion and should be assessed by the distortion required, or by success under a common valid budget [21]. Neither the norm and stopping settings nor achieved distortions are reported for these benchmark runs. The table therefore cannot establish that its attack rows represent equally constrained adversaries. The code finding in Section 6.6 reinforces the need to validate distances, but does not resolve the historical configuration. The additive-noise row is a useful control condition; without its distribution, budget and number of trials, its small changes do not establish a comprehensive random-search check.

6.8.4 Sensitivity to the Scoring Rule

Figure 6.5 subtracts each attacked accuracy from that configuration's clean accuracy. It makes the score dependence visible without changing the original results. Under PGD, the activation detector has the smallest Δ , zero, but its attacked accuracy is 12.9 percentage points below adversarial training. Similarly, the middle autoencoder has a smaller loss than adversarial training while its attacked accuracy is 2.4 points lower. A smaller loss alone therefore does not identify the more useful defended classifier.

For paired clean and attacked examples, let C denote an initially correct prediction and C' a correct prediction after attack. Expanding the two accuracy terms yields

$$\Delta = \Pr(C \cap \neg C') - \Pr(\neg C \cap C'). \quad (6.13)$$

Both accuracies contain the correctly classified observations in $C \cap C'$, which cancel. Thus, Δ measures the net change in correctness. It is not the conditional fraction of initially correct examples made incorrect. Equal numbers of harmful and corrective changes can leave $\Delta = 0$ despite changes in individual outcomes. The aggregate table cannot determine those two counts.

Constant-prediction result. For any constant classifier $h_c(x) = c$ and label-preserving attack a , $h_c(a(x_i)) = h_c(x_i)$ for each observation. Therefore,

$$A_{\text{clean}}(h_c) = A_{\text{attack}}(h_c) = \hat{p}_c, \quad \Delta(h_c) = 0, \quad (6.14)$$

where $\hat{p}_c$ is the evaluated class fraction. The statement holds for each attack and any weighted average over attacks. Independent opponents cannot eliminate this property of the score. The result establishes an indifference to prediction utility when ranking only by loss; it does not imply that the constant classifier correctly classifies the other classes.

For attributes, choose a fixed vector m using training-set majority labels. Its held-out accuracy is

$$A_{\text{attr}}(m) = \frac{1}{40} \sum_{j=1}^{40} [m_j \hat{p}_j + (1 - m_j)(1 - \hat{p}_j)], \quad \Delta(m) = 0, \quad (6.15)$$

where $\hat{p}_j$ is the prevalence of attribute j in the evaluation set. When the training and evaluation majorities agree, each summand is $\max(\hat{p}_j, 1 - \hat{p}_j)$. This explains how a fixed attribute vector can have high average accuracy on an imbalanced corpus. The baseline labels must be fixed before evaluation; selecting them from hidden test labels would use unavailable information.

These identities support reporting clean utility together with attacked performance and a matched constant baseline. Prediction distributions and paired error transitions provide additional checks. None of these quantities can be reconstructed uniquely from an unchanged accuracy column, so the activation result remains an observation rather than a second demonstration of the SD collapse mechanism.

6.8.5 Comparison with Constrained PDF-Malware Features

D7.5 reports a separate partner study of PDF-malware evasion that illustrates domain-specific input constraints [38]. It uses 10,000 benign and 10,000 malicious Contagio files, with 21 PDFiD tag counts as features. The split assigns 45% to defender training, 45% to proxy training and 10% to adversarial crafting. A neural proxy generates examples transferred to neural, SVM and random-forest classifiers without target queries.

Tag counts must remain integer-valued and cannot decrease. For an original vector $v \in \mathbb{N}_0^{21}$, the reported constraints require

$$v' \in \mathbb{N}_0^{21}, \quad v'_j \geq v_j \quad (j = 1, \dots, 21). \quad (6.16)$$

A perturbation bound applies in the standardised representation, followed by inverse transformation and rounding. These feature constraints do not establish that a corresponding

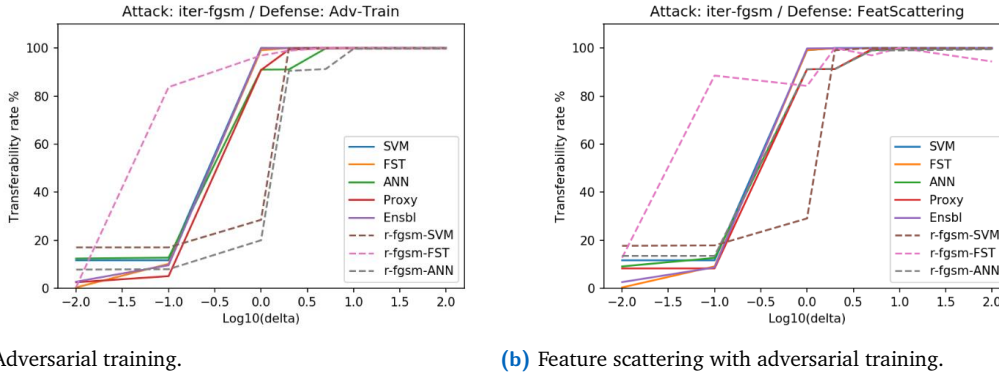


Figure 6.6. Original iterative-FGSM transfer curves from the partner PDF-malware study, D7.5 Figure 15, p. 23 [38]. The horizontal axis is the source's $\log_{10}(\delta)$ perturbation budget, distinct from the accuracy loss Δ . The published feature-space results contain no uncertainty bands.

modified PDF was constructed with its malicious functionality preserved, as required by the problem-space distinction in Section 2.5.

The legends identify the SVM, random forest (FST), neural network (ANN) and ensemble (Ensbl). The prefix r-fgsm denotes adversarial retraining with FGSM examples; Proxy is the attack's source model. The feature-scattering comparison also retrains the SVM and random forest with adversarial examples [38].

Figure 6.6 reports model-dependent transfer success across perturbation budgets. Some curves are non-monotone: they describe the tested searches, not a worst-case envelope over all admissible perturbations. The comparison supports budget-dependent reporting under explicit feature constraints, but not a general ranking of defence families. For Android, the constraints must describe valid application changes and their effect on the detector's representation.

6.9 Threats to Validity

Measurement validity. The contest combines task-dependent accuracies with a net-change score. The identity and attribute objectives require different success criteria, and the targeted software output uses target-label agreement. Aggregate accuracy loss cannot identify induced errors, useful classification or detector alarms. The analytical results in Sections 6.7 and 6.8.4 identify these limits; they do not recover unreported individual outcomes.

Attack coverage and budgets. No independent attack submissions entered the contest. Its fixed gradient-based baselines did not constitute a documented adaptive assessment of every defence. The SD development account includes transfer attacks, but the official table provides no per-attack breakdown. The benchmark additionally mixes attack families without the full configurations needed for comparison. A later evaluation would require valid perturbations in common coordinates and attacks adapted to the complete defended system, with convergence checks [21, 20].

Generalisation and attribution. The four participating teams belonged to SPARTA, so the contest did not test participation beyond the consortium. The evaluated image tasks also differ from Android malware analysis in their inputs and validity constraints. The public PDF-malware study narrows this conceptual difference through constrained count features, but supplies no Android result. Its experiments and the partner defence algorithms retain their attribution. D7.4 is a confidential deliverable; the figures and empirical comparisons in this chapter use public reports.

A focused subsequent study could compare an ordinary classifier with a training-selected constant baseline and the same classifier under output thresholding. For detectors, paired base predictions and alarms would test Equation 6.11. Budget curves and per-example attack aggregation would then measure how the conclusions change under validated perturbations. Those experiments remain separate from the historical evidence presented here.

6.10 Individual Contribution and Project Attribution

The candidate's contribution is stated at section level, following his account of the collaborative project. He wrote D7.1 Sections 2.2.2 and 2.3, and D7.2 Sections 2.1 and 2.2. For D7.2 Sections 2.4 and 2.5 he contributed the concept and definitions. He edited and refined partner text in D7.3, with no authored content claimed for that plan. He wrote Sections 3.1 and 3.2 of D7.4, Chapter 5 of D7.5, and the whole of D7.6 [37, 170, 162, 38, 39]. His D7.4 contribution is recorded as part of this account, without using the confidential report as an empirical source.

Within SAFAIR, the candidate contributed to organising the competition and to developing the reusable benchmark framework. The contest toolkit and benchmark source code are published on GitHub under the candidate's name [40, 163]. The evaluated defensive algorithms remain partner methods. In particular, the histopathology configurations originate in the Vicomtech study of D7.2 Section 2.3.2; the PDF-malware curves belong to the partner study in D7.5 Section 2.8. Authorship of the reporting deliverables does not transfer those experiments or algorithms to the candidate. The mathematical analysis and source comparison in this chapter examine their implications for the evaluation framework.

6.11 Chapter Summary

The SAFAIR work addressed RQ4 through a competition and a reusable benchmark framework. The competition evaluated submitted defences on hidden samples under explicit input constraints. SD's reported constant-prediction submission exposes a weakness of accuracy-loss ranking: a constant predictor can obtain zero loss despite limited predictive utility. Section 6.8.4 proves that this property persists under any label-preserving attack. Logit thresholding raises a separate evaluation problem because it obstructs direct gradients while preserving attribute decisions.

The framework provided a common evaluation procedure with task-specific interfaces. Vi-comtech used it to evaluate histopathology defences, which documents an application outside the contest's facial-image tasks. The resulting table illustrates why the scoring choice matters: a configuration with smaller accuracy loss can have lower attacked accuracy. For detector-based systems, the comparison also depends on the response to an alarm; unchanged aggregate accuracy alone does not identify the underlying mechanism.

Evaluators should therefore report clean and attacked utility together and test attacks against the complete defence. Per-example outcomes and valid perturbation budgets are needed to interpret the scores. The contribution combines a documented evaluation design with an analysis of its measurement properties. The available aggregates do not establish a general relation between clean performance and adversarial robustness, and experiment-level reproduction remains unresolved.

Cross-Contribution Discussion

Chapters 3 to 6 contribute methods for representing security-relevant context and procedures for evaluating learning-based decisions. This discussion relates their design choices to the evidence each study provides. The comparison between Hybroid and HGANN-Mal is conceptual; empirical comparisons remain within the corpus and protocol of each study.

7.1 Contributions of Contextual Representations

The IoT service-graph design makes communication partners part of the detection decision. Each node stores observed relations as its reference and permits an operator to approve new legitimate interactions [32]. In the industrial work, NADICS represents network connections through protocol features, while the SWaT detector analyses process measurements in multivariate time windows. Chapter 3 thus connects representation design to the unit of observation available in each setting.

Hybroid studies context across evidence sources. Its static graph describes program structure, and its flow features summarise network behaviour observed during execution [35]. The within-system comparison in Table 4.6 reports higher detection F_1 for fusion than for either branch alone. For category classification, fusion improved the ensemble results, while the decision tree favoured static features. The contribution includes a direct measurement of the value of combining these observations for the tested tasks and learners.

HGANN-Mal studies context within the application package. It groups methods by call neighbourhood and semantic similarity; slice-aware hyperedges add relations around security-sensitive call sites [36]. These groups define which methods contribute together to a hyperedge representation. The hypergraph configurations achieved higher reported accuracy than the pairwise baselines for binary detection and for the dataset-specific classification tasks: families on Drebin and categories on CICMalDroid (Tables 5.2 and 5.3). These are comparisons of complete configurations. Comparisons with non-attentive operators vary both construction and weighting; an attention-only effect requires a controlled ablation.

The operator analysis in Section 5.8 gives a separate account of what learned weighting can represent. Different hyperedge groupings can produce different attentive propagation even when they produce the same fixed propagation matrix. This capacity depends on the grouping retained during construction: attention adjusts existing memberships. The distinction connects the model's mathematical properties to the program relations made available by static analysis.

The SWaT comparison adds an operating-objective perspective. Full process inputs gave the highest reported recall, while the PCA configurations reduced the false-positive rate (Table 3.6). The preferred configuration therefore depends on the relative importance of detecting anomalous observations and reducing false-positive observations. Across these studies, the useful question is which context serves the prediction task, with the comparison made under a stated protocol.

7.2 Observation Requirements and Analysis Cost

Static and multimodal analysis place different requirements on data acquisition. Hybroid’s evaluation reused traffic supplied by CICAndMal2017 alongside each application package. Its fused representation therefore requires a recorded execution. HGANN-Mal constructs its representation from the package, so it can be applied without arranging a runtime capture. This separates the effort of obtaining an additional observation from the computational work of modelling static structure.

Chapter 5 also provides a computational account of higher-order representation. The hyperedge cardinality cap bounds retained memberships, and the sparse representation grows linearly with the numbers of methods and slices (Equation 5.22). Construction adds call-graph searches and comparisons between semantic embeddings. The bounds in Section 5.13 distinguish this stage from propagation and make their scaling assumptions explicit. An end-to-end runtime comparison would require measurements under common hardware and workloads.

Observation requirements also define the evaluation cohort. Hybroid compares modalities on the same applications with both representations available, which permits a matched comparison on that cohort. CICMalDroid supplies applications retained by its authors’ dynamic-analysis procedure [160]; HGANN-Mal then extracts static representations from these packages. The processing requirements of a detector and the acquisition history of its benchmark are therefore separate properties. Recording both makes explicit which applications and operating conditions a reported result describes.

7.3 Evaluation Principles Across the Contributions

Representation design determines the evidence available to a detector. Predictive evaluation measures how effectively it uses that evidence, while adversarial evaluation examines its decisions when an attacker deliberately modifies the input. The SAFAIR contest and benchmark in Chapter 6 contribute a procedure for investigating this second question under explicit attacker assumptions.

Reusable software provides a second connection between the contribution chapters. NADICS places estimators behind a common interface, so that researchers can change learning methods without rewriting the surrounding pipeline [33]. SAFAIR applies a corresponding separation to evaluation: attack implementations share an execution procedure, while task-specific components define the model and data interfaces [39]. Vicomtech’s application of the framework

to histopathology defences demonstrates reuse on a prediction task outside the facial-image contest [39, Section 3.1.3.1]. Both frameworks separate components that researchers can vary within one configured procedure.

The Android studies provide within-system comparisons under their own protocols. Hybroid holds the applications and learner fixed when comparing input modalities. The HGANN-Mal evaluation supplies common call graphs and node features to its relational configurations. Matching inputs and protocols makes these contrasts interpretable; their reported margins remain point estimates. Time-ordered evaluation addresses the additional question of performance on later observations [12]. Each comparison can thus be read against the conditions it was designed to examine.

Chapter 6 adds analytical criteria for interpreting evaluation scores. A constant predictor has zero accuracy loss under label-preserving attacks (Equation 6.14), even when its classification utility is limited. Clean and attacked accuracy therefore answer different questions and should be reported together. Under the benchmark's binary label-reversal response, Equation 6.11 relates the change in task accuracy to the correct and incorrect base predictions that trigger an alarm. These derivations make the scoring procedure part of the scientific contribution. Their assumptions can be examined independently of a particular benchmark, and they specify outcomes needed to assess the complete defended system [21].

7.4 Implications for Android Malware Analysis

For Android analysis, the distinction between acquiring evidence and structuring it gives two design choices that can be investigated together. A common-cohort evaluation could compare static relational encodings and then measure the additional value of traffic for the same prediction task. Acquisition effort and construction time would give those performance differences an operational interpretation.

The SAFAIR evaluation design provides a basis for extending such a comparison to an explicit adversary. Android-specific tests would use valid package transformations that preserve malicious functionality and would assess the complete pipeline, including fusion where applicable [21]. The Android studies establish predictive results; resistance to those transformations remains a question for that evaluation. Together, the contributions relate contextual modelling to the evidence and measurement procedures needed to assess a security-analysis system.

Conclusion and Future Work

8.1 Answers to the Research Questions

RQ1. *How can behaviour in networked and industrial systems be represented to support anomaly detection?* Communication graphs represent service relationships, while protocol records and process windows support industrial anomaly detection. Chapter 3 addresses gap G1 through these implementations.

RQ1.1. *How can service relationships be represented as communication graphs to identify unexpected interactions?* A node-local graph stores relations observed during service start-up. Unexpected relations trigger an alert and can be approved by the site owner as the deployment changes.

RQ1.2. *How can a modular learning framework analyse network and process features for anomaly detection in industrial control systems?* NADICS separates feature preparation from interchangeable learning components. On the session-disjoint S7-300 evaluation, a random forest achieved 94.50 per cent attack F_1 at a record-level false-positive rate of 0.484 per cent. The GAN-based extension with 24 PCA components achieved 92.00 per cent timestamp recall at a false-positive rate of 5.12 per cent on later SWaT test data.

RQ2. *To what extent does fusing static code-graph representations with dynamic network-flow features improve Android malware detection and category classification compared with either modality alone?* On CICAndMal2017, fusion improved reported detection F_1 by one or two percentage points over the better single modality. For category classification, the difference ranged from minus two to plus three points across learners. These matched comparisons address gap G2 within the evaluated corpus and protocol.

RQ3. *To what extent do attention-weighted hypergraph representations of static Android program structure improve binary malware detection and malware classification over pairwise graph models and non-attentive hypergraph models?* On Drebin and CICMalDroid, all hypergraph configurations achieved higher reported accuracy than the pairwise baselines with common call graphs and node features. HGANN-Mal exceeded the strongest non-attentive baseline by 1.2 to 4.1 percentage points in three of four task and corpus combinations; on CICMalDroid binary detection, it scored 97.6 against 97.7 per cent. These results address gap G3 through comparisons of complete systems. Attention-specific attribution requires a controlled ablation.

RQ4. *How can an explicit adversary model be translated into a reproducible benchmark for adversarial attacks and defences, and what does such a benchmark reveal about the relation*

between clean predictive performance and adversarial robustness? SAFAIR translated attacker assumptions into a contest protocol and a benchmark with task-specific interfaces. The implementation and scoring analysis address the design and measurement aspects of gap G4. Any constant classifier has zero accuracy loss under label-preserving attacks. In Vicomtech's histopathology benchmark, the activation detector had zero loss under PGD but an attacked accuracy 12.9 percentage points below adversarial training. Clean and attacked accuracy therefore need to be reported together, under an explicit prediction objective and valid perturbation constraints. The four contest rows do not establish a general relation between clean performance and robustness.

8.2 Summary of Contributions

The Android studies form the main contribution. Hybroid combines static program-graph representations with observed network flows and measures the value of fusion on the same applications. The candidate designed the overall pipeline and traffic analysis and conducted the evaluation; his contribution to graph processing was partial. HGANN-Mal represents context within the application package through groups of related methods and learned membership weights. Chapter 5 formalises the construction and analyses the operator's representational capacity. Different hyperedge groupings can produce different attentive propagation even when their fixed propagation matrices coincide.

The industrial contribution is a modular framework for network and process observations, supported by the candidate's extractor and labelled corpus, with a GAN-based extension for SWaT. His contribution to the IoT study concerns the communication-graph concept and measurement design. SAFAIR contributes a threat-guided evaluation design and reusable software with task-specific interfaces. Chapter 6 analyses accuracy-loss scoring and the effect of binary label reversal after an alarm. Partner defences provide applications of the framework.

8.3 Future Work

The first priority is a controlled HGANN-Mal ablation on both corpora. Comparing learned and fixed weighting over identical fused hyperedges requires common preprocessing and a fixed normalisation convention, with repeated runs. A separate comparison with topological hyperedges alone would assess the contribution of construction. A common Android cohort with time-ordered splits would permit comparison of static relational encodings and the additional value of traffic, as proposed in Chapter 7. Acquisition effort and computation cost should be measured separately. Adversarial evaluation would then apply valid package transformations that preserve malicious functionality, under stated attacker access and budgets. It should measure clean and attacked performance of the complete pipeline, including fusion.

The dissertation connects contextual representation design with the evidence and evaluation procedures needed to assess learning-based security decisions.

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Feature Importance of the NADICS Extraction Fields

Figure 3.4 in Section 3.4.1 names only the fifteen most important fields of the NADICS extraction schema. Table A.1 lists all 63 in rank order, with the importance of each field and the cumulative importance down to its rank. The cumulative column gives the thresholds of panel (c) of Figure 3.4 exactly: the first ten fields hold 54.47 per cent of the total, the first 24 hold 80.72 per cent and the first 36 hold 90.41 per cent. The remaining 27 fields share 9.59 per cent. Fields of equal importance keep the order of the source record, and the highest process field, `setpoint_change_count`, stands at rank 19.

Table A.1. Importance of the 63 fields of the NADICS extraction schema in the random forest of Section 3.4.2, in rank order. Importance is the normalised share of the total, and the cumulative column sums the importances down to that rank. The letter of the identifier gives the group of Table A.2, and the bar colours follow Figure 3.4.

Rank	ID	Field	Group	Importance (%)	Cumulative (%)	
1	C2	s7_function	Protocol	11.93	11.93	
2	D6	role_violation	Relational	7.95	19.88	
3	B6	period_deviation	Timing	6.76	26.64	
4	B11	unanswered_req_ratio	Timing	4.77	31.41	
5	C12	err_rate	Protocol	4.57	35.98	
6	D1	is_known_edge	Relational	4.37	40.36	
7	C5	write_read_ratio	Protocol	3.78	44.14	
8	A9	rate	Flow	3.58	47.71	
9	D3	n_distinct_dst_1min	Relational	3.48	51.19	
10	C9	addr_entropy	Protocol	3.28	54.47	
11	C11	s7_error_code	Protocol	2.88	57.36	
12	B3	iat_cv	Timing	2.58	59.94	
13	B7	cycle_count_gap	Timing	2.49	62.43	
14	D4	n_distinct_ports_1min	Relational	2.39	64.81	
15	B10	resp_latency_max	Timing	2.19	67.00	
16	C4	n_write_items	Protocol	2.09	69.09	
17	B8	cycle_jitter	Timing	1.79	70.87	
18	B9	resp_latency_mean	Timing	1.79	72.66	
19	E4	setpoint_change_count	Process	1.69	74.35	
20	C10	n_distinct_db	Protocol	1.59	75.94	
21	C15	n_tsap_tried	Protocol	1.29	77.24	
22	A3	service	Flow	1.19	78.43	
23	A4	state	Flow	1.14	79.57	
24	C3	n_read_items	Protocol	1.14	80.72	
25	A1	duration	Flow	1.09	81.81	

continued on the next page

Table A.1. (continued)

Rank	ID	Field	Group	Importance (%)	Cumulative (%)	
26	D5	role_pair	Relational	1.09	82.90	■
27	C6	s7_area	Protocol	0.89	83.80	■
28	D7	dcp_identify_count	Relational	0.84	84.64	■
29	A14	src_loss	Flow	0.80	85.44	■
30	B5	period_power	Timing	0.80	86.23	■
31	A5	src_pkt	Flow	0.75	86.98	■
32	B2	iat_std	Timing	0.75	87.72	■
33	A10	src_load	Flow	0.70	88.42	■
34	C1	s7_rosctr	Protocol	0.70	89.12	■
35	A6	dst_pkt	Flow	0.65	89.76	■
36	D2	edge_age	Relational	0.65	90.41	■
37	A7	src_bytes	Flow	0.60	91.00	■
38	B13	is_new_session	Timing	0.60	91.60	■
39	C18	opcua_session_rate	Protocol	0.60	92.20	■
40	A15	dst_loss	Flow	0.55	92.74	■
41	B4	dominant_period	Timing	0.55	93.29	■
42	A8	dst_bytes	Flow	0.50	93.79	■
43	D9	new_mac_seen	Relational	0.50	94.28	■
44	A12	src_mean_sz	Flow	0.45	94.73	■
45	B1	iat_mean	Timing	0.45	95.18	■
46	C7	s7_db_number	Protocol	0.45	95.63	■
47	E2	value_delta	Process	0.45	96.07	■
48	A11	dst_load	Flow	0.40	96.47	■
49	E5	value_freeze_len	Process	0.40	96.87	■
50	B12	session_age	Timing	0.35	97.22	■
51	E3	value_out_of_range	Process	0.35	97.56	■
52	A13	dst_mean_sz	Flow	0.30	97.86	■
53	C8	s7_start_addr	Protocol	0.30	98.16	■
54	C14	cotp_tsap_slot	Protocol	0.30	98.46	■
55	A2	protocol	Flow	0.25	98.71	■
56	C13	pdu_ref_gap	Protocol	0.25	98.96	■
57	C17	opcua_sec_policy_none	Protocol	0.25	99.20	■
58	A18	tcp_rtt	Flow	0.20	99.40	■
59	C16	opcua_msg_type	Protocol	0.15	99.55	■
60	D8	arp_conflict_count	Relational	0.15	99.70	■
61	A16	syn_ack	Flow	0.12	99.82	■
62	E1	tag_value	Process	0.10	99.92	■
63	A17	ackdat	Flow	0.08	100.00	■

Table A.2 gives the group totals behind panel (b) of Figure 3.4, together with the importance per field that Section 3.4.1 discusses and the rank of the highest field in each group.

Table A.2. Importance by field group. Both shares are percentages of the 63-field total. Importance per field is the group share divided by the number of fields in the group, and the last column gives the rank of the group's highest field in Table A.1.

Group	Fields	Importance share (%)	Field share (%)	Importance per field (%)	Highest rank
A ■ Flow base	18	13.32	28.57	0.74	8
B ■ Timing / periodicity	13	25.84	20.63	1.99	3
C ■ Protocol semantics	18	36.43	28.57	2.02	1
D ■ Relational	9	21.42	14.29	2.38	2
E ■ Process derived	5	2.98	7.94	0.60	19
All groups	63	100.00	100.00	1.59	